

Bio 320 Essentials of Endocrinology

Numbered Transcript

BIO 320 • Dr. Benjamin Bikman

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Lecture 1: The Science of Signaling

At a glance — Bikman opens on endocrine signaling — what hormones are and how cells communicate across the body.

- 1.1 I am absolutely thrilled to be your temporary professor, the topic of which is perhaps the most fascinating you will have ever heard—endocrinology. This is the study of hormones.
- 1.2 In the time we have together throughout this series, we're going to review— make sure that you leave with a basic understanding of what hormones even are, where do they come from, what do they do, why do we even have them? Then we're going to get into some specific parts of the classic endocrine research or studies, like the thyroid gland. You're going to be experts on how thyroid gland works. The adrenal glands, even the gonads, so how the testes and the ovaries not only become such distinctly opposite types of endocrine glands themselves, but then what they do in turn. Then we're going to talk about the pancreas and some fat tissue. Endocrinology is the study of hormones. You're going to be experts.
- 1.3 To get started, we have to cover some basics. Firstly, we're going to discuss, as I alluded to, what hormones are and why we have them in the body. Then you're going to learn about where they come from and how we classify them, because the body has different types of hormones, and the different types are made in different places. They move through the body in different ways, and that's the transport. How they then send the signal, what happens when the hormone gets to its destination—that alone is quite unique across all these different hormones. Then the nature of sending the signal. That's going to be the topics that we cover within this first lecture, which we'll just call Endocrine Basics.
- 1.4 Let's discuss what are hormones. Hormones are messengers, and they have the ability—it is the message that is being sent through the body from one cell to another. It is essentially how cells communicate. There are no nerves directly connecting every part of the body to the other. So there must be a way of sending a signal from one part of the body to some distal part of the body. Hormones are that messenger system.
- 1.5 The classic view of endocrinology, indeed, even within my own lifetime as a student, when I first took a class in endocrinology, the professor only discussed the classic glands, as they were called. These are the things— and the big ones—they're the things like thyroid gland, adrenal gland, etc. But what we now know is that every single tissue in the body is an endocrine tissue.
- 1.6 Every tissue—muscle, secretes a bunch of hormones. We're going to talk about some of them. Fat tissue secretes a bunch of hormones. The gut secretes hormones. We'll talk about some of them. It's not just these classic glands that we look at as part of the endocrine system. It is literally every tissue of the body that is participating in

endocrinology, in sending signals from one part of the body to the other, and we need those signals.

- 1.7 After all, how could the fat tissue know what the muscle is doing? When we get up and go on a run, how's the fat tissue expected to know that the muscle is moving and that there's an increased energetic demand? Well, part of it is going to be that the muscle will start sending a hormone to the fat, for example.
- 1.8 It's like a jazz band, if you will, where within a jazz band, you have individual players—you have the trumpet player, you have the bassist, the piano, and the drums. And what I love about the jazz band analogy is that each one isn't waiting to be told what to do. They're feeding off of each other. They're playing off of each other. That's part of the beauty of jazz, where the trumpet player will know, "Okay, I'm going to do something right now," and then the drummer is going to start changing the way he's drumming based on the way she's playing the trumpet.
- 1.9 And so there's a lot of direct communication, not one single conductor leading the music. There are some instances where there's a conductor. In some ways, the brain acts as a bit of a conductor. But then there are other instances, like the example I mentioned a moment ago, where the muscle is able to communicate with fat tissue without the need for a central conductor to tell those tissues what to do. So why have hormones? It's a way for various tissues and cells to send a message to other tissues and cells. It is the messenger system.
- 1.10 Now, let's cover an example as we start diving into some more of the relevance of all of this. You are all very familiar with glucose. Blood glucose is always there. In fact, blood glucose is so important that you don't even need to eat it. The body is going to make enough of its own, and we're going to talk about that now. The consequences of having blood glucose get too low is something called hypoglycemia.
- 1.11 Throughout these lectures, you're going to hear me use the words hypo- or hyper-. And then even another term here to remember is the -emia. When you hear that suffix-emia, it means "in the blood." So hypoglycemia, that's not going to be ideal. The person can start to manifest with some classic symptoms, like they start to get a little irritable, they may be a little nauseous, they may have a headache. There are some tissues like red blood cells that must have glucose, and so thank goodness, the body makes it.
- 1.12 Now, what happens when we start to exercise? Well, there have been classic studies that have existed for almost 100 years, where you can have someone get on to an exercise cycle, so it's easy to continue to monitor everything that's happening in their bodies. And during the course of this intense cycling bout, you could detect that the amount of glucose going into those muscles that are pedaling that bike so vigorously is going up by 20 times. So the muscles just become these massive glucose hogs chewing through all the blood glucose.

- 1.13** These are big muscles—in fact, as you can see on me, they are quite big—and so you're thinking, that glucose is just going to be plummeting, and the person, of course, is going to become hypoglycemic. And yet they don't. How is it that when we're measuring the glucose going into the muscle, we see that it's going up by 20 or 30 times, and yet, as we're sampling blood glucose, it's totally normal. It's essentially rock steady. In fact, if anything, sometimes, depending on the intensity and the length of the workout, it actually may be even going up a little bit. How on Earth is that possible? How can we reconcile this hungry consumption with a rock steady plasma glucose?
- 1.14** Well, it's because there's signals. And in this case, just within this example, the liver starts to get a signal. And it's balanced. So the muscle is eating the glucose, and the liver starts to sense that reduction. Why? Because a hormone tells it that something's happening. The moment there would start to be a risk of glucose going low because the muscle is eating so much, a hormone comes to life.
- 1.15** This is a whole new organ. So this would be the liver is the drummer, the muscle is the trumpet player, and now we have a separate one entirely—who haven't I mentioned? The bass player. Now we have a new organ that gets introduced here that's monitoring the situation and responding accordingly, and that would be the pancreas. The pancreas starts dumping out this hormone called glucagon. Now, the specifics aren't important. All of this is just the example that now we have a messenger where the glucagon has sensed, "Hey, muscle is eating a lot of glucose. Liver, you may need to step in here in case the muscle keeps eating all this glucose."
- 1.16** So now glucagon starts to tell the liver to start dumping out more glucose. Thus, the muscle can be very hungry and consume as much glucose as it wants, and the liver is able to meet all of the needs to keep the blood glucose humming along like normal. So now we have that example that reflects how the body is able to keep things running normally because of the signals that are being sent from one tissue to the other.
- 1.17** Now let's discuss the very nature of the hormones themselves, now that you're a little more familiar with what the role of the hormones are. Hormones will fall into three different classifications. Peptide hormones being the most common. That's what we'll talk—probably of all the hormones we discuss, peptides will be the most common of all of them. But then there are very important but fewer amine hormones built on an amino acid backbone. Then lastly, we have the classic steroid hormones, all of which will have one thing in common, as you'll see in a minute.
- 1.18** With peptide hormones, these are the classic—any hormone that's coming from the brain— in the next lecture, we'll talk about the hypothalamus and the anterior pituitary. All of those are peptide hormones. Insulin's a peptide hormone, glucagon's a peptide hormone, growth hormone, which is coming from the brain, that's also peptide. A lot of the big ones, when you think about hormones, are peptides. These are, like every protein, which is what peptide is referring to, just a bunch of amino acids, like a string of pearls, where every little pearl is an amino acid linked together.

- 1.19** But it's not just an actual little string. We produce the amino acids all linked together in a peptide, and then almost as if we had that string of pearls and we were laying it down on a table, and as we laid it down, it was wrinkling up. That will happen as well, where there is what's called a post-translational modification, where we have a string of pearls or a string of amino acids in a peptide chain, and then we start folding it up. Depending on how we fold it up and how we maybe clip it apart, we give birth to specific hormones. We'll have lots of examples of this.
- 1.20** Those of you who already watched my Basics in Biology will be somewhat familiar with this from a previous series I did. But within the cell, within the nucleus, we have the gene, and then that gene will be transcribed into a message. So we take the DNA message and we transcribe it into an RNA message that can come out of the nucleus and into the cell. That's transcription. Then that RNA message will be translated into the proteins. And it's those proteins that will then become the peptide hormone.
- 1.21** Let's leave peptides behind and briefly discuss the amine hormones. What all of the amine hormones have in common is that they're built from specific amino acids. Now, not a big long chain of amino acids like the peptides, but rather just one or two. The most common, certainly within the context of the series that we're doing in endocrinology now, is going to be tyrosine.
- 1.22** Tyrosine is the most relevant amino acid within the context of our conversations, but there will be other amino acids used as well, like phenylalanine or tryptophan, but tyrosine is the main one. In fact, tyrosine is the big one that we're going to focus on when we start talking about thyroid hormone, where what we call thyroid hormone is this protein that's taken some amino acids and then started adding some iodine onto it. That's the amines. Pretty simple. It's a smaller group of hormones, but boy, do they pack an outsized weight because, of course, thyroid hormone is a very powerful hormone. We'll have a whole lecture devoted to thyroid in just a bit.
- 1.23** The last one is one of the more famous ones because it's steroid hormones. When you hear the word steroid, it actually is a very specific term that is derived from what's called the steroid nucleus. What that has in common with any steroid hormone or the foundation of that is, in fact, the much maligned molecule called cholesterol.
- 1.24** Within the molecule of cholesterol, there are four rings: hexagon, hexagon, hexagon, pentagon. Those four rings create what's called the steroid nucleus. Every single steroid hormone will have taken a cholesterol—the cell will take a cholesterol molecule, and then it will start clipping it apart a little bit and manipulating it through the actions of various enzymes, these proteins that will take the molecule, rearrange it, pass it to another enzyme, who will then take it, rearrange it, and pass it to another enzyme.
- 1.25** Then as a result of all of this modification from just the humble cholesterol molecule, we get every steroid molecule, like cortisol or aldosterone—hormones that come from the adrenal cortex. We'll talk all about them. Or the famous androgens and estrogens

from the ovaries and the testes. We'll talk all about those as well. Even vitamin D, which is far more hormone than it is vitamin—we never should have ever called it a vitamin—is also built on this steroid nucleus from the much maligned cholesterol.

- 1.26** Lest we ever decide to wage war on cholesterol—which we have done certainly clinically over the past several decades—perhaps it's no surprise that everyone's suddenly low T or low testosterone and low every other hormone, low vitamin D. That's a consequence of vilifying this molecule of life. We must have cholesterol. It gives us the steroid nucleus, which all steroid hormones are built from.
- 1.27** Now, briefly, a little more on that, and we're going to touch on this when we talk about how the body makes cholesterol or the sex steroid hormones like androgens and estrogens. But you'll have a cholesterol that is being carried through the blood on LDL, another molecule that is always maligned and vilified. We have a collective fear of LDL cholesterol, and yet it is, in fact, the carrier. It's the bus.
- 1.28** LDL is the bus that is carrying a lot of cholesterol. It would pull up to a cell and say, "Hey, testes or ovaries, you need some cholesterol to make your sex hormones. So here, cholesterol, get off the bus, get into the gonads, the testes or ovaries, and become the sex hormones that they need you to become." That actually happens within a part of the cell called the mitochondria.
- 1.29** The mitochondria has a protein called the steroidogenic acute regulatory protein or StAR, S-T-A-R, steroidogenic acute regulatory protein. This is the gatekeeper, if you will, where the bus is pulled up. It's dropping off the cholesterol. The cholesterol moves into the cell and the mitochondria will say, the StAR protein on the mitochondria will say, "Hey, cholesterol, I need you to come in here." Then the mitochondria are actually going to facilitate the production of the steroid hormone that we need.
- 1.30** Just because I'm going to say this later when we start talking about some of the specific hormones—that first conversion is taking the steroid molecule, trimming it down to its basic nucleus and creating a molecule called pregnenolone. You're going to hear me say that again, so I'm just floating it out there right now.
- 1.31** So now you know the class of the hormones, and you know how they're each made across these three classes. Now, let's briefly discuss how the cell stores them, if it stores them at all. When it comes to peptide hormones, the cell will make a bunch of the hormone and have it on hand in what's called vesicles. A vesicle is a little bubble that has a bunch of the hormone already made.
- 1.32** For example, insulin, a hormone that anyone who's viewed anything I've ever created knows that I talk a lot about, including the Peterson Academy, Why We Get Sick series. Insulin is a very relevant peptide hormone. The pancreas in its beta cells will have a lot of insulin ready to go, usually, and it stores it in these little sacs called a vesicle. When there's a need to release the insulin, the vesicle will be the first responder. Every

peptide hormone, to some degree, will be stored in a little vesicle, a little bubble or sac within the cell.

- 1.33 The amine hormones also have a storage, but as much as we lump the amine hormones together, the biggest, most famous ones, like catecholamines—you guys have heard of the term adrenaline? We call it—that's the English term; here in the U.S., we refer to it as epinephrine, but it's the same thing. So epinephrine will be stored in a little vesicle, just like a peptide hormone will.
- 1.34 But thyroid hormone is actually stored in a more complicated vesicle called a follicle. We'll talk about follicles later, both in the thyroid context and in the ovaries, when we get into the endocrinology of the ovaries. But it's a unique, just another way of storing it. But it is stored and there's a bunch there in that little sack ready to go.
- 1.35 Then lastly, the steroid hormones don't have storage because they don't play nice. Because the steroid hormones are fatty, built on that cholesterol molecule, you can't hold them in place. They can just go wherever they want. They can slip through any little membrane. And so even if you tried to store them, it's just like greasy hands and you're dropping them out. And so the cell couldn't hold on to them. You don't store them; they're just made on demand. So when the gonads, the testes or the ovaries need to make some sex hormones, they just make them. If the adrenal glands need to make a bunch of cortisol, then it will make it on demand. It doesn't have a set stored amount.
- 1.36 Now, the differences persist. Beyond the synthesis and the storage, the nature of the hormone also will determine how it moves. So once it gets into the blood, which is the classic endocrine signaling— something is dumped into the blood, it moves through the blood, and then sends its signal to something else— the nature of the hormone will determine how it moves.
- 1.37 Within this, there's two classifications. We have the water-soluble hormones, and then we have the water-insoluble or the lipid-soluble hormones. That's purely a matter of chemistry and the presence of various polar groups like hydroxyl groups or carboxyl groups. I'm not a chemistry professor, so we won't get into the chemical structure. But suffice it to say that the structure of the hormone will determine whether it's water-soluble or whether it is lipid-soluble.
- 1.38 If it's water-soluble, which includes all the peptide, every peptide hormone is water-soluble, no exception. But it also includes one of the amine classes, the catecholamines, which is epinephrine and norepinephrine, a.k.a. adrenaline and noradrenaline. Those are the water-soluble. So when the cell releases those hormones into the blood, they can just flow freely in the blood. There's no escort carrying them around. And that escort actually has more of an effect than you think.
- 1.39 The lipid-soluble hormones, they don't play well in water. This would be like you have a dish of water, you drop a little bit of oil in it, and it's going to stay alone. It's going to

bundle up to itself in fear of the water. It is hydrophobic. It doesn't like the water. It wants to stay with its own fatty little friends, and so the little fat bubble stays together. That won't work in the body. We need it to be dissolved and spread through the bloodstream. And so in order to help that—this would be almost like an emulsifier.

- 1.40** Well, you've noticed in peanut butter, various peanut butter, if it has an emulsifier in it, all of the peanut oil is nicely, uniformly sprinkled through the peanut butter itself. If there's no emulsifier, then of course, it has that fat layer and you got to mix it all in. That's not too far from what we see in the body, where this hormone needs to be spread out and has a carrier, almost as if it's an emulsifier. In that case, as I've gone a little too long winded on that, with the lipid-soluble hormones, it has a carrier. It will have a transport protein.
- 1.41** Every time when we start going through specific endocrine organs, talking about its hormone product, if it is a steroid or a lipid-soluble hormone, I will mention it has a carrier. Then most of that hormone will be moving through the blood bound to a carrier. That actually has a very interesting physiological effect. The thyroid hormone is a really, really good example. To state it another way, every lipid-soluble hormone will primarily be moving through the blood bound to a carrier.
- 1.42** In fact, that is what we can call the bound pool of the hormone. There will be a certain amount of testosterone or the main estrogen, estradiol, which we'll talk a lot about later, there will be a certain amount of that hormone, thyroid hormone as well, that is in the bound pool, that we're counting it, but it's bound to its carrier protein. Then there will be a fantastically small amount of that hormone that is free. It's only that free hormone that is actually signaling and sending the signal that we expect that hormone to do.
- 1.43** Let me give an example. Thyroid hormone is a great example. In pregnancy, there is a fantastic difference in the amount of bound thyroid hormone—you're going to learn why that happens when we get to thyroid as a topic and we look at the consequences of what happens if a baby is developing in the absence of sufficient thyroid hormone. It is catastrophic to the fetus. So this is a really clever mechanism within the female body, that during pregnancy in the increase in estradiol and progesterone throughout the pregnancy, which we'll talk all about later, it's increasing the amount of the carrier. So that's a thyroid-binding globulin, but we'll just call it the thyroid carrier for now.
- 1.44** So the bound pool of thyroid hormone in the woman is more than doubling. It gets enormous. But the free pool stays the same. And that's nice because let's say during the course of the pregnancy, she somehow had some thyroid damage and wasn't able to continue to produce thyroid hormone, which is a critical hormone for fetal development. It's one of the many hormones that play into the cocktail of pregnancy.
- 1.45** But the nice thing about this is because she has so much—it is orders of magnitude, thousands times more is bound than is free. Literally, a thousand times more thyroid

hormone is circulating in her body in the bound form than it is in the free. But remember, it's only the free pool that's actually acting like we expect thyroid hormone to act. All of the bound hormone, not only is it protected from degradation because it's bound, but it's waiting until the protein will allow it to come off, and now it can have the signal that we expect to happen.

- 1.46** Back to the gal—if she is pregnant, she's making a lot of bound. Her bound pool is growing and growing and growing. Then if during the course of the pregnancy, she lost her thyroid gland, she still has more than a thousand times the thyroid that she has free, which gives her a lot of buffer because now all of that bound can start to trickle out into the free pool and then act like we expect thyroid hormone to act. It's a very, very clever way to have a bit of a storage.
- 1.47** As much as a moment ago, I described that we don't store steroid hormones—remember? We don't store steroid hormones. Now you could look at here and say, "Well, we do in the blood," and that's true. We can't store it in the cell, but we can have a big source of it circulating in the blood, and that source is going to be the bound pool of the hormone. I hope that makes sense. To sum that part of it up: within the lipid-soluble family of hormones, we have a little bit that will be free, and we'll have a lot more of it, potentially thousands of times more of it, that is bound waiting to be used.
- 1.48** Okay, now, how is the hormone actually used? This brings us to the signaling part of it all, where a hormone can act and send its signal in one of three ways as it plays across all of the cells that are making it. It can act in an autocrine or a paracrine or an endocrine. I know what you're thinking. You're thinking, "Ben, this is a class about endocrinology. Why are we even talking about the other ones?"
- 1.49** Well, let's start with that one. Endocrine refers to the classic idea of one cell over here releasing a hormone into the blood and moving through the blood to a cell over here. And what's important there is the mediating process of getting into the blood. That is endocrine.
- 1.50** Paracrine is when you have a cell that is located directly next to another cell. When we talk about the pancreas—we'll talk about the alpha cells and the beta cells, two very different cells, but are literally bumped up right against each other—they actually send hormones back and forth. That's paracrine because it's not getting into the blood. The alpha cell, which makes glucagon, doesn't need the insulin from the beta cell to go in the blood to come back and tell it what to do. It's just passing the message right back and forth. It's two naughty little kids in class that are just handing messages back and forth. That's paracrine.
- 1.51** But then there's even one last one, which is autocrine. This is very often something that you see in immune reactions, where you'll have one cell producing a hormone that is stimulating itself and its identical neighbors. So you'd have one single cell, like a

macrophage of the immune system. It's releasing hormones, and they are in turn affecting itself. So it's autocrine. It's stimulating its own response with its own hormone.

- 1.52** So we have autocrine—a cell making a product, stimulating itself. Then we have paracrine— one cell handing a message to a cell right next to it, literally, physically right next to it, without the need of sending it into the blood. And then endocrine is when you have the message actually flowing through the blood. We'll talk about examples of each of these.
- 1.53** The differences still aren't done between water-soluble and lipid-soluble because their nature, their chemical composition, determines how they then affect the cell once they get there. Remember, the point of the endocrine system is sending a message from one cell to another cell. So when you get to that cell, what happens to the message?
- 1.54** Water-soluble, like peptides and catecholamines, they can't squeeze through the fatty membrane of the balloon of the cell. They can only come to the surface. So what they'll do is come to the surface and knock on a door that's specifically designed for that hormone. So that will be a receptor. And then once they knock on the door, then the cell will respond. But the hormone itself never goes past the door for all intents and purposes.
- 1.55** In contrast, the lipid-soluble hormones, they have no problem. They're like a ghost that can just slip right through the wall of the cell. They don't need to wait and knock on a door. They can move right in. And once they move right in, they can then go to the nucleus if they want to and do whatever they want, whatever the hormone wants to do. We'll describe some examples of that later.
- 1.56** The water-soluble must knock on the door. Then once it knocks on the door, it will have these things called second messengers. The door gets knocked on, and then there's effect, effect, effect. Those other effects are what's called second messengers. The hormone itself knocking is the primary or first message, then it's second messengers. But the lipid doesn't need any of that. The lipid-soluble hormone just comes right in and does what it wants to.
- 1.57** With those in mind, with those ideas in mind, there's, if you will, a complicated taxonomy of signaling. But I just want you to be familiar with some of the basics of the most important ones that we'll talk about when we discuss some of the signaling. One of the common signaling forms for a water-soluble hormone that can't get through the cell itself is what's called the tyrosine kinase. Insulin is a really good example of this, where the hormone will come and bind to a receptor, at which point the receptor will dimerize or activate, and then it will phosphorylate itself. This is requiring—for the sake of time, I won't repeat myself too much, but we discussed that in the Basics in Biology course where I talk a lot about these messengers—it activates itself by

phosphorylating. It adds this little phosphate group, and that is activating the signal. Then we have the hormone doing what we would expect it do.

- 1.58** One of the most relevant, in fact, of all the hormone receptors we talk about over this series, the G protein-coupled receptor system is going to be the most common. In fact, it is so relevant to human biology that the guys who discovered it won the Nobel Prize in Physiology and Medicine. It's that fundamental to our understanding of human biology. So the G protein-coupled receptor will be this receptor on the surface of the cell, where the molecule will come and bind to and turn it on.
- 1.59** But linked to that receptor are these little subunits: alpha, beta, gamma. And when that receptor gets activated with whatever hormone comes and binds to it, we have this conformational change in the structure of these subunits, the alpha, beta, gamma subunits, that they get pulled apart. Then the alpha gets separated from the beta and the gamma. By pulling these apart, we then get the signaling effect that we want. We're going to talk about a lot of examples of that.
- 1.60** But of all the ones that I've mentioned by way of signaling, the G protein-coupled receptor is going to be the most relevant of all of them. That's going to be the one we come back to the most. Then a critical part of that, when the G protein-coupled receptor is activated and the alpha, beta, gamma units separate, we're going to have the production of a molecule called cyclic AMP. Cyclic AMP is part of that canonical pathway. Every time you think of a G protein-coupled receptor, it is almost certainly going to be the case that, give it a step or two, and you're going to get a cyclic AMP molecule. That cyclic AMP molecule can then go on to activate other processes. We'll give examples of that throughout some of the various tissues that we talk about.
- 1.61** The G protein-coupled receptor isn't a one-trick pony. In addition to activating cyclic AMP and its effects, it also can activate another pathway that I'll refer to called phospholipase C. Phospholipase C will start messing around with other molecules within the cell, giving its own signaling pathway. Again, I'll have examples of this throughout the series. But for now, I want you to remember that within the G protein-coupled receptor, we have both the adenylate cyclase and its actions and forming cyclic AMP, and then we have the phospholipase C.
- 1.62** As I mentioned, the steroid hormones, the fat-friendly hormones, don't need a membrane receptor. They can just go right in. Like sex hormones, for example, or cortisol, these steroid hormones will go straight to the nucleus. Then when they get to the nucleus, there's going to be an escort that will say, "Okay, here is the nucleus, here is the genome, go and turn on the expression of the genes you want to turn on." That's how the steroid hormones are going to act.
- 1.63** So we have discussed a handful of points so far, including the methods that the cells communicate—autocrine, paracrine, endocrine—the nature of the synthesis across the peptides and the amines and the steroid hormones, and all of the variety that comes

from that, not only how they're made, how they're stored, how they move, and how they signal.

- 1.64 One last comment—something I've spoken about in the Basics in Biology—is remembering this idea of negative feedback. Negative feedback is the process where there's a stimulus, so it turns the body on to do something, and then the body will respond, so a hormone will respond in this case, and its intention is to turn itself off.
- 1.65 Look at the blood glucose, how we started this small lecture—blood glucose goes up. Insulin will sense that, and then insulin will push the glucose back down. When the glucose goes back down, insulin, having done its job, retreats back into the background and insulin levels go down. So that's negative feedback. There's a signal, there's a response, and the response is to turn that signal off. Negative feedback.
- 1.66 Now, we will discuss one instance because it's so rare, but there's also something called positive feedback, where more of a signal actually creates more of itself. The most famous of this is in the complication, the wonderfully complicated world of female endocrinology. Women are more complicated than men, and the endocrine system makes that perfectly clear.
- 1.67 We will talk about two examples. One is the ovulation cycle, the ovulatory cycle, which we'll discuss later, and then pregnancy itself, where a signal has changed in such a way that what was once a negative feedback becomes now a positive feedback, where more of the signal is actually amplifying, resulting in more and more and more of the signal. So it's a way of amplifying a response. It doesn't happen very often, and it's always under very strict conditions, but that is the counter to negative feedback.
- 1.68 So as we wrap up this Basics in Endocrinology, you now are experts at this foundational level in understanding why the body would need an endocrine system, the ability to send messages, how those messages are sent across the various hormones, h

Lecture 2: The Axis of Control

At a glance — The axis of control — how endocrine feedback works without a central conductor.

- 2.1** All right. Earlier in the endocrine basics lecture, I mentioned that there's no central conductor required in a lot of the endocrine systems and signaling, but that doesn't mean there isn't one. If there is a central conductor—which some systems within the endocrine system, in a larger sense, actually do require—it would be the hypothalamus and the pituitary. Where the hypothalamus is like the CEO of the company of endocrinology, and then the anterior pituitary is like the right-hand guy or gal that is then going to go and send the signal on further.
- 2.2** But to say all this another way, what we're going to talk about now is the hypothalamus-pituitary axis. Now, when I say axis, we're actually going to include some other tissues in that depending on the axis. This would be something like the hypothalamic-pituitary-thyroid axis or the hypothalamic-pituitary-adrenal axis, and so on. We'll go through all of those examples. But throughout this, you'll get a better appreciation of the feedback across the hypothalamic-pituitary gland or other tissue axes, how they play across each other. Then we'll talk firstly about the posterior pituitary, and then the anterior pituitary, and then some disorders at the end of it all.
- 2.3** As a reminder, a critical thing to keep in mind as we talk about the hypothalamic-pituitary axis is this idea of feedback. Just to grease the skids here with our understanding, as an example, I'd mentioned the hypothalamic-pituitary-thyroid axis. I'll introduce all the characters here. But for now, just appreciate that the hypothalamus is going to send a thyroid-specific signal to the anterior pituitary. Then the anterior pituitary is going to send a thyroid-specific signal to the thyroid gland itself. This is the end of it. That's the end of the axis. But now this is going to act as a signal that turns these two off.
- 2.4** That's coming back to that negative feedback. You'll need to appreciate negative feedback. When we go through the specific disorders later, like when we get to thyroid gland endocrinology, or adrenal gland, or gonadal axes, that idea of negative feedback will become more and more important.
- 2.5** One final point: when you think about the hypothalamic-, pituitary-, then gland or whatever the tissue is, axis, when there's a problem, there's some specific terminology in endocrinology that we use. If the problem is at the tissue itself, like in this example, the thyroid gland, we would call it a primary problem. If the problem is at the anterior pituitary, the middleman, it's a secondary problem. If the problem is originating at the hypothalamus, it's a tertiary problem. Although that one's a little tricky, and we'll talk about why in a few minutes. To flip that around, if a person manifests with an endocrine disorder, you would determine where it is and then give it a classification.

It's a tertiary problem, hypothalamus; a secondary problem being the pituitary; or the tissue, the gland itself, in this case, the thyroid gland.

- 2.6** Let's first start with the posterior pituitary, because it actually— I have to talk about it because it's part of the pituitary, and yet it isn't part of the axis because the pituitary gland actually is a little marbled. And it's a little stalk branching down from the hypothalamus, deep within the brain, wrapped very protectively by the skull. It's so important. But the pituitary gland actually has two parts. It's segmented into a posterior, or at the rear, and an anterior, or at the front.
- 2.7** The posterior pituitary makes two hormones—they're important—oxytocin and antidiuretic hormone, ADH, also known as arginine vasopressin or AVP. We'll talk about both of them in a little more detail. But what's quirky about the posterior pituitary, not only because it only releases just a couple of hormones, but it's a literal branch of the hypothalamus itself. So it's not separate. Whereas the anterior pituitary, the frontal part, is in fact separate from the hypothalamus, it's connected through blood vessels, but the posterior pituitary is a literal offshoot of the hypothalamus. So it doesn't quite fit into this tertiary, secondary axis that I mentioned a moment ago.
- 2.8** So now that you're familiar with the posterior pituitary and its quirky anatomy within the brain, let's just talk about those two hormones in a little more detail. AVP, arginine vasopressin, is synonymous with antidiuretic hormone. You'll see why it has those two names, because it was discovered in two different places at two different times with two very different effects, and only later they learned that it's actually the same hormone. That's actually not uncommon within endocrinology.
- 2.9** But then oxytocin, of course, is the famous hormone. Let's start with that one because it's so clearly relevant to pregnancy, where oxytocin not only helps with the contraction of the uterus, which helps the baby come out—all of which we'll talk about when we get to pregnancy and parturition, or the process of childbirth—but also with nursing and the actual contraction of the breast tissue to get the milk out to nurse baby. That's oxytocin's role. We'll come back to it in just a bit.
- 2.10** I just wanted to add that there is, in fact, some evidence to suggest that oxytocin may also have a role in muscle health, which is so reflective. Part of the problem with endocrinology and even biology is, I swear, it's the science of chaos. When you think you understand it, it's not as tidy and neat as something like physics, perhaps. Biology is much messier than we'd like to think it is. Oxytocin affecting muscle tissue and even enhancing muscle regeneration is a totally unexpected effect of it. All right, that's it for oxytocin for now, but we're going to talk more about it later.
- 2.11** Now, what about AVP/ADH? Why on earth does it have two names? Within the first term, you hear the word—the first title we could give it is arginine, which just reflects one of the amino acids on it. But then the next part is what tells us what's happening: vasopressin. Vasopressin suggests this effect at controlling or pressing in on the vessels,

the vaso- part of it is the vessel. It, as I'll show you in a moment, is affecting blood vessels.

- 2.12** However, at the same time, it has receptors for this hormone within the kidney, and it affects the production of urine, or diuresis. It has this specific diuretic or an antidiuretic effect, while at the same time, because of where its other receptors are expressed, it can affect the size of blood vessels. Now, let's just start with a view that addresses that kidney aspect, because that's the most famous part. When people think about antidiuretic hormone, they're thinking of the kidney because you think of uresis or the production of urine.
- 2.13** Within this system, there are two important parts. One is the carotid baroreceptor. If you think about the word baro-, it's pressure, like a barometer. A barometer is sensing the pressure in a given space or in the atmosphere. Well, the body has its own barometers that we call baroreceptors. Within the carotid arteries, where you want to try to feel someone's blood pressure, you can feel the carotid vessels. If blood pressure is going too low in this case, these baroreceptors are sensing the change in pressure. That is a primary trigger, and indeed, it's the primary stimulus for ADH/AVP, that hormone. So when blood pressure goes low as sensed by the pressure receptors, it will turn it on.
- 2.14** Now, there's a secondary signal. Within the brain, within the hypothalamus, there are these things called osmoreceptors, which sense the concentration of the blood, how salty the blood is. It senses the osmolarity. We have the baroreceptor sensing pressure. That's the main regulator. Then there's a secondary or backup regulator in the hypothalamus that is sensing how salty the blood is. What is the blood concentration? If blood is getting too salty or too concentrated, it turns it on. So low blood pressure turns it on, or high blood salt will turn it on.
- 2.15** Then in this case, just within the sake of the kidney, it's going to look at both of those signals, the end result of which in the kidney is going to be the recovery of water. So we're bringing water back in. Remember, low blood pressure was the primary cause. If we bring water back in, that water comes back into the blood. Most of what we call blood is water, after all, and that can help restore the blood pressure. At the same time, if the secondary trigger is that the blood has become too salty, by bringing water in, we dilute the blood; we make it a little less salty. The kidney is going to be involved in both of those effects, restoring blood pressure and reducing how salty the blood is.
- 2.16** None of this talks about the vasopressin aspect or the effect on the blood vessel. Well, AVP/ADH will bind to a specific receptor called the V1 receptors. This is found at smooth muscles that line our blood vessels. When AVP comes and binds to the V1 receptor, it activates one of the receptors that I told you about, namely a G protein-coupled receptor. This is the first example of this receptor that we've talked about, and there will be dozens more over the time we spend together.

- 2.17** But after a series of signaling, once the G protein-coupled receptor is bound, it will result in a bunch of calcium being in the cell. Where you have more calcium when you have muscle cells—this goes back to my physiology lectures with the Peterson Academy from before—but more calcium within a muscle cell results in more contraction. We start to contract around the blood vessel, and as we push it in, we bring blood pressure up. That helps us restore the mean arterial pressure or the blood pressure in general.
- 2.18** But then separately within the kidney, we have the V2 receptors that are binding to ADH/AVP. This is another G protein-coupled receptor, and the result of this is going to be the expression or the production of these little pores called aquaporins. It's these aquaporins that ultimately allow the water to come from what's becoming urine in these tubules of the kidney. So the kidney has been filtering out water, and then we want to get that water back into the blood. And it's the aquaporins that are opening these little doors for the water to come back in. And all of that is, as I said, under the control of the V2 receptor.
- 2.19** Now, let's talk a little bit about some problems that can happen here. What if, for example, a person is making too much ADH or AVP? The most common cause of this would be a tumor. You have a tumor that's just dumping out a lot of ADH/AVP. The result of this is going to be a patient who has a lot of body water. They're getting waterlogged. A lot of that water is in our blood, and so the blood pressure can be really high, so the patient can manifest with very high hypertension or high blood pressure. But at the same time, they're producing very little urine because we keep pulling all the water back in.
- 2.20** So that is a problem called the syndrome of inappropriate antidiuretic hormone, SIADH, which is uncommon. All of these problems are relatively uncommon, and slightly less common than that one is a condition called diabetes insipidus. Later, when we get to the pancreas, we'll talk about diabetes in the way you're thinking. That is diabetes mellitus. Now, the term diabetes means lots of urine production.
- 2.21** It comes from the Greek word for "siphoning." You think about someone siphoning water out of a vessel, you're dumping a lot of water from this vessel through this little hose as we're siphoning it out. That is somewhat similar to what's happening when the body—or you could see how they'd use that term to refer to someone who just can't stop urinating. They're peeing a lot. That's what happens if you don't have enough antidiuretic hormone. If antidiuretic hormone is too low, you've lost this signal that's recovering water.
- 2.22** So all of that water is staying in the urine or in the bladder, going to the bladder, producing a phenomenal amount of urine. It can be up to 20 liters a day. The person just cannot stop, literally cannot stop their production of urine. The result of this, beyond just the obvious effect of needing to urinate, is that their blood pressure will be very low because they're always dumping their water from their blood, resulting in low

blood volume, which leads to low blood pressure, and they're pushing it all into their urine.

- 2.23** Now, there's different types. Let's imagine you, future clinicians or someone listening to this, wondering, how might this look in a patient who comes to the clinic? Well, they would feel lethargic because their blood pressure is so low, they may have headaches, and they would say, "I just can't stop urinating. I'm going to the toilet 30 times a day to urinate." Then you'd say, "All right, let's look at the problem. It's going to be one of three problems. It's either a problem with your neurons, they're not producing enough ADH. Or it's problem with your kidneys where they're not sensing the ADH. You're making it, but it's not getting sensed in the kidneys. Or it's a problem of you just not being able to stop drinking water.
- 2.24** That's something called psychogenic. There are known neurological disorders where the person just feels the need to drink. They will often say, "I need to purge myself of demons" or "I need to purge myself of toxins," and they have this compulsion to keep drinking. Well, that's, of course, a separate issue. That has nothing to do with ADH. So psychogenic is different.
- 2.25** Now, if you have the patient come in, though, and you notice that they're urinating and they fit all the signs and symptoms of having this diabetes insipidus—but I realized now, I didn't explain the difference. Diabetes mellitus is when a person is making a lot of urine, but it attracts flies like honey. That's mellitus, which we'll talk all about when we get to the pancreas. So diabetes mellitus is a lot of urine and it is honeyed. Not because it tastes like honey, but it attracts flies like honey. That's the origin of the term.
- 2.26** Then diabetes insipidus—a lot of my professor friends, they don't have any taste in their wardrobe. I obviously have very good taste, so I am not insipid. Their taste in attire is insipid. It is tasteless. So diabetes insipidus is when the urine is being produced in excess and it has no particular draw like honey. The flies will leave it alone because it's just pure water for all intents and purposes. So that's the difference. Diabetes mellitus, the honeyed urine, and you're making a lot of it, rich with glucose, and then diabetes insipidus, it's just a lot of water.
- 2.27** That's what we have here: diabetes insipidus. Now, the patient comes in, you're worried they're producing a lot of urine, and they are thirsty all the time, and their blood pressure is so low. How can we determine what the nature of the problem is? Well, we restrict water. Under clinical supervision, we say, "You know what? Let's have you stop drinking water for several hours and let's see what happens." If they stop drinking water and you find that they now suddenly stop urinating so much and the urine is less watery and it's starting to get more concentrated, clearly, nothing's wrong. They're just drinking too much water. That's the psychogenic origin of the diabetes insipidus.

- 2.28** However, let's say we restrict their water and they still continue to produce dozens of liters of urine. Of course, there's going to be a complication, which is why you do this under supervision, because if you stop water coming in, but you can't stop the water coming out, their blood pressure is going to go so low that they go unconscious. That's why you're watching them. Let's say you restrict the water and they continue to produce a phenomenal amount of urine.
- 2.29** What you do is actually give them an inhaled dose of ADH/AVP. Why inhale it? Because it can permeate through the upper roof of the mouth and get to that hypothalamus where the ADH is being made in the first place. So they inhale it in, and now we watch and see what happens. If after administering the synthetic ADH/AVP, we notice that, "Hey, they're not urinating as much." Now we know the problem. We know that the problem was neurogenic. Those neurons of the posterior pituitary are not making ADH because when we put it in, everything went back to normal. So that's neurogenic. It's the neurons of the brain, of the posterior pituitary that are not working well.
- 2.30** However, let's say we give them the ADH, the synthetic inhalation, and it doesn't work. They continue to produce a lot of urine. Then we know it's not a problem of the neuron, it's a problem of the nephron or the functioning unit of the kidney. So we would say that it is nephrogenic.
- 2.31** Okay, that's it for the posterior pituitary. Let's move on to the slightly more complicated anterior pituitary. This is big because so many important processes are influenced by the anterior pituitary, and it's a little more complicated because the anterior pituitary, as I said before, is connected through blood vessels, but it's not a direct extension of the hypothalamus. And so the hypothalamus will send a separate hormone signal to the anterior pituitary, which in turn will send its signal out to the body. This includes a lot of very prominent hormones.
- 2.32** You will need to remember these. Everyone listening here in a home, you will need to remember, at some point, you'll want to draw this out where you have the brain sending a signal. That first signal from the hypothalamus in no particular order will be a signal called gonadotropin-releasing hormone. Within that term, you hear the word gonad. That's good. So gonadotropin-releasing hormone, GnRH, will come from the hypothalamus to the anterior pituitary, and then it will induce the release of two hormones: follicle-stimulating hormone, FSH, and LH, luteinizing hormone. We'll talk all about all of these later. So right now we have one line. We have the hypothalamus producing GnRH, which in turn produces LH and FSH.
- 2.33** Now, let's go to the next line over: we have thyrotropin-releasing hormone—you hear thyroid in there, TRH. TRH will come from the hypothalamus, go to the anterior pituitary, and tell it to release thyroid-stimulating hormone, TSH. Now, we have another line: corticotropin-releasing hormone, CRH. Then CRH will induce the release of adrenocorticotropin hormone, ACTH. Then we're almost done. Those all will go on

and have their own signals. LH and FSH will go to the gonads. TSH will go to the thyroid gland. ACTH will come to the adrenal cortex of the adrenal glands.

2.34 The other ones, prolactin and growth hormone, are the end. They don't need to tell any other hormone to be released; they have their own effects. When you think about prolactin, it has its own effects on the body. When you think about growth hormone, it has its own effects throughout the body. Whereas the other ones that I just said—ACTH, TSH, and then LH and FSH—have to go to their target organ for them to do what they want to do. That's to the gonads, to the thyroid, and to the adrenal glands.

2.35 In the midst of all of this complication is one more wrinkle—I hate to say it—which is prolactin. In every case of these lines of hormones coming from this axis—the hypothalamus to the pituitary—they need to be told to be turned on. If you want to say, "I need the anterior pituitary to make more TSH, to stimulate more thyroid hormone." Well, to do that, you need more TRH. So everyone is turning on the one below it, with the exception of prolactin. Prolactin is totally anomalous from the anterior pituitary because it does not need to be told to be released. In fact, it needs to be told to stop.

2.36 So prolactin will be released—if we came and carved out an anterior pituitary but kept some blood flow to it somehow and put it in a little petri dish, it would stop producing every hormone because it would have lost that hypothalamic signal except for prolactin. The anterior pituitary must be told to stop. So it doesn't fit this normal balance because the signal coming from the hypothalamus is called prolactin inhibitory factor or PIF. It's actually the same molecule as dopamine. So dopamine is a hormone that when it comes from the hypothalamus, will go to the anterior pituitary and tell it to stop making prolactin.

2.37 So prolactin is its quirky little hormone. Maybe it's a reflection of just how important it is. If a gal has some damage between hypothalamus to anterior pituitary over the course of childbirth, she still is making prolactin to enable her to create milk for the baby. So it could just be this clever system of ensuring the survival of offspring. Regardless of the speculative reason, prolactin needs to be told to stop, whereas all the other hormones—growth hormone, ACTH, TSH, LH, FSH—need to be told to turn on, like we typically think of when we think about the glands and the axis here.

2.38 Okay, as a reminder, when we start talking about this hypothalamic-pituitary-gland, we have the tertiary level, the secondary level, and then the primary level. So let's just go through each of these in a little more detail. Firstly, within the anterior pituitary, we have a class of cell called gonadotropes. The gonadotropes are the cells that make follicle-stimulating hormone and luteinizing hormone— LH and FSH.

2.39 Upstream of them, like I said, is the signal coming from the hypothalamus, which is gonadotropin-releasing hormone. We're going to come back to every one of these views when we get into these other tissues later within this lecture series. But for now,

just appreciate gonadotropin-releasing hormone, GnRH, will come to the anterior pituitary and induce the release of LH and FSH. As we'll get to later, leptin, the little hormone from our fat cells, is critical in that process. If you take away the leptin, you take away the GnRH. It is that important. We'll talk about it, and you're going to love it. It's super interesting.

- 2.40** Within the thyroid axis or signaling, we have the hypothalamus producing TRH, which will go to the thyrotropes, the specific cells of the anterior pituitary that make TSH. Then TSH will go to the thyroid gland. We'll come back, we'll talk about this. But some of the signals that will turn it on at the hypothalamic level would be something just as simple as cold. Cold exposure will turn on TRH. But stress, even something like sleep deprivation, actually turns it off. There are these non-hormone signals that can dictate the origins, the turning on or off of the pathway.
- 2.41** We also have that cortisol axis where the hypothalamus will make CRH, which will then go to the corticotropes of the anterior pituitary, which will in turn release ACTH, the adrenocorticotropin hormone. So ACTH, which will then go to the adrenal glands and have its effect. Low glucose and stressful situations will turn on the CRH at that upper level. So once again, we have those non-hormone signals.
- 2.42** What about prolactin? I told you that prolactin is the quirkiest of all because the lactotropes of the anterior pituitary are just going to release prolactin unless they're told not to. Now, prolactin is its own hormone that has its own effects. It doesn't need to go to some other tissue and tell it to release another hormone. Like TSH, the only point of TSH is to get to the thyroid gland to tell it to make more thyroid hormone. But prolactin is the end of this pathway. There's no primary. It just went from tertiary to secondary, and now it's done. And so prolactin is the end product and thus provides its own negative feedback.
- 2.43** So as prolactin levels are going up, it would come to the hypothalamus and say, "Hey, there's too much of me." Remember, in every other path, the hypothalamus would turn off a pathway by just producing less of the hormone. But in this case, it has to tell prolactin to stop. That's where we have the prolactin-inhibiting factor, PIF, P-I-F, which is the exact same molecule as dopamine. So dopamine will come from the hypothalamus to the anterior pituitary, telling prolactin to go down. We'll talk more about prolactin when we get into the pregnancy and lactation section.
- 2.44** Then the last one is growth hormone, where you have growth hormone-releasing hormone coming from the hypothalamus, going to the somatotropes. The somatotropes are the cells that make growth hormone from the anterior pituitary. Then we're going to talk more about growth hormone when we get actually into the puberty lecture. It seems to fit there well because one of the key stimuli for that whole pathway is, in fact, the sex hormones. When estrogens and androgens, like estradiol and testosterone, start to go up during puberty, they will really accelerate that growth hormone pathway, stimulating the growth hormone-releasing hormone to stimulate

more growth hormone. Then growth hormone will have its myriad effects. Again, all of which we'll talk about when we get to, in this case, the puberty lecture where it really fits well.

- 2.45** Within this axis, one aspect of all of this is our inability to measure hormones that are coming from the hypothalamus. All of those hypothalamic hormones only exist in this microscopic blood vessel going from the hypothalamus to the anterior pituitary. All of the anterior pituitary hormones are flowing through systemic circulation. So we could draw a blood sample from a blood vessel in the arm, take it to the lab, and measure FSH and LH, and ACTH, all of those products, prolactin, growth hormone, all of them. But the precursor signal coming from the hypothalamus, you can't detect it in systemic circulation. It's all but gone. Its levels are so low to absent. How can we know whether there's a problem here? Let me just rephrase all of this.
- 2.46** Let's imagine that someone came in with symptoms of low cortisol. We'll talk about that when we get to the adrenal glands. They have low cortisol. Let's say we measure, "Okay, cortisol is low." Now we're going to measure your ACTH, and we see that your ACTH is also low. All we know at that point is that the problem isn't with the adrenal glands, because the adrenal glands are listening to the signal coming from the anterior pituitary. But how do we know whether it's a problem of the anterior pituitary or the hypothalamus? In other words, how can we determine whether it's a secondary signal or a tertiary signal if we can't measure that tertiary signal? Does that make sense?
- 2.47** Just within the cortisol line, they have low cortisol. We can measure cortisol, we can measure ACTH, and let's say they're both low. All we then know is that it's not the problem of the cortisol because it's listening to its boss—ACTH. But how do we know whether ACTH is listening to its boss or not? This one might be really high, after all. We don't know because we can't measure it. So we do what's called the Hypothalamic Factor Stimulation Test.
- 2.48** The patient—let's just stick with the cortisol. They have low cortisol, low ACTH, and now we're going to say, "All right, we don't know what CRH is—the corticotropin-releasing hormone—so we're going to inject a bunch." So we will infuse a bunch of CRH into their bloodstream and wait and see what happens. Now, let's say after we inject that hypothalamic hormone, the CRH, after injecting, let's say that we actually see that the ACTH levels go up, that the anterior pituitary woke up, and now it's producing a bunch of ACTH. Whose fault was it?
- 2.49** Remember, we're trying to determine: is it a secondary problem or is it a tertiary problem? We inject the tertiary hormone in, and all of a sudden, the secondary hormone is working great. Well, then it wasn't the secondary hormone's fault. It's not the anterior pituitary's fault. The problem lies with the hypothalamus. So we would say that this is a tertiary problem.

- 2.50** However, upon injecting that hypothalamic signal, let's say that as we continue to monitor blood levels of ACTH and cortisol, they didn't change. They stayed low. Now we know that even when we bumped up that tertiary signal all the way, the secondary signal still didn't respond. The anterior pituitary stayed quiet. It's the anterior pituitary's fault. That's how we get to that differential diagnosis, where by injecting that hypothalamic signal, we can know, if we see a response, it's a tertiary problem. It was the hypothalamus that's not producing because the anterior pituitary responded perfectly well when we kicked it and prompted it. But even when we gave the signal, an alternative outcome would be we didn't see anything happen, so the problem then lies at the anterior pituitary.
- 2.51** So that is a way. The Hypothalamic Factor Stimulation Test is a way to help us tease out what level a problem is if a person comes in with low levels of a hormone, and then we see that the low levels also persist, if they also persist at the anterior pituitary. If a person comes in and they had low cortisol, and the problem was at the adrenal cortex, and we'll talk about an example of that later, then what would we expect? There would be no negative feedback to turn anything off because there's no signal. So what would we expect ACTH levels to be from the anterior pituitary if there's nothing to turn it off? We would expect it to be high.
- 2.52** It's easy to determine primary problems or problems at the gland itself. Just with cortisol as the example, again, if someone came in with low cortisol and we measure that ACTH is really high, well, then we don't need to go any further. We already detected that the problem was, in this case, the adrenal glands themselves.
- 2.53** With all of that in mind, let's just mention a couple of pituitary disorders. The first one is when the pituitary gets wiped out because of an infarction. An infarction is when there's too little blood flow. One of the complications of childbirth is the risk of hemorrhaging and blood loss, resulting in the pituitary not getting enough blood. That's why I mentioned the quirky aspect of prolactin a moment ago. In fact, it's so possible that it has a specific name called Sheehan syndrome.
- 2.54** Sheehan syndrome is when the pituitary dies, or at least a part of it, because it doesn't get enough blood flow. Anything could cause that, but childbirth is one of the more common ones within this relatively uncommon problem. But of course, the end result of this would be that the individual is not getting enough of a hormone signal because the anterior pituitary is not sending this signal on anymore. Then the solution, of course, is that the person has to go on a lifetime supplementation of the hormones that they're now missing.
- 2.55** On the other end—so that's hypopituitarism when the pituitary has infarcted—but then we have the hyperpituitarism, too. That is overwhelmingly a result of an adenoma, a slow-growing, well-encapsulated tumor. What I'm not saying is adenocarcinoma; we're not talking about the biology of cancer. An adenoma is a well-encapsulated, very slow-growing, not metastasizing tumor. An adenocarcinoma is

cancer and it's spreading and wreaking havoc. This can be wreaking havoc too, because the cells are similar enough to what they came from, to what they mutated from, that they still may be producing a lot of hormone. And so the person could manifest with the signs and symptoms of having too much of a hormone because there's too much of that anterior pituitary hormone coming.

- 2.56** So let's say a person has hyperthyroidism, and we do some CAT scan or an MRI and detect, "Oh, there is a pituitary adenoma. That's got to be the source." Sure enough, we see that TSH is through the roof, the thyroid-stimulating hormone, and then, of course, thyroid hormone is through the roof as well. Often, the person will start to manifest with other symptoms that are not a result of the hormone, but a result of the mass that is growing on the anterior pituitary. Those are going to be headaches and vision problems because of the proximity to the optic nerve.
- 2.57** If you have a tumor growing on the anterior pituitary, it will start to impinge and pinch on the optic nerve, creating this progressively narrowing field of vision. That will be another, a pretty strong differential opportunity to say, "Okay, you have hyperthyroidism. Is it actually a thyroid problem?" "Oh, but doctor, I'm also noticing that I have headaches all the time and I'm losing my vision." All right, now you're immediately going to think that it's a problem of the anterior pituitary.
- 2.58** Then what do you do if you need to get it out? Actually, what's interesting is that because of its proximity to the mouth or the nasal cavity, you just go up through the nose and crack into the bottom of the skull and resect or remove the tumor and then patch everything up. It's actually, heaven forbid, any of us need to have it done, but it's not as bad as you might think if you think about having to get into the brain. It's a lot easier to come up from the bottom when the pituitary gland is at the bottom of the brain. You'll recall that I said it's wrapped into the skull, which means we can just go right through that nasal cavity.
- 2.59** All right. In this lecture, you now are very familiar with the axis of the hypothalamus and anterior pituitary, as well as the quirky role of the posterior pituitary in that pathway. But I encourage you, before you dive into the remaining content where we go through the thyroid gland, the adrenal glands, and the gonads, and the pancreas, less so the pancreas, more so those other three, that you remember they fit within this axis, that there's a hypothalamic signal, and there's an anterior pituitary signal relevant to all of those.
- 2.60** Thank you.

Lecture 3: The Thyroid Axis

At a glance — The thyroid axis and its regulation of metabolism.

- 3.1 Now that we've discussed the basics of endocrinology, the very nature of hormones and what they do in a general sense, and you're familiar that if there's any central system, it would be this hypothalamic-pituitary system. Let's go beyond it and start getting into the body. It's, in this case, some of its most famous endocrine systems.
- 3.2 Let's start with thyroid. Thyroid hormone is unique, and because of that, the thyroid gland is a bit unique because the thyroid hormone will affect every cell of the body. Every cell is going to have some response to thyroid hormone. It's just such a popular hormone to talk about because of its influence in metabolism. You will become very familiar with those effects because after all, if I'm a scientist in anything, it is metabolism. So I can't help but always come back to that to some degree.
- 3.3 In the lecture discussing thyroid hormone, now, we're going to talk about its role in that hypothalamic-pituitary axis, where it fits and how it all plays together. We'll talk about how thyroid hormone is synthesized—it's very quirky—and how it moves through the blood, which is a little simpler, and then how it acts—its signaling. Then lastly, we'll end with some thyroid disorders, just to bring it into a real-world, even, dare I say, clinical context.
- 3.4 Within the broader axis—you'll recall when we spoke about the hypothalamus and pituitary axis—we have the hypothalamus, which in this case releases thyrotropin-releasing hormone, TRH. Then TRH will go to the thyrotropes of the anterior pituitary, resulting in the production of thyroid-stimulating hormone, TSH. Then TSH would ultimately come to the thyroid gland, telling it to do everything we're going to talk about.
- 3.5 As I noted in that previous lecture, and as you'll see as a somewhat of a theme, there still has to be something to turn the hypothalamus on in the first place. In this case, cold is a very powerful stimulus for this process, and you'll learn why in a minute. But when the thyroid gland gets that TSH signal, now this is where the magic happens because we want to get to the thyroid hormone itself.
- 3.6 As you'll learn, the thyroid gland is a bit—I don't want to say confused, but it likes to keep things complicated because it will make two forms of thyroid hormone. It will make a thyroid hormone called T4, and it will make a thyroid hormone called T3. As you'll learn, the difference in numbers is how many iodines are bound to it. One has four iodines, one will have three iodines. It makes it in a heavily skewed ratio of about 20 to 1. So 20 times more of the hormone coming from the thyroid gland will be T4.
- 3.7 That's odd because T4 isn't the active; it has a very low activity, as we'll discuss. Then only a 20th level is going to be T3, the very active form. Well, there might be some

reasons for that as we'll get to, but just appreciate the fact that the thyroid gland is going to be producing two different forms of thyroid hormone, T4, which will be coming out at 20 times higher levels than T3.

- 3.8** Then one other comment or thought on this—when we start talking about thyroid disorders and the nature of goiters, when the thyroid gland is getting too big, in order to help you understand that topic when we get to it, you need to appreciate that when TSH comes to the thyroid gland, yes, it stimulates the production of thyroid hormone, and we'll talk all about it, but it also stimulates the growth of the cells themselves. So TSH has this dual effect. This is not the same effect, although you could see that they're complementary.
- 3.9** TSH will induce the synthesis or activate the synthesis of thyroid hormone. And separately, it'll stimulate the growth of thyroid cells, making the thyroid cells both bigger and more abundant. So in other words, activating both hypertrophy of the cells and hyperplasia of the cells, which is going to have an obvious effect on the size of the thyroid gland. Okay, when the thyroid gland releases the hormones, T3 and T4, it's going to go through the body and have its various effects. One of the main effects is going to be changing T4 to T3. Just keep that in mind, that as much as T4 isn't the active form, it's a potential source of T3 in a process that we'll get to.
- 3.10** Before we get into all of that, let's zoom into the thyroid gland and then look at what happens when the thyroid gland receives that pituitary signal. You'll recall the signal coming from the anterior pituitary is TSH. So when TSH comes to the thyroid gland, it binds a G protein-coupled receptor, and it creates cyclic AMP, which activates this process of producing a critical protein called thyroglobulin. Sometimes it's just called thyroglobin, but we'll call it thyroglobulin. Thyroglobulin is critical in what's going to become thyroid hormone. We just have to start adding on some tyrosines, like I said, and then putting on some iodines before we get there. But thyroglobulin is going to be the base molecule or the precursor, if you will, to what will become thyroid hormone later.
- 3.11** Earlier when we talked about the storage of hormones, I mentioned that thyroid hormone is a bit quirky because it is stored in follicles. We're going to talk about a follicle again when we get to the ovary endocrinology, but what they both have in common is little fluid-filled bubbles surrounded by cells. That's different from a vesicle. A vesicle will have its balloon, its sac is formed with lipids and maybe some proteins, not cells. But in this case, you have cells linking together to create this little sac. That's a follicle, and you have follicles within the thyroid gland.
- 3.12** These follicles will be composed of something called colloid, which is the middle, and that's a lot of water, but then a lot of thyroglobulin. All the other molecules, like tyrosine, amino acids—commonly just we say tyrosine residues—it'll have tyrosines, it'll have some iodide. We'll talk all about that and the difference between that and iodine.

But the colloid will be the substance, the watery mix of all these things that's going to result in the alchemical production of what we call thyroid hormone.

- 3.13** But around it are all these follicular cells. We have follicular cells lining the follicle. These follicular cells are going to be critical in the synthesis of thyroglobulin, like we just mentioned a moment ago. The follicular cell will make the thyroglobulin and then dump it into the colloid, that center part. That's where it's going to be just sitting and waiting. At the same time, we need to get the iodide from the blood. We eat the iodine or iodide, and then we move it into the—we have to get it through the follicular cell into the colloid where we're going to make the thyroid hormone itself. That's another component to this follicle, the movement of the iodide in.
- 3.14** At the same time, we need this, as you'll see, absolutely fantastically complicated and talented enzyme called thyroperoxidase or thyroid peroxidase, TPO. I call it thyroperoxidase, TPO. TPO, as you're going to see, is involved in essentially every single step, multiple steps, of making thyroid hormone. Then we ultimately produce the thyroid hormone itself, and then we've got to get it out. We'll go through this whole process in more detail. Each step we'll talk about in more detail.
- 3.15** Firstly, we know that we're making thyroglobulin, but we also need to get the iodide in there. You'll notice I'm using that word very specifically. Elemental iodine is, of course, a thing. That is what you're putting on wounds back in the day, the colored little dark ointment. That's iodine. That's an elemental form that the Earth just naturally has. We don't eat iodine. When we eat iodide—or when we get it, we eat it in the iodide form, whether it's from iodinated salt or whether it's from seaweed or some natural source, it's iodide. So we eat iodide, and that's how it stays in our body for now.
- 3.16** That's circulating through the blood, and that's circulating through the blood vessels of the thyroid gland. Then we need to get it from the blood into the colloid. There's a two-step process because we have this first access point from blood side of the follicle cell. We got to get it into the follicle cell in the first place. Then we got to get it out of the follicle cell on the back end. This is collectively called iodide trapping. Iodide trapping is the term for taking the iodide from the blood and locking it into the cell because it is a one-way path in.
- 3.17** The first path takes advantage of sodium's transporting. When we did the Basics in Biology course here within Peterson Academy, I talked about this transporter called the symporter. This is an example of a symporter where sodium will come in, and when sodium sneaks in, it will open the door. Or let me not say—sodium doesn't sneak in. Sodium is flowing in very readily and down its concentration gradient. So it's a passive, easy process. The iodide actually sneaks in with the sodium. It's a sodium-iodide symporter. So iodide sneaks in with the sodium.
- 3.18** But that's not enough. There's also an active iodide transporter. From the Basics in Biology course, you'll recall that active transport is one that is using energy in the form

of breaking ATP. And so ATP will be used to actively pump in the iodide. And whenever you're actively moving something, you know that you're pushing it against its concentration gradient. You're moving it into an area that already has a lot. Nature doesn't want to do that, so you have to force it, and the ATP allows you to force it. So we have these two mechanisms: the active iodide transporter and the sodium iodide symporter, both of which are pushing the iodide into the follicular cell.

- 3.19** But we're not done. Remember, we need to get it into the colloid, and so we have one last specific transporter called pendrin. Pendrin will then take that iodide from the follicular cell and squeeze it into the colloid where we ultimately need it. Now we have trapped it into the colloid because once it gets in, the iodide cannot get out. It's going to come out on its own in a different form later—or not on its own, rather. It will come out later in a different form once it's bound into the thyroid molecule. But never again will it come out in its own form.
- 3.20** But we're not done with iodide yet because in its current iodide form, we can't use it. The chemistry of it would be that it has basically one too many electrons. If you're looking at the chemical structure here or the nomenclature for it, rather, it's I^- . Well, we can't use that. We need to oxidize it. This is the first time we use that enzyme that I said is multi-talented: TPO, thyroperoxidase.
- 3.21** Its first action now is to take hydrogen peroxide— that's where the peroxidase part of it comes in—it takes hydrogen peroxide. Yes, even the very hydrogen peroxide you would pour on a wound, although that's not what's in the cell, but it is the same molecule. But pouring it on your wound isn't going to get it into your cell. We make it. But hydrogen peroxide will now interact with the I^- , the iodide, and oxidize it and give us iodine, the elemental form now is residing within our bodies. So the iodide trapping happens first, then we have the oxidation, and it's the first time we see this critical enzyme, TPO. But it's just the first of several times we're going to see this.
- 3.22** With regards to iodine and iodide as we eat it, it is absolutely essential for thyroid function. There is no thyroid hormone production in the absence of iodide in the diet. This is why within so many countries now, especially as our diets have changed, perhaps, we may become iodide deficient. After all, it's not common here in Arizona for us to be eating a lot of ocean fish that could have iodide in it. We're certainly not eating a lot of seaweed, which could have a lot of iodide in it. So wherever we are, the further we get from an ocean, the more likely it is that we're not getting enough iodine.
- 3.23** So to address that, very often, iodine is just added to salt, which is a good thing. I don't ever suggest that's a negative. It's a good way to make sure you're actually getting iodide. You can eat too much. Iodide is a mineral that the body has a hard time regulating. It can regulate it to a degree and excrete some extra. But if a person really goes overboard, they will be making a hyperthyroid issue potentially.

- 3.24** But this whole idea is what prompted my friend and remarkable scientist, Dr. Stephen Cunnane, in positing this theory of human evolution. All of evolution is theoretical. That doesn't mean it's wrong, but I just feel inclined as a scientist to remind us that we don't actually know how we humans came to be who we are. All evolution is a theory, and one of them is the shore-based theory where Dr. Cunnane posits that part of what made us so unique and these big hungry brains that have such a massive reliance on thyroid hormone—as you'll see when we talk about what happens to a fetus developing with too little thyroid, it is catastrophic—but he said that we evolved by eating, living on a shore of an ocean, where we were eating a lot of iodide to enable the production of a lot of thyroid hormone. At the same time, he also talks about the omega-3 fats and how important omega-3 fats are for brain development. But that's a bit of an aside. But anyway, I just want to impress upon you how relevant this idea is of iodide being so essential for thyroid hormone production, which in turn is so essential for everything, our big, hungry, complicated brains included.
- 3.25** All right, now we're not done with iodide. Now we have this process of organifying it, in other words, actually bringing it into contact with an organic molecule. So far, it's just this boring little mineral, albeit in an iodide form, that's flowing through the blood, not doing any good. But now within the colloid of the follicle, of the thyroid follicles, TPO isn't done yet. TPO, yes, it oxidized the iodide into iodine. Now we want to organify it.
- 3.26** The organification of iodide is when we take the thyroglobulin with its tyrosine—and tyrosine is a unique amino acid, not totally unique, there are others that do this as well, where it has these little aromatic rings, these little hexagonal rings, these little cyclical structures. We'll just call it a ring. And on that ring are two docking stations for the iodine as we organify it and bring it onto an organic molecule. Organic molecule being a molecule with carbons and hydrogens and oxygens.
- 3.27** So the first step on this first ring on its thyroglobulin that's bound to tyrosine is, we place one iodide on it. At that moment, it is something called a moniodotyrosine, M-I-T. It has only one single iodine on it. We can stop there, but we usually will go one more. And now we have a diiodotyrosine. But that is just one single ring, and we never stop with just one ring. We have another ring that will get bound from another tyrosine amino acid. And so we've stacked on another one.
- 3.28** So we have an iodine on this part of the ring, an iodine on this part of the ring. Now we stack on another ring on top, and we can then keep going, and we can iodinate the next ring. So on this first ring, we are iodinating the 5 position and the 3 position. And the way you're looking at it will be the way it's accurate. And so we have the 5 and the 3. And then we stack on another ring. So we have a diiodotyrosine right now. And then we can stack on another ring that will either have a 5' and a 3' together, or we'll just have one.
- 3.29** But you can see if we have one diiodotyrosine coupled—and that's what this is called, by the way, coupling. We're coupling these iodotyrosines together. If we have only a

monoiodotyrosine, we're left with T3, the active form of the thyroid hormone. But remember, it's the less common form. The more common form is T4. That's when we have a diiodotyrosine coupled with another diiodotyrosine. You'd never guess which enzyme is actually mediating all of that coupling. It is, of course, TPO. Once again, thyroperoxidase is critical in the process of creating these two forms, these most prevalent forms of thyroid hormone—T4, and at a 20th of the level, T3. Again, it was the coupling of both.

3.30 Now, we're not done because now that we've made the thyroid hormone, it's not doing us any good by keeping it in the colloid of the thyroid follicle. We need to get it out. So we wrap it up in a little vesicle, and it undergoes this process of exiting from the colloid but maintaining a protected state through the follicle. In fact, there's a little more trimming around, ultimately getting us into the formal thyroid hormones that we think of, T3 and T4. That process is called the endocrine phase.

3.31 It's interesting to say endocrine because the way I described it, we are exiting the colloid, so you'd want to use the term exocrine. But the correct way of looking at it is it's now coming into the blood. And so we invoked the term the endocrine phase. So once we've made the thyroid hormone, we have the endocrine phase of thyroid hormone production, that whole complicated process that we just got done going over. So it's endocrine, meaning we're moving the thyroid hormone into the blood.

3.32 But preceding all of that would be what we call the exocrine phase. Although I didn't mention it like that at the time, but the exocrine phase are the components of everything that became thyroid hormone as they're leaving the blood and moving into the colloid. This would be like the movement of the iodide from blood into colloid. Then the synthesis of the thyroglobulin within the follicle cell and moving it into the colloid. Before we ever had the endocrine phase of moving thyroid hormone from follicle into blood, we had the exocrine, which is basically moving all the building blocks from blood into the colloid. Those two phases, of course, are complementing each other, allowing the whole process to run.

3.33 As you remember from the earlier lecture, thyroid hormone is lipophilic. The nature of its rings in the presence of the iodines make it so that it's not very suitable in water. It's lipophilic. It's afraid of water. It loves fat, and thus it's going to have a transport protein. It's going to be carried in the blood. That's where you'll recall the example of the big bound pool where there's a thousand times more of the hormone bound than there is free.

3.34 What is it bound to? It's bound to thyroxine-binding globulin. Thyroxine-binding globulin, or just sometimes thyroid hormone binding globulin. That's going to be the main carrier. In fact, about 80% of all of the thyroid hormone that is bound is on that binding globulin. Then there's another carrier called transthyretin, which is about 10%. Then lastly, the other 10% is bound to the protein in the plasma called the albumin.

- 3.35** Then there's just going to be a fantastically small amount that is free. That is when it's shaken off its escort and then actually binds to a receptor to do what we think of. Because, again, before it can actually act and move into a cell, it needs to get off of its carrier protein. It says, "Thanks for escorting me through the blood. I'm home. I'm going to go in now." The carrier is the gentleman, and the thyroid hormone is the lady being dropped off with perhaps nothing more than a chaste kiss at the doorstep.
- 3.36** All right, so once the thyroid hormone is brought into the cell, if it's already in its T3 form—because remember, T3 is one of the hormone products, it's ready to go—and it will just do what thyroid hormone wants to do. We'll talk about the signaling in just a moment. But if it's not T3 and it's being dropped off in its T4 form, it's not really ready to do anything yet either.
- 3.37** This is where we have a process called deiodination, where we take the T4 and we're going to take off one of the iodines. At that first ring, we have a 5 and a 3, and it's just called 5 and 3. Then on the next ring, we call it 5' and 3'. So it's still in the same positions, but we say the prime to indicate that it's on that next ring up as we coupled those diiodotyrosines together.
- 3.38** When the T4 comes into the tissue, and this can happen at essentially every tissue of the body— muscle can do this, fat tissue can do this, the liver can do this, where all of these tissues can take in a T4 and then convert it into T3 to dump it back out. But it won't always be the same type of T3. The predominant form of this is going to be the deiodinase I and II. What it does is take the 5 and the 3, the 5' and the 3', and it strips off a 5'.
- 3.39** Now we're left with the actual classic T3 like we think of, which has its very, very powerful effects, where as much as when the thyroid gland is making T4— it's making 20 times more T4 than it is T3—T3 is 100 times more active than T4 is. When you think about what thyroid hormone is doing in the body, it's T3 that is always the one actually getting the work done and doing what we think of when we think of thyroid hormone.
- 3.40** Okay, so again, we have T4 coming in, and then we can undergo this process of a singular monodeiodination, stripping off one of the iodines. In fact, it's the 5'. So we're left with the first ring with its iodines, and then the next ring has had one stripped off, giving us T3. However, you also have the potential where when you deiodinate it, you actually end up stripping off one of the original iodines. Now we're left with a version of T3, which isn't fully functional. In fact, it's called reverse T3.
- 3.41** Reverse T3 is like the villainous version of Superman, the evil clone of a bizarro where it looks a little bit like the original T3, but it's not working the same way—reverse T3. It doesn't make a lot, but it is essentially inactive. It is not doing what we want T3 to do. So it begs the question: Why would the system be designed to even make this inactive form? What a waste of T4.

- 3.42** If there's any reason—and by the way, that's something called teleology, when you wonder at the nature of something. And as a biologist, I like to joke with my students that we can never answer "why," if you'll allow me to philosophize with you for a moment. A scientist can never answer "why." All we can do is answer "how" and "what." "Why" is, I say with my students, God only knows. Why was the system designed— or invoke evolution if you'd like. I'm comfortable with either—why would it have evolved that way or been designed one way? God only knows. I don't know. A scientist can only speculate, but let's speculate a little bit because it doesn't mean it's not fun.
- 3.43** Making reverse T3 could be a way to maintain normal thyroid function but not want all of its effects. As we'll get to in a minute, one of the thyroid's effects is to increase metabolic rate. Well, what if we're starving to death but we don't want our thyroid tissue to die and atrophy due to disuse? We still want it to be producing some hormone, lest it really would atrophy and die. We want to keep the thyroid gland alive and busy, but we don't want our metabolic rate to be too high because we're starving.
- 3.44** One reason why reverse T3 may be a thing is because it actually has been shown in severe starvation to go up. So the person will have less of the active T3 and more of the reverse T3. So that theory fits with what we actually do observe in human biology. Of course, it is just speculation. But that doesn't mean it's not fun. As I mentioned earlier, T3 is the active form, and it is the form that is going to come into the cell and actually have the effect we want when we think about T3.
- 3.45** We have covered a lot of material. Those that are listening, I encourage you to pause the lecture and go back and refresh the whole idea of that axis, the movement of the iodide and the iodide trapping, the oxidation, the organification, making the iodotyrosines, mono- and di-, and then them coupling together, all of which is mediated through this multi-talented enzyme TPO.
- 3.46** Now we've made thyroid hormone, so what? This is where T3 really shines because as I said, it is 100 times more active in the body than any amount of T4 is. It is a lipid-soluble hormone. Because it's lipid-soluble, it would come to the wall of the cell. Whereas its water-soluble counterpart or mates would have to knock on the door, it just sneaks right into the home like a ghost. Then what it does is bind to something that's going to escort the thyroid hormone into the very nucleus and bind to what's called a response element on our very DNA, on our very chromosomes.
- 3.47** It's going to say, "Here's a docking station for thyroid to come and dock." Once it has done so, it initiates the transcription of a lot of genes. I'm going to just touch on a few, and you're going to notice what they all have in common is that they're somewhat central to metabolism. But I don't mean to imply that thyroid hormone is only relevant to metabolism. It is fantastically involved in a remarkable number of processes, none of which are as interesting as metabolism. So we'll just focus on those.

- 3.48** With that in mind, we know that there's an effect of thyroid hormone on mitochondrial function and overall energy expenditure. We know the thyroid hormone has an effect on fat cells in a very fascinating way. And then even the very growth and activity of all cells, including things like cholesterol production and neurons, as we'll get to. So firstly, I want to talk about each of these in just a little bit of detail.
- 3.49** I like to joke that I am a fat scientist because I study fat tissue more than any other, even to the point that in my lab, in my university, we actually will take fat biopsies from people's bellies. In fact, I hope none of my administrators at my university are listening to this. When we were first optimizing that process, we'd never done it before, and so I had my students practicing on me. It was me hunched up on this little bed and me telling them, "Oh, you went too deep. You have to go in a little more shallow." So we numbed it, we cut it, and then we took out my own little fat. My students loved every minute of it, perhaps a little too much. But we were able to master the process.
- 3.50** So we study fat tissue a lot, and thyroid hormone has a fantastic effect. One is when you have fat cells that see a thyroid hormone, thyroid hormone will powerfully activate a process called lipolysis, which is simply a term for breaking down the stored fat. And so fat is stored as a triglyceride, where there's three fatty acids bound together. But when thyroid comes knocking on the door, or rather, it doesn't knock on the door, it just barges into the house, like I said before, it will activate the expression of the genes and enzymes that are involved in splitting off those fats.
- 3.51** So what happens is as thyroid hormone goes up and as lipolysis goes up, we have a lot more free fatty acids coming into the blood. Free fatty acids are the form that fat takes when it's being broken down from its stored fat. And so free fatty acids go up anytime thyroid hormone is going up.
- 3.52** At the same time, we also have a different type of fat in our bodies called brown adipose tissue. Brown adipose tissue is located in very distinct sections of the human body. Typically, it's up around our clavicle and thoracic area. But we have it. Humans do have brown fat, and it is unique because it has a very high metabolic rate, easily 10, 20 times higher than the metabolic rate in most of our fat.
- 3.53** The fat that we can pinch and jiggle is called white fat. Having seen my own fat come out of my own body—the color of your skin, mind you, has no influence on any of this — it's a very whitish yellowish fat. It looks like Crisco almost. It's just this glumpy little white fat tissue. But the fat that is here, the brown fat, is very dark and in fact has a brown color.
- 3.54** Why is it brown? Because it is so loaded with the powerhouse of the cell, the mitochondria. These mitochondria are very, very unique because they have what's called a lot of uncoupling. That has nothing to do with the coupling of thyroid tyrosine residues, but it has a lot to do with the coupling of how efficient are the mitochondria. In brown fat, the mitochondria are not efficient. They're just burning a lot of energy.

That actually makes it favorable to someone who wants to lose weight. If you can have your fat tissue burn a lot more fat, you're going to lose weight more readily. That's what happens with the mitochondrial uncoupling.

- 3.55** So brown fat is designed to waste energy. It's like an engine, you're revving the engine, but you're keeping the car in neutral. You're not going anywhere. The speedometer is not moving, even though the RPMs are moving. You've uncoupled the burning of the fuel with the movement of the car. In fat tissue, where it's more coupled, not only are you not really moving the engine, but you only move the engine, you're only revving the engine, when you actually want to get something done. White fat doesn't want to get a lot done, so white fat has a very low metabolic rate. Brown fat has a very high metabolic rate.
- 3.56** Thyroid hormone affects both. It not only turns on brown fat, making it even run hotter, but it also starts to tell the white fat to act a little more like the brown fat. So it increases the mitochondria in the white fat, making it what's called beige. It's not as dark as brown, but it's darker than it was, and it has a lot more mitochondria, using a lot more energy. All of that energy use is that phenomenon referred to as mitochondrial uncoupling.
- 3.57** I'd mentioned that thyroid hormone is critical for the brain and neurons. And here, it not only helps with the actual differentiation of neurons, taking stem cells and telling the stem cell, "Hey, you need to become a neuron." So thyroid hormone will help activate that process. In the process, it will also stimulate this phenomenon called synaptogenesis. When you have one neuron budding up to another neuron to continue to send the signal—signal comes down this neuron, connects to this neuron, and continues—that's the synapse. Well, thyroid hormone helps organize all that, so it facilitates synaptogenesis.
- 3.58** But even before you get here, we have another one, which is the process of myelination. Myelination is when you take the neuron and you start wrapping it with these sheaths, what's called a myelin sheath. That not only protects the neuron, but it accelerates the rate at which the signal is moving. You have the myelination—thyroid hormone does that too. Otherwise, you end up having a very slow signal. The less myelin you have, the slower and more frayed the neuron can get.
- 3.59** So whether it is the very synthesis of the neuron, whether it is the structure of the neuron and the synaptogenesis, or the myelination and in sheathing or in wrapping the neuron to accelerate the signal, after all, you want to respond and think quickly, thyroid hormone is involved in all of it. But that is also why it is so important during development, because if that little fetal brain is developing in an absence or too little thyroid, that can be irreversibly damaged, resulting in catastrophic lifelong disability.
- 3.60** You, at this point, are an expert in the thyroid gland. We reviewed the axis, the hypothalamic-pituitary-thyroid axis. We talked about how TSH signals, which of course

was a G protein-coupled receptor, which are so common. Then we went through the very complicated process of how thyroid hormone is made. Then the simpler process of how it moves and how it signals. Remember—it is lipophilic. So despite my stumble earlier of it coming and knocking on the door, no, it goes right through the door and then goes into the nucleus and induces the transcription of genes involved in lots of things, many of them metabolic, with distinct effects at fat tissue and neurons and many, many other tissues.

- 3.61** Let's now talk about what happens when the system gets a little confused. As a reminder, we have this axis of hypothalamus being tertiary, anterior pituitary being secondary, then the thyroid gland itself being primary. But then as another reminder, TSH from the anterior pituitary has two effects at the thyroid gland—it both stimulates the production of thyroid hormone by activating things like TPO, and it stimulates the growth of the thyroid cells, the tissue itself. That's important to remember.
- 3.62** With two of those signals, take the secondary signal, TSH, and the primary signal, T3, the active form of thyroid hormone, and imagine, if you will, a grid, because this is what helps us. We can use this grid to think through the interplay between TSH, secondary signal, and T3, primary signal. If a person has a normal amount of TSH and a normal amount of T3, we would say that they have euthyroid or euthyroidism, in other words, normal. When you see that prefix eu-, that means normal. So we would say they're normothyroid because TSH is in a normal range and so too is T3.
- 3.63** But think, if you will, what would we call it if TSH is low, but T3 is high? First of all, if T3 is high, we would say that it is a hyperthyroidism. So T3 is high. The patient has hyperthyroidism. But when you measure TSH, you see that it's low. Someone isn't behaving here. So what would we call that? Where's the problem? Is the problem at the secondary level or is the problem at the primary level?
- 3.64** Remember, the problem will always—the guilty party will manifest because it's not listening to the one that's upstream of it. So who's at fault? What do you think? Primary. Yeah, so this would be a primary hyperthyroidism because the gland isn't listening to its boss. Remember, CEO-manager-employee.
- 3.65** Now let's flip it around. What if TSH is really high and T3 is high? I want you to think about this one for a second because you're going to think you have it obvious, but there might be a little wrinkle to it. If you want to get it right in one, which, of course, you want to do. So TSH is high and T3 is high. Now, we know it's a hyperthyroidism because T3 is high. At the same time, if T3 is obeying TSH, what do we know it's not? We know it's not primary because if T3 is listening to TSH, which it is, he's not at fault. The employee isn't at fault. But what do we not know? We know what TSH levels are. What can we not measure? What's the one upstream of it? What's it called? Do you remember? It's thyrotropin-releasing hormone, TRH. We don't know what TRH levels are. Does that make sense? Because you can't measure it in systemic circulation.

- 3.66** Just to say all that again, the secondary level hormone from the anterior pituitary, TSH, is high. We know the primary level hormone from the gland itself, T3, is also high. Because this one's listening—this one is listening to this secondary, it's not the primary's fault. He's obeying the orders. But what we don't know, is this one the origin of the problem—secondary—or is it listening to the hypothalamus? That's what we cannot distinguish. So we're left concluding what? Anyone want to guess? Tertiary or secondary. That's exactly right. We cannot identify between the two. We need to do some other type of scans or tests.
- 3.67** Let's keep going. Let's imagine now that TSH is high, but T3 is low. First of all, what would we call it? Don't give me the secondary, primary, tertiary. If T3 is low, what do we already know part of this is going to be? Hypothyroidism. We know you have hypothyroidism. And thank goodness, because one isn't listening to the other and we can measure it, whose fault is it? The one that's not listening. So what would we call it? What type of hypothyroidism? The gland isn't listening to the anterior pituitary—primary. So here we have a primary hypothyroidism.
- 3.68** And then lastly, to go around the corners here, we end up having both of them being low. So they come in with hypothyroidism—T3 is low, but wouldn't you know it, TSH is low as well. Who do we know is not at fault? Primary. We know it's not the gland's fault because it's listening to the boss's orders. But we don't know whether this boss is listening to its boss. And so once again, we have a hypothyroidism, but it is secondary or tertiary. We can't tell with the data that we have.
- 3.69** Now let's talk about problems of the thyroid. I introduced the topic by going around that grid in the scores that we did. But let's just go through some examples of hyperthyroidism and then hypothyroidism and talk about times where the person may have a goiter and then you being careful at not assuming too much.
- 3.70** Hyperthyroidism is a condition that's generally known as thyrotoxicosis. Thyrotoxicosis is a name for hyperthyroidism. The clinical manifestations, some of them are going to be general to anyone. Like, they're going to notice some weight loss. Not surprising because thyroid hormone accelerates metabolic rate in the body, generally, and it directly accelerates metabolic rate in fat tissue.
- 3.71** They will also manifest with a degree of heat intolerance. So they'd walk into a room and just say, "Oh my gosh, it's so hot in here. I got to take off my jacket." And they're going to be thirsty because they literally will be panting. They're so warm, panting like a dog. They're breathing faster. So they're going to be thirsty, especially in a place like Arizona or depending on where anyone's watching, the drier the environment, the more you're losing water through your breath. But also the heart is going to be beating faster. The entire cardiovascular system is amped up a little bit. So pulse rate is going to be higher. Pulse pressure is going to be higher. Blood pressure is going to be higher to the point that the person may even be manifesting with arrhythmias, spasmic little beatings of the heart.

- 3.72** The neuromuscular junctions are going to be also a little over excited. So the person may find that they have those little muscle tremors and shivers that are happening at any moment. And of course, they're warm. The skin is going to be a little moist because that's how humans primarily cool off. Although I mentioned the panting dog a moment ago. Why does the dog pant when we just sweat? It's because we have exposed skin. Putting a little bit of water on the skin and having it evaporate, as I discussed in the Fundamentals of Physiology, that's the best way for the human body to cool off.
- 3.73** Now, we're going to talk about a specific form of hyperthyroidism in a moment called Graves disease, where it has even more signs and symptoms, and we'll come to that one in a moment. But any time a person has too much thyroid that they're making, you can give them something that inhibits TPO. That is the nice thing. The nice thing about having one enzyme to rule them all when it comes to thyroid hormone production is that if you can develop a drug that simply inhibits that one single enzyme, you can turn down the entire process. So that's the good news of hyperthyroidism, that if you give them a drug like propylthiouracil, you slow down the activity of TPO, and so you can block all of that production.
- 3.74** A moment ago, I mentioned Graves disease. It is actually the most common of the thyroid, of the hyperthyroid disorders. The next most common is a little tumor, an adenoma—remember, it's not an adenocarcinoma; it's not spreading. It's a small little encapsulated tumor that's generally playing nice, but it may be secreting too much thyroid hormone. Then there's some uncommon ones, like actual malignant tumors, a pituitary problem that's creating too much TSH, which is then stimulating the thyroid gland too much. Then the most rare of all of them is the struma ovarii, which is when the ovary tissue spontaneously starts developing thyroid tissue in itself. Isn't that fascinating? You have this little marbled thyroid tissue within the ovary.
- 3.75** Let's go through the first one—Graves disease. With Graves disease, it is an autoimmune problem. It's important for you to change how you view autoimmunity. When you think about autoimmunity, you think about the immune system destroying something. After all, that's what commonly happens. That's what happens with type 1 diabetes—the immune system is destroying the beta cells. But when it comes to Graves disease, the immune system is stimulating the thyroid gland; it's not destroying it. That is critical for you to accurately understand this disorder.
- 3.76** We have these thyroid-stimulating antibodies, and these thyroid-stimulating antibodies will come and mimic the effects of TSH at the thyroid gland. Both effects. Remember, TSH is both stimulating the production of thyroid hormone and stimulating the synthesis of new thyroid cells. Now, the thyroid-stimulating antibodies are binding to those very same TSH receptors and doing the exact same thing that TSH was doing.
- 3.77** So think through this with me for just a moment. If we have antibodies that are coming in and mimicking TSH, we're going to have a production in thyroid hormone. If we are producing a lot of thyroid hormone, what's going to happen to negative feedback? So if

we were to go back and measure actual TSH levels in this person, their body is flooded with thyroid hormone, what is this axis going to be trying to do to the thyroid gland? Tell me, what do you think? It's going to want to turn it off. So what would you expect TSH levels to be? Very low.

- 3.78** You already can tell me the nature of the problem. If T3 is high and TSH is low, who's not listening? The gland. So what type of problem is it? Primary problem. Yeah, so Graves disease is a primary hyperthyroidism, but it's all because of the darn immune system that's mimicking TSH and thereby activating and stimulating those TSH receptors. So with Graves disease, we not only have all the same signs and symptoms of hyperthyroidism, like the weight loss, the heat intolerance, the cardiac arrhythmias, the spasm in muscle, the skeletal muscle twitching, but we also have unique effects because of the antibodies.
- 3.79** One of them is this phenomenon referred to as exophthalmus. With exophthalmus, the antibodies are, oddly enough, stimulating the growth of little fat pads behind the eyeballs. We have these—everyone has these small little fat pads, maybe a bit of a cushion or something, behind at the back of this orbital socket. Well, these antibodies start stimulating the growth of the little fat pads, the result of which is it starts to push the person's eyeballs further out, and so the person will have eyeballs that are bulging out.
- 3.80** Because of this, they will have a lid lag, where if you are looking at the patient or any one of us who are doing this, and you're holding a finger in front—in fact, look at my eyes, I'm looking at my finger, then as my finger is going down, you'll notice as my eyes point down, my eyelids are coming down with it. However, a person with exophthalmos, you'd think, "Well, I wouldn't even need to do that." You can look at the person and tell their eyeballs are bulging. I know. But this is something that's called a lid lag, where now they're looking down at something, but their eyelids are still tucked up around. They're not passively overcoming the curvature of the eyeball.
- 3.81** So they'll have a lid lag. All of this is a result of the exophthalmos, the bulging, the pressing out of the eyes. It is an absolute rock-steady sign and symptom. If you find someone and they're thin and they have their bulging eyes, you can almost bet that they have Graves disease.
- 3.82** It doesn't end there. Some people—the exophthalmos is a universal feature of Graves disease. Another feature is the dermopathy. Within that term, you hear the word derma or dermis, which is skin, where, as a result of the antibodies, there can be an excessive growth of the dermal layer and these molecules beneath it called mucopolysaccharides, which basically just create this gel-like bulk beneath the skin. This is commonly called orange peel skin. Imagine the bulging wrinkles of an orange peel, and that's what's starting to happen, typically on the lower limbs of the individual. Unique to Graves disease. Just run-of-the-mill, if you will, hyperthyroidism will not

manifest with these two problems. The exophthalmus and then the dermopathy are unique to the Graves disease version.

- 3.83** None of this is describing so far a goiter, or at least not necessarily. So let's talk about how a goiter fits in to the thyroid gland and hyperthyroidism, how we commonly think of it as only being connected to hyperthyroidism, and yet it isn't necessarily. A goiter is any time the thyroid gland is big. Thyroid gland is larger than normal, that's a goiter. But despite what you may think, that can actually happen in low thyroid conditions, high thyroid conditions, and even normal thyroid conditions. So the size of the thyroid gland or the presence of a goiter does not lead you to some conclusion about their thyroid hormone levels. You'd have to do more testing. All we know is that there's something wrong with the thyroid gland.
- 3.84** So let's describe some instances. In the first case, how could you ever have a case of the thyroid gland being big, but thyroid hormone being low? Well, one could be if you don't enough iodine or iodide. If you're not eating enough iodide in the diet, then think about it. Not enough iodide would result in too little thyroid hormone being produced. So they are hypothyroid. Then think back to the whole negative feedback system— if we're not producing enough thyroid hormone, we're not shutting off the feedback. So what would TSH levels be? If T3 is too low, TSH is going to go through the roof because the anterior pituitary is thinking, "Hey, thyroid gland, come on, kick it into gear. We need you to be making thyroid hormone. Why aren't you making it?" The thyroid gland, of course, is saying, "Well, I don't have enough iodine. Here, I need this molecule to make the thyroid hormone."
- 3.85** But even though the TSH is not successful at stimulating the growth of new thyroid hormone, thus the hypothyroidism, it is still successful at promoting the growth of the thyroid gland. Thus, we have the goiter. Now, on the other hand, high thyroid is any time you'd think about hyperthyroidism. Maybe we have a TSH secreting tumor. Maybe we have Graves disease with the thyroid-stimulating antibodies. Any of those would be goiter and hyperthyroidism.
- 3.86** Then normal, the most common case of normal thyroid hormone but big thyroid gland, would be an adenoma, but it would be what's called a nontoxic. So it's growing, but it's not secreting thyroid hormone. So it's still getting big because of the aberrant growth, not because of too much TSH or anything else, it's just a growth. And so you'd see it. Now, usually, the difference in that case is it's going to be one side, where it's not the whole thing being uniform. It's one particular spot that's getting big. So it'll typically manifest in a quirky way that would reflect that it's not a general problem with the thyroid gland, but it's a bigger thyroid, and so it would still fit the bill and meet the description of a goiter.
- 3.87** In one way, we can determine where a problem is, is by actually infusing a radio-labeled iodine or a radioactive iodine. And so you'd put in a little bit of this radioactive iodine, wait a little bit, and then do an X-ray because it's hot. It's going to hit the X-ray

and you're going to see that. If you notice that the entire thyroid gland is really lighting up, that means it's a general or it's a diffuse growth, a diffuse goiter. If it's one particular little node or one particular spot, that is a nodular or it's just one single node. Typically, that's going to be the adenoma or the aberrant growth.

3.88 Before we leave hyperthyroid behind, I would be remiss if I didn't share with you one of the most remarkable examples of this. It's called thyrotoxicosis factitia, when you're taking in too much thyroid. An obvious example of this would be someone who's taking too much of their thyroid medication. So someone is on a thyroid medication and they take too much. They would have this thyrotoxicosis factitia. They're taking in too much.

3.89 But in the prestigious Canadian Medical Journal— that's of course a joke, but I say that as a Canadian. There's nothing prestigious about it. What Jordan, Peterson, and I have in common is that we're both Alberta refugees, much to my and his great delight. But the prestigious Canadian Medical Journal, years ago, documented this case of seasonal hyperthyroidism, where this woman would come in with all the signs and symptoms of hyperthyroidism. It would happen only in one particular time of year. Then when they would measure her six months later, it would be totally normal. Six months later, it would be 10 times what it should be. So she had this bizarrely cyclical pattern.

3.90 What they found out with a little bit of probing is that them living on their farm, they would regularly butcher some of their own cows and harvest the meat from it. The butcher was unbeknownst to the family, putting thyroid tissue in the hamburger. This woman would eat the hamburger at certain times of year when they were getting to the bottom of the freezer, perhaps, having gone through all the steaks, and then she would start to manifest with hyperthyroidism, all because the bovine thyroid hormone is so close to human thyroid hormone that she would manifest with hyperthyroidism. Not because her thyroid is broken, but because she is getting the cow's thyroid tissue with it. Pretty stark and interesting example.

3.91 Now let's shift it all. We're done talking about hyperthyroidism. Let's go to hypo. In hypothyroidism, some of the symptoms are going to be obvious because they're just the exact opposite. If the hyperthyroidism person is hot and high metabolic rate and hyperactive and twitchy, the hypothyroid person is lethargic. Not only in physical movement. If you do a patellar tendon reflex— tap the patellar tendon and the leg kicks out—usually it's immediate unless they have hypothyroidism. You tap it, and then it goes. There's a noticeable delay as the signal is having a hard time moving.

3.92 You'll actually see a physical—the reactions are slower and the mental processing is slower. The heart is beating more slowly. You have a general slowing of the body, including weight gain. As metabolic rate comes down, it makes it easier to gain weight. As opposed to being heat-intolerant, now they're cold-intolerant. Every time they go into a room, it's too cold and they need to bundle up. Everything is the opposite, as you would expect.

- 3.93** But then there's also some quirky manifestations, including hair loss. It's unique hair loss, where you'd have a very young, healthy teenager who's starting to experience hair loss, perhaps just on the crown of the head, not male pattern baldness in all of its glory, a very unique hair loss. But also even the eyebrows, the ends of the eyebrows start to fade away. Gals torture their eyebrows so much that it may be hard to know, but that's another distinct pattern where the first part of the eyebrow is present, but then the lateral portions are gone. So there's some unique quirks to hypothyroidism.
- 3.94** One consequence of this is a phenomenon called mixedema, where you actually— back to the mucopolisaccharides that I mentioned in the dermatopathy of Graves disease, now you have this bizarre expression or accumulation of mucopolisaccharides, particularly around the eye. What is so fascinating about this, despite two totally different origins, is that it ends up resulting in an opposite look from what you'd see in hyperthyroidism.
- 3.95** In hyperthyroidism with Graves disease, you'll recall the person's experiencing a bulging out of the eyes because of an expansion of the fat pad at the back of the orbital socket. In hypothyroidism, with the loss of thyroid hormone, there's an accumulation of mucopolisaccharides around the eyes. And so now the person starts to manifest with these eyes that are sunken in behind a bunch of fluid. So it's the opposite look. The eyes aren't actually moving. They're still firmly in the orbital socket, but you're getting a lot of puffy swelling all around the eye. So it ends up looking in this opposite way.
- 3.96** Now, throughout the lectures, I've alluded to the relevance of thyroid hormone in development and how critical it is for neurons and synaptogenesis and myelination. There is something called congenital hypothyroidism. Once upon a time, it was called cretinism—that is now an offensive term, generally, but it refers to this baby that is born in a low thyroid state.
- 3.97** The good news is, if mom's thyroid levels were too catastrophically low, she would have miscarried. It is so antithetical to normal pregnancy. You must have thyroid hormone for a pregnancy to go full term. So the good news is that even if baby is developing in a slightly low thyroid state, there's enough to get the job done. And then if you intervene within the first few weeks of life, you correct. You can correct any deficit. Maybe there's a little bit of learning loss and a little loss of cognition, but for the most part, you can correct it if you step in within the first few weeks. Because, again, if there's enough thyroid to get the baby to the finish line, then there's enough to make up for it if you intervene.
- 3.98** If you don't intervene, it results in this irreversible, as I've said, catastrophic loss of normal cognition, normal mental function, and even features and body type. So this congenital hypothyroidism that is not corrected will result in an individual who has much much lower cognition and brain function, altered features, and a much heavier body with much more fat accumulation. So thyroid hormone is critical.

- 3.99** Now, what if a person isn't making thyroid? What do you give them? Levothyroxine is the most common. And because this has become so common, as thyroid problems have become so common, I actually just feel inclined to mention that there is, in fact, a best time to take it. Most people are told to take it in the morning, and in fact, it appears that evening is the better time to take it. So anyone who's on a thyroid supplement, this paper published—I can't see the year, I don't have the year there, but it was published just a few years ago—they find that taking it in the evening is superior.
- 3.100** Now what are the causes? I mentioned the most common causes of hyperthyroidism. The most common cause of hypo- is Hashimoto's thyroiditis. With Hashimoto's thyroiditis, you hear within that term the -itis, that suffix. That means inflammation. Anytime you hear -itis, it's an inflammation. So here we have an autoimmune disease, and it's hurting the thyroid gland. It is the most common form of hypothyroidism within the U.S. Usually what we're doing is actually making antibodies against TPO, that critical multi-talented enzyme that's involved in every process from beginning to end of thyroid hormone production. We come in and the immune system is attacking the thyroid production and ultimately the thyroid cells themselves.
- 3.101** One thing that is so interesting about Hashimoto's is just how fantastically connected it is to celiac disease, where it is extremely common. I think I saw a paper once where it suggested 20 times more likely that a person who has celiac disease is 20 times more likely to develop Hashimoto's thyroiditis than an average individual. What this suggests is that like so many autoimmune diseases, there's an environmental trigger. And perhaps in most people with Hashimoto's, the trigger is the gluten that they continue to eat.
- 3.102** I'm not one of those guys who thinks gluten is the worst thing in the world, but some people are obviously sensitive to it and responding poorly. The more they're exposing themselves to this noxious stimulus, the more it's eliciting an immune response, which originally is to defend the body against the gluten, but ends up attacking self, which is what autoimmune disease is—the immune system attacking self, and it ends up destroying the thyroid-producing functions.
- 3.103** Similarly, there's a fantastic connection with type 1 diabetes, too. A person who has a gluten sensitivity, you have almost, not inevitable, but a very likely pathway of developing Hashimoto's and then type 1 or type 1 and Hashimoto's. These two problems very often go together.
- 3.104** To wrap it up, We started by discussing the hypothalamic-pituitary thyroid axis, and you're more familiar now with the terminology of the tertiary, secondary, and primary, and how to identify that. We discussed the complicated TPO-mediated process of producing thyroid hormone, and then how it gets out, how it moves, how it signals, and then we wrapped up by having a clinical bend, which I'm always inclined to do in discussing the thyroid disorders themselves.

Lecture 4: The Adrenal Axis

At a glance — The adrenal axis — stress hormones and their effects.

- 4.1 As we continue our journey through the endocrine system, we now come to one of the other canonical big endocrine systems, namely the adrenal glands.
- 4.2 Over the time that we talk about the adrenal glands, we're going to do a brief review of the adrenal gland anatomy because it's a little quirky. In fact, it has two important parts when it comes to endocrinology. It's like an orange, and we have the peel, which is the adrenal cortex. Then we have the flesh of the fruit that in this case we call the adrenal medulla. We're going to talk about how each of those parts of the adrenal glands is important to the production of the hormones from the adrenal glands because there are a lot of them. We're going to focus on them thoroughly but briefly, and then we'll talk about some of the adrenal gland disorders at the back end.
- 4.3 First and foremost, let's get familiar with the tissue that we call the adrenal glands. As the name suggests, you hear the word renal in there, and that's because they are, in fact, physically located right above both kidneys. Above each kidney is the adrenal gland tucked right against it. But then the adrenal gland, as I said, is a tissue with two parts: the cortex and then the medulla.
- 4.4 When we think about the cortex, as we dive into that, you're going to see a lot of steroid hormones because every hormone that's coming from the cortex that we're going to talk about is a steroid hormone. Then within the adrenal medulla, we talk about the catecholamines. That's going to be epinephrine, mostly, and a little bit of norepinephrine. Let's get into the adrenal cortex.
- 4.5 The adrenal cortex, in fact, is not one homogeneous mass. It actually has three distinct layers. In order of these zones, it's the glomerulosa, the fasciculata, and the reticularis, the G, F, and the R: glomerulosa, fasciculata, and reticularis. Each of those is actually responsible for the production of distinct classes of hormones. So that's how we can identify them, slightly different cells producing slightly different, but very relevantly different hormones.
- 4.6 One mnemonic, as you try to remember the relevance or the role of these different layers—glomerulosa, fasciculata, reticularis—it is salt, sugar, sex. Then you'll pardon me for adding, but it helps you remember, the deeper you go, the sweeter it gets. So salt, sugar, sex. It's hormones involved in salt handling in the body on the outermost layer. Then it's hormones involved in blood sugar or glucose in the middle layer. Then even surprisingly, it's hormones involved in sex. Yes, the production of sex hormones.
- 4.7 Let's just start at the top as a brief overview before we dive in more to the cortex. That top layer, the main hormone that we're interested in, is a hormone called aldosterone. As you'll learn, aldosterone is very important at handling sodium and potassium. Now,

I'm pantomiming that out in a very deliberate way, which will become obvious in just a moment. It handles salts, these minerals within the body. Aldosterone coming from that outermost layer of the cortex.

- 4.8** Then we go one layer deeper: the zona fasciculata. The zona fasciculata makes the glucocorticoids. You hear within that—but I didn't use the term I should have. Aldosterone is a mineralocorticoid. That's the salt layer. Mineralocorticoid on that salt layer. Then when we get to the sugar layer, it's a glucocorticoid with cortisol being the most famous. Of course, that part of it is responsive to ACTH, especially.
- 4.9** Then as we get to the deepest layer, the zona reticularis, that is the sex hormone layer, and it's creating those steroid hormones that we associate with sex: the androgens and the estrogens. All of them are derived from that steroid nucleus that I mentioned previous lectures ago, where you have the hexagon, hexagon, hexagon, pentagon. That base structure of cholesterol is what's used as the scaffolding as it commits down the various pathways.
- 4.10** Just as a reminder, this requires the cell to have a lot of cholesterol on hand. Cells can pull in the cholesterol. The way I described it earlier is you have an LDL molecule driving past the cell and kicking off the cholesterol into a cell. That cholesterol goes in and then can be turned into the steroid hormones. At the same time, a cell may have its own little pool of cholesterol in what's called the cholesterol ester droplet. Well, that can release some cholesterol. Then this is going to play into the mitochondria through the steroidogenic acute regulatory protein that I mentioned so long ago, the StAR protein.
- 4.11** The StAR protein is this escort that's waiting at the mitochondria for the cholesterol to come in, and it brings it in, and then it will create the steroid hormone of interest. This same process applies to every single steroid hormone, including all the hormones we'll talk about in the adrenal glands, whether it's the mineralocorticoids, aldosterone being the main one. Whether it's the glucocorticoids with cortisol being the main one. Or whether it's the sex hormones, androgens and estrogens. They'll all follow this same pattern.
- 4.12** Now, it can be a little more complicated where you can start in the mitochondria and then hand it off to another organelle like the endoplasmic reticulum and then come back to the mitochondria for finishing touches. But just for the sake of simplicity and for the sake of this series, we'll just say it begins, and it certainly ends, at least, in the mitochondria. So we'll focus on the mitochondria as the source of all of these steroid hormones.
- 4.13** So let's start by talking about the mineralocorticoids and aldosterone, that first outermost layer, the zona glomerulosa. Before we talk about aldosterone, we want to know what stimulates it. What is the initiating event? The main initiating event is something that is sensed in the kidneys. That's what is called the granular cells. The granular cells will actually sense a reduction in blood pressure. So you can start to see

that there's a redundant effect with antidiuretic hormone. Antidiuretic hormone also has its primary stimulus in low blood pressure. So too does aldosterone. They both share a similar trigger, albeit being totally different hormones in their own spheres. But the trigger will be the same. So the low blood pressure will be sensed by the kidney, and that's going to have an effect in a moment that we'll get to in just a moment.

- 4.14** But then there is another sensor within the kidney, similar to ADH, where it's sensing the salt content. It's sensing how concentrated are two particular ions—how concentrated is potassium and how concentrated is sodium. I'm starting with potassium on purpose because the aldosterone pathway is actually more sensitive to having too much potassium. Then having too little sodium is a secondary concern. Now, why might that be the case? Why might the aldosterone system in the handling of salts be more sensitive to potassium? Because a little bit of potassium goes a long way. If potassium levels start to climb, it can become very lethal very quickly. It makes some sense that aldosterone is going to be primed to respond to the potassium changes a little more so than it's responding to the sodium. But its end effect is actually going to be manipulating both of them, and we'll get to that.
- 4.15** So I mentioned a moment ago that the first step is sensing the low blood pressure, and that's going to get us—I'm about to describe what is going to appear to be a very complicated process. The end of it, though, is going to be aldosterone being produced. But this pathway, this process is called the renin-angiotensin-aldosterone system. It is one of the more famous systems within endocrinology, but it is a little complicated. It not only involves the kidney, because the kidney is where it's going to start, but then surprisingly, it involves the liver, and it involves the lungs, and it involves the hypothalamus, and it involves the adrenal glands at the end, because that's where the aldosterone is actually going to come from.
- 4.16** But to just remind you of the initial triggering events: low blood pressure will turn it on, low sodium or high potassium will also turn it on. And when I say turn it on, I mean that the kidney is going to release a protein called renin. So that's the first step.
- 4.17** Even before that, though, let me just say, those are the triggering events: low blood pressure, high potassium or low sodium. The outcome that ultimately is going to shut it off will be blood pressure getting back to normal. So we want to increase blood pressure. So keep that in mind as an end effect of this whole busy system—getting to aldosterone, which is the actual adrenal cortex hormone we're talking about, but before we get to aldosterone, we have to go on this adventure through the renin-angiotensin pathway. So low blood pressure, we'll say, just to keep it simple, triggers the kidney at those granular cells that I mentioned a moment ago, and that results in the production and release of renin into the blood.
- 4.18** Renin, as it is now released, will interact with a protein that has its origins in the liver. This protein is called angiotensinogen. Anytime you hear a protein that has the ending -inogen, it's about to turn into something else every time. So renin takes

angiotensinogen, clips off a big chunk of it, and now we're left with angiotensin I. We're not done yet. Angiotensin I will flow through the blood. As it flows through the lungs of all places, the lung expresses an enzyme called angiotensin-converting enzyme. Now, angiotensin I turns into angiotensin II. We're still not done.

- 4.19** However, angiotensin II does have some effects that are worth mentioning because remember, the purpose of all of this is to bring blood pressure back up because that was the main trigger that started it: low blood pressure. So we want to bring the blood pressure up. By the time we've come to angiotensin II, we have some effect, which is, the first one being, it will come to the blood vessel and constrict it. Just like we talked about with the V1 receptors, when we talked about AVP/ADH, we have another redundant effect here, where angiotensin II is squeezing the blood vessels, which if you squeeze it, it will increase the pressure, which is what we want. Remember, that's the end goal.
- 4.20** But angiotensin II isn't done because angiotensin II will also go to the hypothalamus and tell the hypothalamus to make more ADH/AVP. So we're back to the posterior pituitary that we talked about a few lectures ago. Angiotensin II is going to recruit some help saying, "Hey, I know my main action is to increase blood pressure. Well, I may as well recruit this other guy to help because I know he can do that. So then, of course, ADH is going to go to the kidney and tell the kidney to start reabsorbing water, which will increase blood pressure. We're still not done.
- 4.21** Angiotensin II will now finally come to the adrenal gland and result in the production of aldosterone, the molecule that we've been waiting to get to. So all of this was a way—on the journey, we were starting to solve the problem a little bit. Remember, angiotensin II is having some effects, and one of those effects is going to be the production of aldosterone from the adrenal cortex. But even before we got there, we were starting to correct this low blood pressure being the primary problem. Then once we make aldosterone from the adrenal cortex, its target is going to be the kidney.
- 4.22** It's going to go to the kidney, and here we have this movement of water, or molecules, rather. Then the water is going to move with it. It has a two-fold effect in opposite directions. Now, we're coming back to what I was acting out a moment ago. When aldosterone comes to the kidney, it both promotes the retention of sodium while promoting the excretion of potassium. Then with the sodium comes chloride. We have sodium and chloride, which is salt, table salt, and we lump them together. Not that that's lumping together in the blood. But where sodium goes, chloride goes, and where sodium and chloride go, water will follow.
- 4.23** So now we have restored some water back into the blood, thereby increasing blood pressure and then turning the system off; or, the system, having done its job in the negative feedback style, will just simmer back down to normal. That is aldosterone from the exterior layer of the adrenal cortex. And its effect on minerals, gives it the

term mineralocorticoid. Anytime you hear that -oid, think steroid, and indeed, yes, cholesterol was its origins.

- 4.24** Let's move on from aldosterone to cortisol. Let's just briefly discuss the axis here because cortisol fits into a hypothalamic-pituitary-adrenal axis. This is sometimes just called the HPA axis, and it is perhaps the most famous of all of them when you talk about these axes. So the hypothalamus is going to release CRH. CRH will come to the anterior pituitary and then stimulate the production of ACTH. Then ACTH will go to the adrenal cortex, resulting in the production of cortisol. Cortisol is the end of that pathway. So it's cortisol that will provide the negative feedback. As cortisol levels start going up, the brain, hypothalamus, and anterior pituitary are sensing it, then they will reduce their production of CRH and ACTH, thereby turning everything off. It's cortisol that provides the negative feedback.
- 4.25** Then what starts it at the very top, at that highest level of the brain, will be things like low glucose. That is the strongest signal. If there's a pressure or a tendency for glucose to start dropping, it'll turn this all on. CRH will go up, and then the pathway is moving along. Within the zona fasciculata, if we come to what is stimulating all of this, we have, firstly, the production of ACTH. This is important, that actually has clinical ramifications. So before we get to the zona fasciculata in the adrenal cortex, let's discuss ACTH coming from the anterior pituitary for just a moment.
- 4.26** CRH from the hypothalamus will bind to a G protein-coupled receptor, one of those famous ones that are ubiquitous. Then when it makes ACTH, this is what's so important, is that it actually tells the anterior pituitary to make a mother molecule, a precursor molecule called pro-opiomelanocortin, POMC. POMC is the mother molecule that on its way out will get cut apart. One part being ACTH, another part being unexpectedly melanocyte-stimulating hormone or MSH. So now we have some melanin role here, which actually will, as I said, have clinical manifestations when we get to adrenal problems later.
- 4.27** Just to clarify—every body, every skin tone has melanin; it's just different colors of melanin. It's not that one has more melanin than another. It's different types of melanin. Some of us are cursed with all kinds of variety of melanin and have less to cover it in all its glory. But suffice it to say, any skin tone and all pigments are a result of melanin, and melanocyte-stimulating hormone is playing a part in the amount of melanin, if not the color of it.
- 4.28** Having stated that, appreciate, please, that whenever ACTH is going to be produced a lot, it's going to have another sibling coming out with it, and that will be MSH. Again, that's going to have some obvious manifestations when we get to some cortisol or adrenal gland disorders in just a bit. But keep that in mind. So out comes an ACTH, at the same time, out comes an MSH.

- 4.29 Now, we come to the adrenal gland, where ACTH binds to, again, a G protein-coupled receptor, ultimately kicking off the series of events that I've already alluded to, where cholesterol will be escorted into the mitochondria via that StAR protein, the end result of which, after several steps, will be the actual production of cortisol. Then the cortisol, having been produced, finds its way out of the cell.
- 4.30 I know what you're thinking, which is, once it's out of the cell and into the blood, it is a steroid or a cholesterol-based hormone. Thus, it is fat-friendly and fearing water. It is hydrophobic but lipophilic. And so in the watery plasma of the blood, it needs to have a carrier, and indeed, it does. It has a molecule called transcortin, which is just a cortisol-binding protein. And so it will be its escort as it moves through the body.
- 4.31 Then as it moves through the body, it will, of course, start affecting various cells, and it affects many, many cells. As a reminder, I'm a metabolic scientist, so I like to focus on the metabolic effect more than some of the others. Not to say it's more relevant. Well, let's face it, it is because metabolism is the coolest topic. So one effect is going to be the effect on plasma glucose. It wants to increase plasma glucose. That is a primary outcome. And indeed, if a person loses cortisol entirely, they very often—what will be lethal is their inability to make sufficient blood glucose in the absence of glucose coming in. So it's a very, very important process.
- 4.32 There's an interesting timing to it where often cortisol levels will start to climb around 4 AM or so with normal sleep-wake cycles. So the latter part of the morning of the sleep cycle, cortisol is going to start to climb. The reason for that, the teleological background here, could be that it's a way for the body to get ready for the brain to come back online, having been somewhat reduced in function during sleep, and then the body to get up and start moving.
- 4.33 So before we get up and start moving and our hungry brain wanting to eat more glucose, which is a good fuel for the brain, or a fuel that the brain uses happily, it uses glucose and ketones very well, but the brain will say, the body will say, "Hey, let's just start mobilizing energy." So blood glucose will start to climb, all because cortisol climbed first. So it has a very strong circadian rhythm to it.
- 4.34 As I said, that main effect of cortisol is going to be to increase blood glucose. To do this, it will activate gluconeogenesis or the synthesis of new glucose at the liver, mostly from lactate. Then it will also activate glycogenolysis or the breakdown of the stored glucose, which is called glycogen. It'll break down the glycogen into glucose molecules. We'll look at that in more detail.
- 4.35 Then it also stimulates proteolysis. It will start breaking down muscle to take those muscle proteins, release them as amino acids into the blood, all so that the liver can take those amino acids and through some biochemical alchemy, turn it into glucose again. To say all that another way, cortisol is prepared to destroy the body of all of its substance in order to turn it into glucose. Yeah, blood glucose is critical. We need it to

survive, but it comes at quite an expense, stripping the body of its substance in the case of muscle and even proteins from bone.

4.36 Cortisol, when it's elevated chronically, also has a quirky effect on body fat. I mentioned earlier that we have subcutaneous fat. That's the fat that we can pinch and jiggle. Well, different bodies, as we'll talk about in the sex-specific lectures coming up, will store fat in different places. Cortisol doesn't like that heterogeneity. It wants everyone to store fat from pubis to head.

4.37 So when cortisol is elevated, it will activate lipolysis, or the breakdown of fat, on all of the limbs, including buttocks and hips, where lots of ladies will store more fat. It's sexually determined. But it's mobilizing all of that fat and then storing it from all the way from stomach up to at the very top of their head. Even around their neck, they'll start to have a lot of fat all around their face. From head to pubis, being the groin area, they're going to start storing a lot of fat. That is a particular quirk of elevated cortisol.

4.38 At the same time it's doing all of this, it's downregulating the immune system. That's a natural effect. This is why some people who have chronic inflammation will take cortisol-like steroids. If you ever have known someone who has an autoimmune disease and they're on steroid shots, it's actually some version of cortisol they're taking because cortisol inhibits the immune system.

4.39 Let's now move on to the adrenal medulla. We've been journeying through the adrenal cortex, and we talked about the zona glomerulosa and its production of aldosterone and the handling of minerals. Then we got to the zona fasciculata and the production of cortisol and the handling of blood glucose and its effects on destroying the body, all to produce more glucose, even to the point of stripping collagen from the skin, making the skin very bruised and marked.

4.40 And lastly, albeit in a much more modest way, we get to the zona reticularis. That is the layer that I mentioned earlier that actually has the ability to take a cholesterol molecule and, in fact, turn it into sex hormones. That's, in most instances, quite modest, certainly in adults. But during puberty, it can play a role and contributes to all of the elevated sex hormones of puberty, complementing the effects of the testes and the ovaries. Then it also starts to play a more important role in menopause, which we'll get to in a future sex lecture coming up. But for most premenopausal adults, it's a very, very modest role. And there are some distinct times when the gonads, the testes and the ovaries, need a little bit of help, and the adrenal cortex, namely that most inner zone right next to the adrenal medulla will produce some sex hormones for us.

4.41 Now let's take that last step and get into the fleshy substance of the adrenal gland being the adrenal medulla. The adrenal medulla is the site of the adrenal gland that is responsible for creating a class of hormone called catecholamines. It is one of the amines. It's built on this amino acid backbone, just like we talked about a little bit, with thyroid hormone. There are different cells that are going to produce different levels of

epinephrine and norepinephrine. Most of what comes out is epinephrine, and then a little bit of it will be norepinephrine.

- 4.42 We do have norepinephrine in other places. I'll even mention it again. Normally, norepinephrine is coming from nerve endings. It's acting from a neuron, what's sometimes called a neurocrine signal. Well, a neuron will come to the end of a tissue, and it's norepinephrine that's coming across and telling that receiving recipient tissue to do something. What's typically flowing in the blood by way of catecholamines from the adrenal medulla, you can just think that the epinephrine is the main one.
- 4.43 The process of creating epinephrine is built on that tyrosine backbone. Then what's interesting within the adrenal medulla cells, when we take the tyrosine, after a few steps of manipulating it and tweaking it around a little bit, we actually get to dopamine. So a little bit of dopamine that you can find circulating in the blood, albeit a little amount, will be coming from the adrenal medulla. But it was not meant to stop at dopamine. Dopamine then goes through a couple more steps to actually ultimately bring us to epinephrine, which is the main catecholamine that we want.
- 4.44 There are a handful of triggers that will initiate this, and it's going to be very much this neurocrine signaling. So the brain—in the higher brain centers, there will be stimuli like pain or cold exposure or bleeding or low blood pressure, low blood glucose—it will activate the splanchnic nerve, which directly feeds into the adrenal medulla. When that nerve comes in, it will initiate the ultimate production of epinephrine. Now, let me just touch on that point with a little more detail.
- 4.45 So we have this signal—I discuss this in more detail in the Basics and Biology class and the Fundamentals in Physiology with the nervous system discussion—but we have this signal that's being sent down a neuron. At the end of the neuron, at that junction, we'll have these what's called voltage-gated calcium channels—this is not meant to be a real review, but it ends up being a review—where at the end of the neuron, we have calcium flooding into the end of the neuron. That actually will help the neuron kick off these other molecules called acetylcholine. Acetylcholine is actually the chemical that's moving across the junction.
- 4.46 Then on the receiving end, in this case, the adrenal medulla, the acetylcholine will initiate its own events, opening up other voltage-gated calcium channels. And much like the nerve, when it was flooded with calcium, kicked off its molecule, acetylcholine in this case, you have a similar series of events within the adrenal medulla, where it gets flooded with calcium, and now it kicks out its vesicles containing epinephrine. Because you'll recall, this is a vesicle-bound bunch of hormones. We have the vesicles filled with epinephrine waiting for calcium to come in and push it out. Now it's happened once the signal has come down from this nerve originating from the brain.
- 4.47 Catecholamine has a very fast half-life, though. It gets in the blood and out of the blood within just minutes. It has a very strong, very powerful effect. You may be familiar with

EpiPens or epinephrine pens or adrenaline pens. Someone's having a hard time breathing, boom, give them some jolt of epinephrine, and it is going to stimulate what's called the sympathetic nervous system—a topic I've discussed in previous lectures here. But one of those effects will be to dilate the airway, and the person who had a hard time breathing now has an easier time breathing.

- 4.48 All of the effects of epinephrine are mediated by these very specific G protein-coupled receptors that broadly fall into two categories. I've narrowed it down to focus on just the main ones. Alpha-1 is a G protein-coupled receptor with specific effects that we'll review. Then there's a small family of the beta receptors. When you think about the effects of what's happening where, it is all a matter of where these receptors are expressed. The heart will express some receptors, the muscle will express some, and different ones, the liver will express different ones, the fat cell expresses very unique ones. Let's talk about it, actually, because it all deals with what I just said, it deals with metabolism, so I want to talk about it.
- 4.49 The first one is the alpha-1 G protein-coupled receptor, or adrenergic receptor. An adrenergic receptor is something that's responding to adrenaline or epinephrine. So alpha-1, epinephrine will bind to it, less or so, norepinephrine, then it activates those same alpha, beta, gamma subunits that we talked about. Then in this case, we get to the molecule, the enzyme, phospholipase C, which starts to branch out and have a host of effects. That is the alpha-1. I just mentioned the general pathway, and I'm going to bring it into context and tissues in just a moment.
- 4.50 Then we have the beta receptors, and the beta receptors are once again G protein-coupled with the alpha beta gamma subunits, and then they create the cyclic AMP. All of this is going back to the very first endocrine basics lecture. But I mentioned then that I'd be coming back to those, and here's just another example of that. Let's talk about the effects at the heart, the effects at fat tissue, and the effects at the liver.
- 4.51 Firstly, with adipose tissue, all the chub that we can pinch and jiggle, within the adipose tissue are these lipid droplets that when you actually look at fat cells under a microscope, they look like little bubbles because it's these little circles, these little bubbles of fat. They look like air bubbles. They're bright white as the light is reflecting off of it. But it's actually not bubbles of air, it's bubbles of fat. So we have lipid droplet within the fat cell. We want to break it's going to take down those lipids. The beta-3 receptor is going to help us do that.
- 4.52 Whether it is a neuron coming down—what's called the sympathetic neuron—coming down and releasing norepinephrine, or whether it's the blood carrying the epinephrine from the adrenal medulla, either of these, norepinephrine and epinephrine, can bind to the beta-3 G protein-coupled receptor. Having bound that receptor through an adenylyl cyclase—cyclic AMP pathway, we end up creating a gene and then a protein and then activating an enzyme called hormone sensitive lipase. That name lipase reveals the purpose. Lipase is going to take that triglyceride molecule and start popping off the free

fatty acids, just like I described thyroid hormone is doing. This is doing it, too, albeit in a totally different way. Then we have the free fatty acids coming out, we're burning fat. Then that fat is ideally going to go to the muscle to be used for energy.

- 4.53** Now let's shift to the heart. The heart has on it beta-1 G protein-coupled receptors. When these same signals, norepinephrine from a neuron or epinephrine from the blood, come and activate the beta-1, we have, again, G protein-coupled receptor, we have the production of cyclic AMP, and now we have it come to the heart cells and result in this big influx of calcium. Whenever you have muscle cells with more calcium—skeletal muscle, smooth muscle, heart muscle—more calcium means more contraction. That goes back to, again, an earlier lecture where we talked about the role of calcium and the cross-bridge cycling of myosin and actin, a topic I won't revisit now because it is complex, and I've already done it before here.
- 4.54** Again, suffice it to say, anytime you have muscle cells, if you load those muscle cells with calcium, you enhance contraction. In this case of the heart, you're making the heart beat harder and faster, which is something that you notice, of course, with a sympathetic nervous response, where that part of the nervous system, when it's active, you're feeling it and your heart is beating harder and faster.
- 4.55** At the same time, you have on the blood vessels an alpha-1 receptor. I just talked about it in a different context, but on the blood vessel, alpha-1 will result in the constriction. All of this is going to mean, with the heart beating faster and harder and our systemic blood vessels narrowing, we can move blood around very, very quickly, which is good if we have a lot of catecholamines and we need to get something done, like run away from something. It's good to move blood more quickly in that case.
- 4.56** As we are running away more quickly, we talked about one fuel source to fuel the body in that case—the fat, namely in the beta-3 receptor there—but the liver also is loaded with a lot of glucose stored as glycogen. We want to get access to that glycogen, and even still here, these hormones are going to help. Epinephrine is going to bind a beta-2 receptor on the liver, and through cyclic AMP, it is going to activate an enzyme that will take glycogen and start snipping the glycogen apart, giving us what will become individual molecules of glucose. Then that glucose, of course, can come out into the blood and fuel the body.
- 4.57** At the same time, we have the alpha-1 receptor on the liver, which has its own set of effects complementing through a distinct pathway, the same event. So whether it is the β -2 receptor, breaking down glycogen, or whether it's another redundant alpha-1 receptor, it will also break down glycogen through a totally separate pathway. But the net effect is going to be a lot of glucose coming out of the liver very quickly. And indeed, within moments, if you inject someone with epinephrine, you're going to start—literally within a minute, the blood glucose is going to start to climb dramatically.

- 4.58** Cortisol is not that fast, even though they both share that effect. It's not as fast. But it's noteworthy that even though cortisol and epinephrine are two totally different hormones coming from different cells, produced in different ways, moving through the blood in different ways, they both have that primary outcome in common. Both of them will increase blood glucose very quickly. But that puts them at odds with the most famous hormone coming from the pancreas, namely insulin.
- 4.59** Insulin wants to reduce blood glucose, and it does so alone. But insulin is so effective at reducing blood glucose that it takes multiple other hormones to act together to increase blood glucose. But there's an effect of this, which is if there's too much of these so-called stress hormones— because both epinephrine and cortisol are the prototypical stress hormones, they both want to increase glucose—that puts them at odds with insulin who wants to reduce blood glucose. This is why, in my Why We Get Sick series within the Peterson Academy, I mentioned that stress is one of the primary cardinal causes of insulin resistance. Because any time you push cortisol up in the body, whether it's a drug, an injection, or just the excessive secretion, the body will become insulin resistant with too much cortisol.
- 4.60** Same thing goes with epinephrine. Any time epinephrine is too high, it's driving up the glucose constantly, which is meaning insulin has to work harder and harder. No surprise, the body is becoming insulin resistant within a very short order. So that is why, in that Why we get Sick Series, I said stress is a core cause of insulin resistance, and that's because of those stress hormones. Cortisol and epinephrine don't play nicely with insulin.
- 4.61** Okay, that has been the normal physiological role in the endocrinology of the adrenal gland. Having gone through the adrenal cortex and its layers and all of its outcomes— salt, sugar, sex, hormones—and then getting deeper into the flesh of the adrenal medulla and then its catecholamines, primarily epinephrine made from tyrosine and also a little bit of norepinephrine. And now you're familiar with how those catecholamines act at some tissues of the body. Remember, the action is determined by which type of receptor. Is it an alpha receptor? Which type? Is it a beta receptor? Which subtype? That will determine the actual action of the hormone.
- 4.62** With all of that in our minds, let's spend the final time of this lecture discussing some adrenal gland disorders. We'll firstly start with the disorders of the adrenal cortex, like Cushing disease is the most famous one. Then we'll talk about a problem, very briefly, with the adrenal medulla. It's a little simpler.
- 4.63** Let's start out with discussing too much cortisol. When cortisol is elevated, that is going to always be called Cushing syndrome. Cushing syndrome is a term to describe any situation of chronically elevated cortisol or hypercortisolemia. Cushing syndrome. Now, within that broader term of Cushing syndrome, there's a specific type called Cushing disease. Cushing disease is when you have too much cortisol, but it's directly a result of too much ACTH because of an anterior pituitary tumor. You'll have a tumor of the

anterior pituitary producing too much ACTH, which then comes to the adrenal cortex, telling it to produce too much cortisol. It is Cushing syndrome but that is given its own specific term of Cushing disease.

- 4.64** What they both have in common is too much cortisol. It's just Cushing syndrome describes any situation of too much cortisol, whether it's the adrenal gland itself making too much, which would be a primary hypercortisolemia, or even if it's a person who's taking too much of the cortisol drug to control their inflammation in the midst of their autoimmune disease. That would also be a type of Cushing syndrome. Then again, Cushing disease is if it is specifically an ACTH secreting tumor.
- 4.65** How might these two diseases look differently or these two conditions: general Cushing syndrome versus specifically Cushing disease. Firstly, what are cortisol levels going to be in each of them? In fact, reason with me, if you will. Give me some engagement. What will cortisol be in these conditions that we call Cushing syndrome or Cushing disease? Both of them are going to be high. We know that they're going to have that in common. The person will manifest with signs and symptoms of too much cortisol, like they're starting to store all their fat centrally, like they're getting lots of stretch marks on their skin because the skin is becoming so fragile and weak. They have really high blood glucose levels; they're getting pushed into type 2 diabetes because of it. So cortisol will be high, and they'll have all the signs and symptoms you would expect.
- 4.66** Where things start to get a little different: what about ACTH levels? In Cushing syndrome, where the problem is just too much cortisol, not Cushing disease, specifically, so I've separated that one out. If the problem is just too much cortisol, like a cortisol-producing tumor of the adrenal cortex, we know that in this case, the body is flooded with cortisol, it's going to be turning off the production of ACTH. So there's a negative feedback. In general, Cushing syndrome, not Cushing disease, ACTH will be low. In contrast, if it's Cushing disease, remember, that disease is in fact defined by the excessive production of ACTH, so ACTH will be high.
- 4.67** But let's come back to the origins of ACTH. It seems a little strange to ask this, but you understand why: What would MSH levels be? Remember, every molecule of MSH that is produced, it's coming with a molecule of ACTH, so it will match the same because it's all coming from that mother molecule of pro-opiomelanocortin. One ACTH Coming out with one MSH. So no surprise, the levels will follow what ACTH is.
- 4.68** Then this has an effect. Someone who has Cushing disease will often look different in the midst of any other changes that they may have in common than someone with just general Cushing syndrome because they will have too much melanin being produced, potentially giving rise to what is commonly called a coppery complexion. You can see here on this fellow, even under the nail beds, even places of the body— you'd think, well, some ethnicities have darker melanin than other ethnicities. Absolutely, and that would complicate our ability to look at it at skin level.

- 4.69 But there are some parts of the body that have a lack of melanin, which tend to be the same color across any ethnicity, like the nail beds or the gums. And in fact, the gums of the mouth can also be a site where if there's too much MSH, you're producing melanin in a place where you don't normally have it. So you have some other sites where you can start to get the melanin being produced, which can help reveal the problem, even in the midst of all of the beautiful variety of melanin that we have across the planet.
- 4.70 Now, when cortisol is too high, as I've mentioned, it will wage war on the body all in an effort to increase blood glucose. That will be a common variable—the blood glucose is climbing, climbing, climbing. They're becoming more and more insulin resistant. Very, very often, they develop type 2 diabetes quite quickly. They will also start gaining fat very quickly. As we'll talk about in the adipose lecture coming up, when insulin is too high, it wants fat cells to grow. So the person starts gaining weight, and they gain the weight in a very particular way, right here on the trunk of the body.
- 4.71 Also, they're stripping the amino acids from muscle, so they're wasting away their muscle. They're wasting away their bone, all so that the liver can pull in those amino acids and turn it into more and more glucose. They have a hard time recovering from wounds and healing because cortisol is inhibiting inflammation.
- 4.72 Then at the end of it all, one of the most common causes of death in someone who has Cushing syndrome, where they can't really treat it very well, is suicide. The person is so discouraged because of the absolute destruction happening in their body that they just find their own way out. So it's a very, very crushing debilitating disorder. That's it for hypersecretion of cortisol.
- 4.73 Let's just briefly mention Conn's syndrome. Conn's syndrome is when a person is making too much aldosterone. So it's a hyperaldosteronemia. What you'd have to do is just imagine—what is aldosterone wanting to do? Well, aldosterone wants to increase blood pressure. So no surprise, they have hypertension.
- 4.74 At the same time, aldosterone wants to be dumping out potassium. So they have what's called hypokalemia, from the Greek origin of the word for potassium. That's where the K—why do you identify potassium as a K? Because it's kalemia or the potassium in the blood. Hypokalemia, too little potassium. Too little potassium makes for very lethargic, weak muscles because you can't generate the signal strong enough in the absence of that electrolyte. So they're very weak, very flaccid muscles and lethargic in their movements. Those are two of these triads.
- 4.75 Then lastly, if you do a blood test and measure that renin is very, very low, that is the final nail in the coffin. To say you have all of the clinical triad of the symptoms, these three symptoms of Conn's syndrome, you're somehow producing too much aldosterone. And then you just measure aldosterone to confirm, but usually it's going to be some tumor of the adrenal cortex.

- 4.76 Now, I mentioned, and you already remember, the zona reticularis, which is the innermost layer of the adrenal cortex, right next to the medulla. That can become aberrant in its behavior when it comes to producing sex hormones. It can, in fact, be producing too much. There is a form of this called congenital adrenal hyperplasia. In the case of the congenital adrenal hyperplasia, that will be associated almost always with what's called virilization.
- 4.77 It can make a little boy much more manly well before he should, or it can make a little girl in this congenital—meaning it's starting in utero—make a little girl come out with genitalia that appear to be mixing, where it looks like there's some budding penis when it's actually just the over stimulated and overgrown clitoris. Then in some instances, you can have adrenal tumors that are separate and just producing any hormone, androgens or estrogens. But suffice to say, if it's producing estrogens, then it can result in effeminization. That's talking about the adrenal cortex in its over stimulated way.
- 4.78 We also can have the adrenal medulla have an excess production of its product. In this case, it's when the catecholamines are being overproduced. It is a result always of a tumor. This is called a pheochromocytoma, which is a reflective of the name of the cells within the adrenal medulla that are making these hormones. But the end result ends up being quite obvious and quite simple. If you're flooding the system with adrenaline, epinephrine, you're going to feel it. The person is going to be very anxious, very twitchy, fast heartbeat, hard heart rate—they're going to feel their pulse pounding. They're going to be breathing more quickly—all of these signs and symptoms of someone who seems to be having a panic attack. It just doesn't stop. So you can see how debilitating this would become. Then the result would be, the effect, the treatment would be getting that tumor out.
- 4.79 Now, that was all hypersecretion. We can have an instance where there's too little being produced. Addison's disease is the destruction of the adrenal cortex. Usually, it's an autoimmune in origin. But now we're not making enough cortisol, and we may not also be making enough the mineralocorticoids with aldosterone. But the main symptoms are going to be related to the lack of cortisol.
- 4.80 Think through this for a moment. So it's going to have some weakness, some hypotension because of the lack of aldosterone. Then you'd say, well, there's not enough cortisol being produced, so they're going to have a hard time keeping their blood glucose in check. So some hypoglycemia. But think, if cortisol is too low because the end gland isn't making it, there's not enough negative feedback to turn the system off. And so what is ACTH levels going to be? Exactly. ACTH is going to be high because it's thinking, "Hey, cortisol, you're not doing your job." But if ACTH is high, what else is going to be high? MSH. And so this is why you have JFK, who had a famously coppery complexion because he actually had Addison's disease. And so you can look and see this change in the pigment of the individual. That would identify it.

- 4.81 One last comment here is adrenal fatigue. Adrenal fatigue is us stepping outside of the realm of endocrinology, into the realm of speculation, where people will insist that they have adrenal fatigue, and they will accuse it for causing a lot of their health problems, and yet this does not appear to be a real phenomenon, just to pop that bubble. There's no evidence to suggest adrenal fatigue is a real thing, that the adrenal gland has been worn out because of your stress. No, you're just stressed, and you probably have bad lifestyle habits. Not to minimize the stress, but it's not a failure of the adrenal glands. That doesn't appear to be a real thing.
- 4.82 All right, we just finished another one of the canonical endocrine systems, the adrenal glands. You are now experts at the anatomy—the cortex and the medulla, and the various layers of the cortex—the products of those layers and the medulla, where they come from, how they move, and what they do when they get to where they want to get. Then we wrapped it all up by talking about some very clinically relevant case studies or clinical instances of when things go wrong. Thanks.
- 4.83 You talked about the steroid hormones being bound to a protein. Do they have to unbind from the protein to enter the cell? Because you said they freely enter the cell. But is there a process in between and how can that get broken?
- 4.84 Yeah, that's a great question. Yes, a resounding yes. The carrier must let it go. It is something that can freely happen and will based on how much is free in the blood already. There will always be a certain percent, if you will, or a certain amount of the protein that is free. If that free amount starts to go down, let's say because the thyroid is being used or the sex hormone, the testosterone is being used, an equal amount will come off the bound and then be free and then quickly go into the cell and do what it wants to do. So yes, it must be released. The degree to which the carried protein is released will depend on the degree to which the other small pool of free is diminishing. So as the free pool diminishes through its use, the bound pool will refill it.
- 4.85 In fact, when we get into puberty in just a second, you'll see one of the initiating events that allows the little girl and little boy to become so different so quickly is that the liver starts producing less of the sex hormone-binding protein. So they start to have more and more free hormone. The liver plays an interesting dynamic there.
- 4.86 This is a cause and effect question, and you often hear it with low-carb diets, gluconeogenesis and cortisol. From what I'm gathering, is it more the fact that you have elevated cortisol that's driving excess gluconeogenesis as opposed to what we're commonly told that gluconeogenesis is what's causing high cortisol? Does that make sense?
- 4.87 Oh, yeah. It is absolutely the cortisol. If you have to put a cause and effect, high cortisol will contribute to the gluconeogenesis. It's a main cause, a main driver. To put that in context of, say, a ketogenic diet, it is absolutely true. There's human evidence to show in women, not in men, interestingly enough—men appear not to have any cortisol

response. It stays within the normal range. Women have an increase in cortisol around days like three through five, and then it goes back to normal. Of course, that blip gives anti-ketogenic advocates a lot to talk about. All they want to do is say, "Oh, well, ketogenic diet is a stress because cortisol goes up." It does temporarily, and then it goes right back down.

4.88 But the problem, part of the problem in all of this is—and I use this same term—it might not be fair to call these things stress hormones. What if they're just metabolic response hormones or something? Unfortunately, that term stress is so loaded that people will try to weaponize it against one particular ideology or another. But it is reflective of there's less glucose coming in. It appears to be a unique female phenomenon. We need a little more help in order to increase the glucose. Then for whatever reason, it's enough to go away. So it's a temporary effect that has been shown in women only. But then to say that that's undue stress, I don't know. I think that's inaccurate.

Lecture 5: The Gonadal Axis

At a glance — The gonadal axis and reproductive hormones.

- 5.1 As we continue, let's get into one of the last of the really important and big canonical glands when we talk about endocrinology, and that's the gonads. But in the midst of talking about gonads, I want to make sure that I weave in the relevance of growth insofar as it plays such an important part of puberty.
- 5.2 We'll start with the testes and the ovaries, the gonads, talk about how they become what they are and the affiliated or connected sex structures. Then we'll move into puberty, and then inserted in there will be a tangent, talking about the hormonal regulation of growth because it's so obvious and so explosive during puberty.
- 5.3 To start off the lecture, we'll talk about sexual differentiation. How does that early little baby become male or female? We'll talk about that binary process. Then we'll talk about the change into puberty, and then end, of course, with a little bit of growth inserted in there, and then end with the idea of the sex disorders having to do with some change in gonads or some change in other structures.
- 5.4 To set the stage, when we talk about sex—as a cell biologist, sex is the appropriate term. Gender is not a term of biology. We talk about sex, and it can start at the level of the genes or the chromosome. We will have what's called chromosomal sex. That is either XY, which becomes male as we typically think of this, and then XX, which is the female chromosomes. Then that, as you will learn, will have an effect at determining the gonads. As you'll see within the early brand new little fetus, the gonads haven't committed yet. They're bipotential. It's a bipotential gonad we call it. Then the chromosomal sex will, when things are working, determine the gonadal sex: XY will turn the gonads into testes; XX will turn the gonads into ovaries. Then lastly, we have the phenotypic sex, which is the structures, so it will differentiate female or male.
- 5.5 As a reminder, within the fetus, there's not testes and ovaries, and one's going to live and one's going to die. No, it's the primordial gonad, the bipotential gonad that has the potential to go down either pathway. Then separate from that—because this is a very important point—we have internal sex structures. In the early little fetus, in the early baby, we have what's called Müllerian ducts or Müllerian structures. Those are the structures that when they develop, become the structures of the female internal sex organs: the fallopian tubes, the uterus, the cervix, and the vagina. All of that is going to be part of the Müllerian ducts, which become the female structures.
- 5.6 Present also within the fetus are the Wolffian ducts. The Wolffian ducts are those things that will become the structures of the male internal system: the epididymis, the vas deferens, the prostate, the Cowper gland—all of those are part of the structure of the internal male sex system. So again, the baby started with a bipotential gonad that

becomes testes or ovaries, and then it has both sets of structures, which, when things are working well as we'll go over, one will survive while the other must die.

5.7 Now, what determines that? In fact, determine is the right word because on that humble little Y chromosome—and it is a lot smaller than any of the others, but it packs a punch— is a section called the sex-determining region Y, or sex-determining region of the Y chromosome. That small little gene is going to initiate the entire series of events that will tell this little fetus, "You're male; become male."

5.8 So this SRY gene will encode and become a protein called testis-determining factor. Sometimes it's also called the SRY protein. We'll call it TDF, the testis-determining factor. Testis with an "i" at the end being the singular version with testes. So testis-determining factor is the protein product of the SRY gene. That little protein product is going to now go to the bipotential gonad and say, "You are becoming testes." And so it will start to express all of the cells.

5.9 These little stem cells start to become two primary types of cells: Leydig cells and Sertoli cells. We're going to come back to those. But for now, you can just remember, the SRY gene became testis-determining factor, which told that bipotential gonad to become testes. And now the testes have these prototypical testicular cells: the Sertoli cells, and the Leydig cells, each making their very important products.

5.10 In another series of events, you would have a cell, a fetus, with just two Xs. There is no SRY gene, so there is no TDF produced. And in the absence of that signal, the default setting is ovary. So rather than developing Sertoli cells and Leydig cells, we make the two primary cells of the ovary being theca cells and granulosa cells. To state that another way—in the absence of the signal that determined testes, the bipotential gonad will become ovary. That means it will have the substance of the ovary, if not the function.

5.11 For the ovary to be functional—by that, I mean actually ovulating, which we'll get to in a moment, and enable reproduction—you have to have two X chromosomes. We're going to talk about a disorder later on where there's just one X. Well, the substance of the ovary is ovary. It has theca cells and granulosa cells. Those are the cells of the ovary, but they're not capable of making eggs. You have to have two Xs for the ovary to be fully functional. But you don't have to have two Xs for the ovary to be the gonad that is what's left in the body.

5.12 Now that we have identified the origins of the testes versus the ovaries, let's come back to the axis being the hypothalamic-pituitary-gonadal axis. Within the anterior pituitary, we have these specific cells called gonadotropes. The gonadotropes are making two different products. One is luteinizing hormone. In the male—we'll start with guys and then get to gals. I end with the gals because the gal, when it comes to sex and ovaries, and then when we get into reproduction, are fantastically more complicated. We'll get done with the simple guys, and then we do the more complicated gal, female version,

afterwards. So don't think that I'm not being chivalrous here with the gentleman being first. It's just to get done the easy stuff before we do the complicated stuff.

- 5.13** So LH comes from the gonadotropes. That's luteinizing hormone. The name luteinizing hormone is derivative of female physiology as we'll get to. But that doesn't mean it doesn't have a place in men. It very much does. It's just men aren't the primary of interest here because it's so much simpler. But LH will come to the Leydig cells and tell the Leydig cells to produce testosterone. That is the main source of testosterone in the male. Again, LH is what determined it. Now, testosterone is the end hormone here, so no surprise, it is the one that has the strongest negative feedback to turn the system, or to turn the axis off.
- 5.14** But there's another cell there, which is the Sertoli cell. The Sertoli cell is primarily stimulated by follicle-stimulating hormone, FSH. "The testes don't have follicles," you're saying. That's true. They don't. The ovaries do. So that is, again, a term that is derived from female physiology, but it still plays a role in men, and indeed, it's somewhat complementary as it has a role in sperm, which I'll come to in a moment.
- 5.15** At the same time, the Sertoli cell is producing the sex hormone-binding protein, but ultimately—rather, it's coming from the liver. It stimulates the liver to create more. In fact, that's a rule of thumb. Anytime I mention a binding protein, almost every time you can assume it's coming from the liver, albeit the liver is being told to produce it by some other signal, but it'll still be the source. This is no exception. So sex hormone-binding protein will be stimulated from the Sertoli cells to be created from the liver.
- 5.16** Then there's one last hormone that is very, very important in the male, and that is from the Sertoli cell, and it's a peptide hormone called anti-Müllerian hormone. Can you already start to imagine what the effect of that hormone is? Anti-Müllerian hormone from the Sertoli cell is going to have a powerful effect at determining which structures internally of the sex structure survives and which set of structures is going to die. Then no surprise, once testosterone is produced and released from the Leydig cells, it will move through the blood bound to the carrier.
- 5.17** Now, testosterone isn't as singular in its end goal or end effect as you may think, because testosterone can be converted into other hormones. On one hand, and of great relevance to the male, there is an enzyme called 5-alpha reductase. And 5-alpha reductase will take the testosterone and change it into a different version, actually a much stronger version, called dihydrotestosterone or DHT. That'll have an effect that we'll come to in a moment. But separately, testosterone can be manhandled by aromatase and converted into estrogens.
- 5.18** The estrogens—when I use that word, if I'm saying it appropriately or if I can remember to use it appropriately, it's a plural word. There is no singular estrogen. It's the estrogens. In fact, when someone says estrogen, they mean the main estrogens of the estrogens, which is estradiol. But there's estriol and there's estrone, which have more

modest effects. So if I ever say estrogen, it's a slip of the tongue. I should have said estradiol, but I'll try to be specific and just call it estradiol. But that comes from testosterone via the actions of aromatase.

- 5.19** Every single estrogen was once, of the estrogens, was once testosterone in males and females. It's just females convert a lot more of it than males do. But males do, too. Males have estrogens, just like females have androgens. It's just relatively more androgens are being converted to estrogens in the female because of the higher actions of aromatase.
- 5.20** Now, I mentioned dihydrotestosterone a moment ago. It's a much stronger signal than testosterone. It is DHT that is so critical for external virilization. What is it that makes that little baby have a penis and a scrotum? It's DHT. Those external signs of the sex require a strong signal, and DHT is that signal. But then, testosterone is helping promote the Wolffian structures, which we'll come back to in a moment. And of course, it's important for spermatogenesis or the production of that gamete, that reproductive cell within the male.
- 5.21** While I'm talking about androgens and spermatogenesis, I feel inclined to say, in the midst of us wondering so much, why are all men becoming so low T, you hear nowadays, that what the average 25-year-old male has by way of testosterone levels, a generation ago, you'd see that in a 65-year-old man—why are we becoming so low testosterone? I'll revisit that later, actually, and bring it back to the fat tissue, but it could be that we're just not eating enough fat.
- 5.22** There's pretty convincing evidence like this study published in 2005 that found if you have men eat less fat, their testosterone dropped within just weeks by 12%. At the same time, their quality and the motility and the function of the sperm also plummets, as seen in a 2016 study. So one way to help a man be a little less manly, if testosterone and sperm production can define that, is to deprive them of fat. And maybe a part of it is the robbing of cholesterol because cholesterol is the building block for testosterone.
- 5.23** Let's come to the overview of the ovary, and we're still going to go into a lot more detail with the internal structures and then puberty, of course. But we've had an overview of the testes. Now the ovary—as I mentioned, in the absence of the testis-determining factor, the ovary will fill itself with granulosa cells and theca cells. They're the ones that are going to respond to the signals from the anterior pituitary. When LH comes from the anterior pituitary, it will tell the theca cells to produce testosterone. We're going to visit this in a moment, and I'll show you that one cell alone is not enough.
- 5.24** The ovaries have to go and make it complicated where you have to have both cells, whereas in the guy, the Leydig cell is sufficient, it produces. It's nice and simple. In the gal, it gets more complicated. The theca cell will respond to LH by making testosterone. And then it will pass the testosterone to the granulosa cell, and then the granulosa cell

will convert it into the estradiol, the main estrogen. And that's going to be the source. So it's this handoff actually in a paracrin fashion where the theca cell starts, makes a product, hands it off to the granulosa cell, which then finishes the packaging up into estradiol. Then estradiol is going to be the main signal providing negative feedback to turn the axis off in the woman or make sure it stays regulated.

- 5.25** At the same time, we have follicle-stimulating hormone acting on the granulosa cell, but now it's actually helping with the development of the follicle because we'll see that the granulosa cell is critical to the growth of the follicle. It's the layer of the follicle, as we'll get to in a minute. Then also we have the stimulus to produce the sex hormone-binding protein.
- 5.26** I mentioned already that in both sexes, in every body, all estrogens were once testosterone. The ovaries will mediate that process more readily through a higher level of aromatase. But as we'll come to later, there are other tissues that express aromatase, including fat cells. So fat cells can, in some instances, act a little bit like the ovary.
- 5.27** A moment ago, I mentioned that the ovary has to go and make it complicated when it comes to sex hormone production. Well, it is a remarkably complicated system, but the summary of it is how I explained it a moment ago. On one hand, we have the theca cell, which will be stimulated by luteinizing hormone. The end result of that—through cholesterol, through the mitochondria, like I'd shown you previously—will be the production of the main androgen: testosterone. There are some others, so it's appropriate to say androgens. The main one being testosterone, but androstenedione will be one, too, and there's some others.
- 5.28** But then the androgens will be handed off to the granulosa cell. Then it's the granulosa cell that finishes that conversion through its own activation with another G protein-coupled receptor and all the signaling we've talked about previously to get us to estradiol. Then estradiol is going to have a lot of systemic effects like we associate with estrogens. But one important effect, as we're going to talk about with reproduction, is that it's helping with development of the egg or the ova within the ovaries. Then the estrogens are going to move through the blood with their own carrier. Well, in fact, it's the same one that moves testosterone—the sex hormone-binding protein.
- 5.29** All right, all of this has been intended to give us an overview of the gonads and how the gonads pick their direction based on the presence or absence of testis-determining factor. If TDF is there, because the SRY gene is there, then you have Leydig cells and Sertoli cells producing lots of testosterone, and remember, producing AMH—anti-Müllerian hormone. That's going to become very important that you remember in just a moment. But then in the absence of TDF, the bipotential gonad has committed to the path of the ovary, producing theca cells and granulosa cells.

- 5.30** Once the gonad has decided where it's going to go, it then starts to have its effects systemically. First and foremost, on those internal sex structures that I mentioned earlier with the Wolffian ducts and the Müllerian ducts that become the male structures or the female structures, respectively.
- 5.31** Testosterone, as it's flowing through the blood, will bind to androgen receptors throughout the entire body, most especially at the Wolffian ducts. It will tell the Wolffian ducts, "You need to stay, and I need you to develop." It will stimulate the development of the Wolffian ducts. Then DHT, the more active form, will also do that while not only stimulating the Wolffian duct development, but also the external genitalia, pulling down the ovaries—what would have been ovaries, now testes—pulling them down to get them closer to the scrotum, closing what would be external labia in the girl, closing it, fusing it together, turning it into the scrotum—then what would have been the clitoris in the girl—expanding it, now it becomes the penis in the boy.
- 5.32** All of those structures, now externally, that can go either way, get determined because of DHT, and then you have the external sex characteristics of the male in that case. So in all of that story that I just got them telling you, it was forced. Typical guys—they have to be told what to do. So these sex developments, the internal and the external structures, have to be told, "Become male." It's a combination of testosterone and dihydrotestosterone.
- 5.33** And then remember, we have the Sertoli cells which are making AMH. It's important to remember because that is what tells the Müllerian structures to die because we don't want both. If we didn't have the anti-Müllerian hormone, you'd have the growth of the Wolffian, but then you'd also have the Müllerian ducts sticking around. They degrade into nothing because of the Sertoli cell making AMH. That's the male determination as we start to talk about the phenotypic sex or the structural differences between the sexes.
- 5.34** In women, the ovaries are now—they have theca cells, which are producing some testosterone, but not enough. Remember, the granulosa cells at the same time are taking in that testosterone, a lot of it, not all of it, but turning some of it into estrogens. And so there's insufficient androgens to stimulate the Wolffian ducts. And remember, typical guy—it has to be told what to do. Well, there's not enough nagging, if you'll allow me, to tell the Wolffian ducts to develop. And so in the absence of that strong signal, there's just not enough of the signal, the Wolffian structures go away.
- 5.35** And there's no AMH or too little AMH. You'll recall that the Sertoli cells make a lot of AMH, telling the female structures, the Müllerian ducts, to die. But in this case, you don't have the signal telling the Wolffian ducts to grow, so they don't. You don't have the signal telling the Müllerian ducts to die, so they don't. So the Müllerian ducts stick around. And then you have the phenotypic sex of the female.

- 5.36 So the internal structures are female. There's the vagina, the cervix, the uterus, the fallopian tubes, and those little gonads—which started here in both sexes but got pulled down in the boys because the vas deference pulled them down—stay there. What would have been a vas deference is, if you will, its own version, the fallopian tube.
- 5.37 To wrap all of this up, the ovary—we have a very little amount of AMH being produced, but it is produced. But not enough to tell the female structures to go away, so they stick around. The gal doesn't need to be told what to do. It knows what to do unless it's told not to. There's too little testosterone to maintain any of the male structures.
- 5.38 In contrast, on the male side of this binary coin, you now have a lot of testosterone and DHT determining the development of the male sex. And there's a lot of AMH, which determines, on the other hand, to get rid of the female structures. So that sets the stage. So the default setting, as I've been explaining, is going to be female in structure and even content of the gonad, but not in function. But we'll talk about some sex disorders at the end. But keeping that in mind will help you understand those sex disorders and why the person starts manifesting with the particular phenotype or the body that they may manifest with. So the male structures need a signal. The female structures just need the lack a signal, and then they'll go on their own way.
- 5.39 As I've just reviewed that a little bit, let me state it somewhat differently and then even insert a little bit of pathology, where I'm now hinting at how some things can go wrong in this process. First and foremost, let's imagine a male. You have Sertoli cells that are producing AMH, so the female structures go away. You have Leydig cells producing a lot of testosterone, some of which is getting converted into the dihydrotestosterone, so the Wolffian ducts develop and the external male characteristics develop. That's scenario one.
- 5.40 Scenario two would be the female version of this, where you don't have Sertoli cells, you have very little AMH being produced, and there is not enough to tell the Müllerian ducts to die, and so they don't, and they thrive, and they develop. So the internal sex structures will be female. At the same time, there's very little testosterone being produced. And even what testosterone is being produced, some of it's being converted into estradiol. So there's not enough testosterone to promote the maintenance of the Wolffian ducts, thus the internal male structures erode.
- 5.41 Now, we're not done with that scenario because what if we had a chromosomal male, XY, but they have not produced—maybe it was some mutation in that Y gene, and they weren't able to create normal testes. Well, then they're going to manifest with the same pattern that the female does. This is a real phenomenon, and we'll talk about it in a moment. I'm just alluding to it because it's a ripe time.
- 5.42 Then there's even a third scenario, which is separate entirely from those, which is what if there's an XY individual, chromosomal male, who does not have any functioning

testosterone, or as you'll see, a person who can't respond to the testosterone. The receptors are broken. Well, in that place, you'd think through that, and starting with the testosterone and DHT side, even though they're making lots of testosterone and DHT, the body can't sense it. And so the default structures go—or at least the male structures don't develop. So there's no male structures.

- 5.43** However, the sensitivity or the lack thereof only applied to testosterone and the androgens. AMH is not an androgen; it's a peptide hormone. Totally separate. But this person has that SRY gene—those are testes, Sertoli cells, Leydig cells. Leydig cells are making testosterone and DHT, but the body can't sense it, so the Wolffian ducts died. But at the same time, they are making AMH from the Sertoli cells, and it's working. It stimulates the death of the female structures as well so the person ends up with no internal sex structures. We'll come to that later. All right, but those are three outcomes.
- 5.44** Remember, remember, remember—the gonads are not part of structures. At the end of the lecture, and even what I just said, when I said the person would have no internal sex structures, that is not to say they don't have gonads. Gonads are separate. Please keep that in mind, that when we talk about the development or the degradation of the Wolffian or the Müllerian ducts, and them turning into the sex structures, that does not apply to the gonads. They're going to be some form of ovaries or testes.
- 5.45** Now, you understand the origins of sex from chromosomal through phenotypic, meaning the chromosomes decide the gonads, which then decide the structure, the phenotype, the physical characteristics that go along with the genotype. Now we're in a good position understanding how the baby is born, all determined by endocrinology, which is why we're talking about this. But then what happens with puberty? Talk about powerful endocrinology.
- 5.46** I like to say that puberty is determined by a pulse, or if you will, the pulse of puberty. That is because of the unique changes that happen with gonadotropin-releasing hormone, GnRH from the humble hypothalamus. What's interesting here is that the hypothalamus, the source of GnRH, is not what's determining the rate and amount at which LH and FSH is coming out. It's not the amount of GnRH, but rather the pulsatility of it. It's not a baseline level that you'd think, "Oh, well, puberty started, and so the hypothalamus is just belching out a bunch of GnRH." No, that's not right. In fact, as you'll learn in a moment, that's actually catastrophic.
- 5.47** What happens is the frequency, the pulsatility of the GnRH is what matters most. Pre-puberty, it has very little cyclical nature to it, a very modest little pulse. Then once puberty initiates, then it becomes a very loud pulsetile signal. It comes and it stops. It comes and it stops. It's the frequency of the signal that actually determines whether the process starts. Then what determines that?

- 5.48** It's a funny little story. The initiating event that tells the hypothalamus that it's time to initiate the pulsing to initiate puberty is actually because of a protein in the hypothalamus called kisspeptin. It has a very, very funny origin. Nothing romantic, but actually it was discovered at a lab in Hershey, Pennsylvania, where the Hershey's Kiss is made, and that literally is the origin of the word. The kisspeptin is named after the sweet chocolate-producing capital of Hershey, Pennsylvania. So kisspeptin is the protein that will initiate the pulse of GnRH to change. With the changed and more volatile pulsing, we prime the pump for puberty.
- 5.49** One other comment, though, and we'll have a discussion of fat tissue later more explicitly, but the important protein that helps initiate kisspeptin to become more active is actually leptin from fat cells. Leptin is the strongest signal that activates kisspeptin, and then kisspeptin initiates that pulsatility that determines the initiation of puberty. Now, remember, it's not the level of GnRH; it's the pulsatility that matters most when it comes to puberty. That's very important to remember.
- 5.50** But leptin takes it one step further. It seems like I've now hijacked this conversation to talk about fat tissue, but you cannot understand puberty in the absence of leptin. In fact, to put a very fine point on it, in the absence of leptin, puberty is impossible. It cannot happen. Leptin is so critical at this, not only because of what it does to kisspeptin in affecting the GnRH pulsatility, but also it has direct effects at the gonadotropes of the anterior pituitary, stimulating the release of LH and FSH. It also has direct effects at the gonads, stimulating the ovaries to produce more sex hormone, stimulating the testes to produce more sex hormone. It also has an effect at the adrenal cortex, telling the adrenal cortex to even start producing more testosterone. Leptin has many effects.
- 5.51** One final one is actually an effect on the liver, too. I'd mentioned earlier that the liver is involved in—it was a question offline—that the liver is involved in the puberty as well, where the liver is producing the sex hormone-binding protein. But if it starts producing less, then more of the sex hormone is going to be free. Remember, it's the free hormone that's signaling and acting. Even if we aren't changing the amount of sex hormone being produced, we're making it more powerful by making more of it be in a free, unbound state.
- 5.52** Then one last comment—I said that I would bring in growth, but even here with the high sex hormones, as estrogens and androgens are going up, it stimulates the anterior pituitary to start making more growth hormone, which we'll come back to, but that's part of what's contributing to the explosive height change from the child. But let's just talk about growth, but still in the context of puberty and sexual changes before we talk about height and bone changes.
- 5.53** Growth hormone is responsible—and its derivative hormone, IGF-1, which we'll talk about more in a moment—they're both responsible for the growth of the child. Then the estrogens also play a role by influencing when the bones stop growing. Estrogens

will influence height as well. In males, especially, it's relevant. Also relevant in females with height. But in females, estrogens take an additional role by dictating the development of typical female tissue turning into woman, like breast tissue. Also, determining where the body is storing its fat.

- 5.54** Why does little boy and little girl start to look so different? One of the changes is where they're storing fat, where the boy with testosterone is overall storing less fat because testosterone wants to burn fat and just build muscle. In the girl, there's a little bit of that. Part of her muscle growth is because of testosterone, too. But it's more estrogens promoting different pattern of fat growth, more breast and hips.
- 5.55** There's one last comment to make about growth hormone and the balance of the sex hormones, because there is what's commonly called the push-pull, where within growth hormone and its related product—insulin-like growth factor, a growth factor, IGF-1. IGF-1 and growth hormone act as a pull because they actually reduce the sensitivity of the body to growth signals.
- 5.56** If you have more growth hormone and more IGF-1, you'd think, normally, it's going to turn itself off through negative feedback unless you can downplay or turn down the sensitivity to that signal. So the estrogens reduce the sensitivity centrally of that signal. And so the brain isn't getting the message that there's more growth signal, so you keep making more growth hormone. In complement of this, we have androgens directly stimulating more growth hormone. So it's pushing more growth hormone while the estrogens are pulling more growth hormone by reducing the brain's sensitivity to it, so there's less of a negative feedback signal.
- 5.57** One other comment—in puberty, there are the arches of puberty or what looks like arches, but it's the arches, it's from the Greek meaning origins or beginnings, where within the process of puberty, we have something called pubarche, which is the development of pubic hair, and that is an androgen- or testosterone-specific effect. We also have thelarche, which is a female-specific effect, a term referring to the budding and the growth of the breast tissue. We have menarche, once again, a female phenomenon referring to the initiation of menses or the menstrual cycle or the period. Then we have spermatarche, the male equivalent, which is the presence of sperm within the urine, in other words, the development of the sex and the ability to reproduce.
- 5.58** Underlying all of these arches is what's called Tanner stages of development. It's a bit of a complicated process, but suffice to say, there's stage one, which is the child, the pre-puberty individual, the child. Then you start going through stages two through five being in your adult body, essentially. But along that are these arches, which is: To what degree does the child have pubic hair? Is it small or sparse? Is it thicker and more coarse? That's moving through the Tanner stages. To what degree does the boy have a larger penis and scrotum? That's moving through the Tanner stages. To what degree does the young woman have breast development? Well, that's also moving through the

Tanner stages. The Tanner stages are the clinical way of documenting or identifying the progression through puberty.

- 5.59** Then, as I mentioned, one of the big differences that accounts for some of the appearance between little boy and little girl who are almost identical—if you cover the genitalia, you can't really tell the difference. But as they get bigger, you certainly can beyond any height differences. Remember, androgens promote more growth hormone. But you also have differences in where the body is storing fat. That is entirely a function of an enzyme called lipoprotein lipase. Lipoprotein lipase is the enzyme that determines where we store fat. Not necessarily how much, although it still influences that, but insulin also plays into this. But lipoprotein lipase determines where we store fat.
- 5.60** No surprise, estrogens are promoting the expression of lipoprotein lipase at the breast, hips, and buttocks in the prototypical female places. Whereas androgens are reducing LPL in fat tissue and in fact promoting it more in muscle, telling the muscle to burn more fat because it wants the boy to get thicker muscle.
- 5.61** Then lastly, what if there aren't enough androgens or no androgens, like what used to happen to these unfortunate little Italian singers with their beautiful little soprano voices? How could you make that beautiful little soprano voice remain a soprano voice? Well, you could castrate him. So there was this group, a small group, of unfortunate boys who became part of the castrati, and they were castrated. They removed their testicles to ensure they would keep their larynx small, and thus their voices would be squeaky and beautiful, I suppose.
- 5.62** But a consequence of this is you still have growth hormone, so they would get tall. In fact, in the absence of those sex hormones, they would be taller than normal because estrogens, with puberty, will actually dictate the closing of bones, the growth plates of the bone. So the bones, especially on the limbs, would get longer than normal. So they'd be taller but longer-limbed, a little gangly. But estrogens also have an effect on the structure of the rib cage. So the person would often be somewhat barrel chested, giving them a remarkable lung capacity. And I imagine they sang beautifully while suffering the consequences of having no testicles. So quite a trade-off, if you'd ask me.
- 5.63** That's it for puberty. And then as a reminder, the critical event in puberty, initiating it and maintaining it, is the pulsatility of the GnRH; not the level. It's not the GnRH went from low to high, it's that it now is pulsing in a particular pattern. It was kisspeptin and leptin that was contributing to this. What if you go into someone's body that is on the cusp or has started the pulsatility and you just ram GnRH into the system and keep it high. That is, in fact, the mechanism of preventing puberty with so-called puberty blockers.
- 5.64** These are hormones or chemicals that were traditionally used for what's called chemical castration, where if you wanted to stop someone's gonads from working, you

would give them these chemicals, perhaps as a punishment in a prison for any particular crime. But now it's being used to block puberty. I know this is an extraordinarily sensitive topic, but I hope I can discuss it not only with compassion, but also with precision, because as a scientist, that's what I want. I want to seek and understand truth in this physical realm. Let's just briefly describe the mechanism of puberty blockers and then mention some consequences that are very often overlooked.

- 5.65** So again, we have the normal pre-puberty signal of GnRH, which is a very slow pulsatility, very mild. And then when puberty is starting, or rather, it starts because of a change in the pulsatility. So if you flatline that pulsatility by giving the individual GnRH as a direct drug, you stop the production of LH and FSH. So you take the pulsatility and you flatline it, albeit at a higher level. But it's not the level that matters, it's the pulsatility that matters. So when you lock in the GnRH at a consistent level, you stop the production of LH and FSH from the anterior pituitary. Then there's no stimulus to the gonads. In the absence of that stimulus, you're not developing sperm and eggs, and you're not making sex hormones. So you will stop puberty from happening.
- 5.66** Now, there's a consequence to this, which is as much as we often look at puberty as singularly relevant to the development of genitalia, that's not fair. That puberty and that dramatic change in sex hormones also has effects on the development of the brain and the development of the bone, in addition to many other effects. It's just within recent history, we have substantial evidence that documents those two changes.
- 5.67** In particular, with the brain, it has been documented that when you put a teenager onto a puberty blocker, it can result in an irreversible lack of white matter of the brain. It is dependent on sex hormones. In the absence of those sex hormones, the brain literally does not finish developing, and it is something that appears to be irreversible; you can't make up for it. So if you don't allow puberty to happen, when it's supposed to happen, you may be depriving the brain of critical development.
- 5.68** Then at the same time, there's very convincing evidence showing that you also fail to prevent the bone, the skeletal system from developing, where individuals who are put on puberty blockers as adolescents, you stop their bones from developing and they have weaker, more frail bones. Those are consequences to puberty blockers. They are not often discussed, but I think ought to be considered, that you're doing more than blunting the development of the genitalia, which itself is a very, very impactful phenomenon, of course.
- 5.69** Now, throughout the discussion of puberty, I've mentioned growth, where when estrogens and androgens start to go up, one of its effects will be to trigger the somatotropes of the anterior pituitary to make more growth hormone. So let's just talk about growth hormone. Growth hormone, as I mentioned at the very beginning with the hypothalamic-pituitary axis lecture, is its own product. Growth hormone is a hormone from the anterior pituitary that can go around the body and do everything it wants, unlike TSH, for example.

- 5.70** TSH is coming from the anterior pituitary, and it's only as good as its ability to come to the thyroid and tell the thyroid to produce thyroid hormone. Growth hormone is not like that. Growth hormone has its own systemic effects influencing many, many tissues of the body. As I noted, the sex hormones are a key trigger in that event.
- 5.71** Also, sleep is an important trigger. In fact, the sleep cycle is massively important. When you look at a younger healthy person, you can see that growth hormones will go up by 5 to 10 times during the night time compared to during the day. So night time is when the child is growing and growing a lot. So it's important to let them get to bed and maybe even sometimes give them a little protein and healthy fats before they get to bed.
- 5.72** But when you compare that effect in an older person, where if you look at a person in their 60s and 70s and compare their growth hormone pattern to a kid in his or her early to mid 20s, it will get to a fraction of the same level. Here they both are at their low waking state. The older adult maybe will get to about a few times over resting, and then the young person will be multiple times, four to five times higher than in the older person. So you start to lose that sleep cycle of growth hormone as we get older.
- 5.73** And then even during exercise, when you start to exercise, that's a trigger for growth hormone, in effect, in an interesting way that connects to the bone that I'll show you in a minute. But when you exercise, growth hormone goes up. Well, that's a nice adaptation for ways that we'll talk about, particularly with fuel mobilization, helping the body with its metabolic demands. But once again, it sucks to get old because when you compare the growth hormone curve to the same intensity of exercise in an older person, it is a fraction of what you see in the young person. Maybe you would say, "Well, Ben, that's no surprise. The older person is done growing." Well, so too is the 27-year-old here. But you can see, even in the human 27-year-old, growth hormone levels during exercise go up a lot more.
- 5.74** Well, that does give some physical advantages, but not in ways most people think. In fact, when we think about growth hormone, the effects that we think of with growth hormone are usually actually more a consequence of a hormone called insulin-like growth factor I or IGF-1. That's important. When growth hormone comes from the anterior pituitary, one of its effects will be to come to the liver, and it will tell the liver to make IGF-1. So IGF-1 will be a further downstream product. And then IGF-1 will also provide a negative feedback signal to the brain.
- 5.75** But let's talk about growth for just a moment and focus in on the bone. When you look at the bone, you have both osteoblasts and osteocytes—or rather, chondrocytes. We have osteoblasts, which are cells that are involved in the girth of the bone. Then you have the chondrocytes, which are involved in the length of the bone, making someone taller, whereas the osteoblasts are making the bone a little thicker and stronger. Well, growth hormone has a stimulating effect of both of those, helping the bones be a little thicker, stronger, and longer. But it pales in comparison to the effect of IGF-1.

- 5.76** If you want to look at what's making the bones grow and the person get taller, it's not growth hormone, it's IGF-1. IGF-1 has a substantially stronger effect. So again, even in the adolescent going through puberty, much of their growth isn't the growth hormone; it's the product of the growth hormone, namely the IGF-1. But then that begs the question what the growth hormone is doing.
- 5.77** But before I mention that, let me just say that within exercise where you have the muscle attached to its bones, when you exercise the muscle, the muscle actually produces IGF-1 that can feed to the bone to help the bone be stronger. Even if the bone plate has closed, so you're not getting longer, you still want that IGF-1 to help with the osteoblasts to help the bone be strong. So that is one direct endocrine mechanism that explains how exercise can help bone health.
- 5.78** Often we will say that the benefit of exercise is purely mechanical—you're just putting pressure on the bone and it's responding to get stronger. Well, that might be part of it, but we also know there's an endocrine aspect to it as well. The exercising muscle releases IGF-1, which facilitates bone mass.
- 5.79** Now, the question I mentioned earlier, let's come back to it, which is, well, if growth hormone isn't primarily driving growth, if it's more IGF-1, which it is, then what is growth hormone doing? Growth hormone is affecting fuel mobilization, where essentially you could say IGF-1 is the main signal promoting growth of the body—muscles, bones, and tissues in general—but that growth requires fuel. That's actually where growth hormone comes in.
- 5.80** Growth hormone will stimulate lipolysis at the fat cell, breaking down the fat, putting that energy in the blood. Growth hormone will stimulate glycogenolysis, telling the liver to break down its glycogen. Growth hormone will also inhibit the breakdown of muscle, keeping those amino acids in the protein. Then bringing it back to puberty, lest you think I've forgotten the topic, it also reduces sex hormone-binding protein, helping there be higher levels of the free testosterone and the free estrogens.
- 5.81** Then what about the bodybuilder, those guys who are taking growth hormone? What's interesting is that there's a human study from 2003 that actually had weightlifters go on an injected protocol of growth hormone, and there was no difference in muscle size or strength. So it appeared that growth hormone does not do the muscle growth that many people may think.
- 5.82** But that's not to say there isn't a role for growth hormone in human health, and that might be part of a pharmacotherapy that people maybe would consider as they get older. If they just are frustrated by the consequences of aging, there are some who believe, and I think it's a double-edged sword, where maybe some growth hormone might help soften some the consequences of aging.
- 5.83** We have gone through a lot already within this lecture, but we're not done because I wanted to end, like I alluded to, with these disorders. These are very, very delicate

topics. I, in all seriousness, hope I can approach this topic with nothing but respect because these are—when you start messing with sex and genitalia, or rather, when there has been a disorder of sex development, I cannot imagine how hard that would be for the individual. So more than any other topic, when we start talking about disorders here, I approach this one with some trepidation and tremendous respect.

- 5.84** The first one—the first ones we'll talk about are a couple instances of sex chromosome aneuploidy. I actually discussed this in a genetics lecture in the Basics in Biology. It's just appropriate to fit back in now. What if an individual is born with only one X chromosome because of some problem in the development of the gamete from mom or dad where the egg or the sperm lacked the X chromosome that it was supposed to? Then they come together and it only got the one X from the other good gamete, egg or sperm. The individual would grow with only one X chromosome.
- 5.85** You'll recall that this is a condition called Turner syndrome. Remember, the internal structures and even the external genitalia—the vagina and the labia that we associate with female genitalia— they don't need a hormone signal; they don't need any signal to develop. So the internal structures in the external genitalia of the woman born with Turner syndrome will be female.
- 5.86** But you'll recall that for ovaries to function, you need two X chromosomes. Well, there's only one, so these ovaries do not function. They do not produce enough sex hormones to go through puberty, and they're sterile. They will not have oocytes, or eggs, to develop. They will not go through puberty unless you intervene and give them hormones to help them go through puberty, which you can, to have her get a little taller. Although the girl will often be quite short, you can get her a little taller, have her develop some breast tissue, have the more female phenotype, but she will be sterile. Those ovaries will not be functional. They will not release eggs and have normal ovulation.
- 5.87** Now, we have another version of this where the aneuploidy is that the individual, an XY individual, has one extra X. This would have been a result, once again, from the parents gamete having a non-disjunction. In the process of producing the gamete, the X or the Y or the two Xs were supposed to split, and then they didn't; they stayed together. That's what came together to create the child here, the fetus.
- 5.88** In this instance, the presence of that extra X— but that Y carries a punch. Like I said before, that singular Y chromosome is sufficient, and those Sertoli cells making AMH, is sufficient so that all of the phenotype, the structure, internal and external, are male— genitalia and male sex structures inside. But that extra X does muddy the water a little bit, where the person will have a smaller than normal penis. They may have some breast fat storage at the chest, making it look like some pseudo small breasts. Also, the change in estrogens result in bones that got a little gangly and long. In fact, they're a little bit in their phenotype like the castrati would have been. Even though they have

their testes and they can even produce sperm and be fertile, they have the phenotype that is starting to blur the line a little bit because of that one extra X.

- 5.89** Now let's leave the chromosome disorders behind and come right back to genetics, which we have mostly been talking about when we talked about the determination of sex. The first one to mention is the SRY gene not being where it's supposed to be, which is, in the process of developing the new genes and the chromosomes, the Y chromosome from the SRY gene from a Y chromosome was translocated onto an X. This is a person with XX chromosomes, so genetic female, but that SRY gene was somehow translocated onto an X chromosome in all of the chaos of biology, and it packs a punch.
- 5.90** So that SRY gene creates TDF, and it told those gonads—which should have been ovaries because of the two Xs—it says, "No, here's TDF. I need Sertoli cells, and I need Leydig cells." And they work. They have functioning Sertoli cells and AMH. They have functioning Leydig cells and testosterone and DHT. Now, they do not make sperm at all. You cannot. There are other genes on that Y chromosome that are absolutely essential for normal sperm production. So there's no sperm that is produced. So they are totally sterile, not infertile. That's not the right word. They're sterile. But they look male. They have all the hardware of males. But I got a little ahead of myself.
- 5.91** So internal—what would be the structures? They have Sertoli cells producing AMH, which would have what effect? What would happen to the Müllerian ducts? Gone. So no female structures. Now, what do the male structures need to grow? Testosterone. And there is testosterone because they have Leydig cells and so the internal structures are male. So all of them—the prostate, Cowper gland, vas deferens—all of those are there. Then external—same thing. When you've promoted the growth of the male structures, you end up also having the testosterone pulling the gonads down, making DHT from that testosterone, and thus having external male genitalia.
- 5.92** However, the person does have a much higher risk than normal at being born with something called hypospadias. Part of the process and the evolution of the shift in scrotum from what would have been just labia pulling down, and then the expansion of the penis, is that the urethral meatus, the actual exit of the urine, should be, of course, at the tip of the penis. But in the process of this development, it didn't make it there. It could be, as you see, anywhere along the shaft of the penis, including up, even separating the very scrotum itself. So it's a very mismatched urinary system here with the hardware of the genitalia.
- 5.93** Now, there can be another form where you have an XY, so a chromosomal male, but the Y chromosome has experienced an aberration called a chromosomal deletion, where unfortunately, the SRY gene gets deleted. In the absence of the SRY gene, there is no testis-determining factor produced. And so those gonads cannot become testes.
- 5.94** You'd say, "All right, well, then they default to female." But remember, for ovaries to be functional, what do you need? You need two X chromosomes. So whether it was this

instance or even the earlier one I mentioned, the Turner syndrome, there end up—they're called gonadal streaks. Even though they have the substance of the ovary, like a granulosa cell and a theca cell, they're loaded with fibrotic or fibroblasts as well. So they're very fibrous, dense, abnormally structured gonads.

- 5.95** So the person will be sterile, and they will also not go through puberty because the gonads aren't working well enough to produce sex hormones. So there's no puberty, and often you won't notice that there's anything wrong until puberty because they won't go through it. They'll be growing. She would be growing—let's talk about that. Let's talk about internal and external.
- 5.96** There's no SRY gene—that means no testis-determining factor. There's no Sertoli cells. In the absence of the Sertoli cells, there's no AMH. So what happens to the Müllerian ducts? So internal is female. She has the vagina going to cervix, uterus, fallopian tubes, although the fallopian tubes are collected—or rather loosely connected to the fibrous masses, not really ovaries in any functional way.
- 5.97** But then what about external? External requires a lot of testosterone and DHT. And remember, this individual is not making a lot of sex hormones. So what's external? Female, yeah. So the baby would have been born with just normal female genitalia, and you would think this is just a healthy little female until she would have gone through puberty. Her girlfriends are, but she's not.
- 5.98** Now, you can intervene. She has all the hardware, if you will—I don't mean for that to be a crass term—all the structures are there of a normal healthy female. The ovaries aren't, but naturally, she may say, "Well, I want to go through puberty. I'm a girl. I feel like a girl." And so you give her the sex hormones to allow her to go through the development of a woman and develop breast tissue and have a female figure as opposed to just a tall girl figure.
- 5.99** There's another disorder that can happen, and that is the absence of 17-alpha-hydroxylase. This is the enzyme that is critical for the development of all sex hormones. This is a person who would not be making any sex hormones. Then that begs the question: How would the person appear? Well, it depends on the sex, of course. But in the absence of any hormones, at a minimum, they're never going to go through puberty. They're not making any sex hormones. Then the nature of the structure, whether external or internal, actually depends on the sex of the individual. It depends on the genotype.
- 5.100** If the person is XX—let's reason through this. They cannot make any of the steroid sex hormones. XX. They're also already making very little AMH. Remember, AMH is not a sex hormone. You'd say, "Well, it is. It's involved with sex." Yeah, but it's not one of the steroid sex hormones. So AMH is not a part of this. They can't make androgens and estrogens in the absence of 17 α -hydroxylase.

- 5.101** So in an XX individual, what would the internal structures be here? Internal structures? Female, because she's not making enough AMH to get rid of the female. At the same time, she's not making enough or any testosterone to promote the growth of the male. Remember, male needs the signal. Female needs a signal to tell it not to happen.
- 5.102** Then external? What about the external genitalia? There's no DHT. Female, yes. She has all all of the substance, all of the structures for a female. She'll just never go through puberty, but you can make it. In fact, she has healthy ovaries, even with eggs in them. She is female, so you can not only induce puberty, you can even induce ovulation and normal menstrual cycles. You just have to be very thorough with it.
- 5.103** In contrast, what about in the XY individual? So this is a person who's not making any sex hormones. But what is not a sex hormone? AMH is not a sex hormone. It doesn't count. So they are making AMH, but they're not making testosterone. So what are the internal structures going to be? None. Let's come back to that in just a second. Because what about external?
- 5.104** With external, you need a lot of DHT to promote the growth of the scrotum and the penis. We don't have it. It's gone. We don't have any estrogens either. But remember, you don't need a signal to promote female development. It needs to be told not to. We already talked about the internal. External doesn't have a not to signal. And so the baby is born with the external characteristics of the female, which is to say, what does it look like if there's nothing else?
- 5.105** There's the external labia. So you have the normal curvature of the little bum coming around in the little baby girl. That would be the external labia. And there is a vagina. The vagina is actually considered more accurately an external structure. There's just nothing beyond it. It's what's called a blind pouch. It ends. It curves in and goes in, and then it stops. And so then there's no other hardware there—I don't mean for that word to be crass or casual here; there's no other internal structures. There's no cervix to the uterus, to the fallopian tubes. It's not there. Then puberty, of course, you could initiate it with hormones.
- 5.106** Nearing the very end, what about a lack of androgen receptors? What happens if you have an individual who's making testosterone, but you can't sense it? It's the structure of—you'll note here that this is particularly relevant in an XY individual. I'll compare that to an XX individual in just a moment. In an XY individual, they have Sertoli cells making AMH. Because of that—actually, let's reason through that one.
- 5.107** Sertoli cells making AMH, so what do we know they don't have? They don't have any female structures. But you'd say, "Well, what about the male?" They are, after all, making a lot of testosterone and a lot of DHT, in fact, they're making more than normal. Why do I even say they're making more than normal? There's no negative feedback. It can't sense it because there's no receptors.

- 5.108** So the body is actually flush with androgens and even estrogens because the more androgens you have, the more aromatase is converting it to estrogens. So this person actually has really high levels of both. The estrogens do signal, but the androgens don't. To bring it all back to the internal structures, if the androgens can't signal, the male structures are gone. We knew that there's AMH present, so the female structures are gone. So once again, we have a blind pouch, no internal sex structures.
- 5.109** Then externally, what's interesting about this phenotype is that while the person has higher levels of both androgens and estrogens, they can't sense the androgens, but they can sense the estrogens. And so she will—when this XY individual has puberty and the sex hormones go really high—the person will get taller. She will start to store her fat in her breasts and her hips.
- 5.110** In fact, the irony of androgen insensitivity syndrome is that this individual will have supermodel status. I don't mean to be silly about this, but they'll often be taller in a much more dramatic female form, like larger breasts. It's a quirky feature of this because there's so much estrogen during development, she ends up being like a superwoman, if you will.
- 5.111** At the same time, the lack of androgen signaling will mean that she has very little body hair, too, because remember, pubarche and body hair is an androgen phenomenon. She'll be taller than normal, have a very female form, and have less body hair. But of course, those are not ovaries. Even though they're up there, they're not bound to any fallopian tubes. Those are testes in structure—sertoli cells, Leydig cells—but, of course, not making any sperm.
- 5.112** Now, the last one we talk about is the most uncommon of all of them. But before I mention that, I just need to say, sometimes students will hear about these sex disorders and believe that explains how they may feel in their own body. I just need to reiterate the important point here that these are fantastically uncommon situations. Very importantly, they're not getting more common. So as we see societal trends indicating differences in sex—or as they would say, gender, which is not a term a biologist uses appropriately—this is not an explanation for any societal trends that we see. These are very uncommon and are not getting any more common.
- 5.113** Then the least common of all of them—I just have to show this because it begs the question: What about the ovotestis? This is what was once called true hermaphroditism. That's not a delicate term, so it's not used, but you'll know the term. This is an instance where there has—usually, this scenario is a marbling of the gonad, where there has been this quirk in development that within one single gonad, you have this marbling of both testicular tissue, meaning Leydig cells and Sertoli cells, and ovarian tissue, meaning you have granulosa and theca cells.
- 5.114** What's important, one other part of this is that the person can then manifest with a marbled phenotype where they may have both structures. They may have testes that

pull down somewhat, but they're locked in the labia, perhaps, or the clitoris did grow, and so there's a micropenis there. Suffice it to say, it's a very delicate, very confusing situation.

5.115 But importantly, I always have students that are curious, "Well, could they get themselves pregnant? Could you take an egg and take a sperm?" And there's no recorded instance of both gametes being produced in an individual. That's important to know that even though they may have a marbling of testis with ovary, that would just be a marbling of the cells, not the gametes or the actual germ line or the reproductive cells. Only one is going to be functioning, and overwhelmingly, it's female. Often, this is called an XX disorder. You can have XY, but it's even less common in what is already a fantastically uncommon situation.

5.116 All right. In this lecture, we have covered a lot. We went through the differentiation from chromosomal sex to gonadal sex to phenotypic sex. We then expanded that into puberty and the growth that happens in puberty and the development of these secondary sex characteristics— those are the changes that happen during puberty. Then lastly, we finished with these very delicate, fascinating, and even heartbreaking disorders of sexual development.

5.117 Thanks.

Lecture 6: Hormones of Life

At a glance — Hormones of life — how the gonads promote health and reproduction.

- 6.1** Now that we have discussed the nature of the gonads and the gonads working well and doing their job to promote puberty, one of the points of puberty is for the child to become an adult and to become fertile and reproduce as a species. So today or now, we talk about the endocrinology of reproduction. And then when it comes to the female side of this—pregnancy, with a little bit of lactation thrown in there as well at the end, just round it out. These are, of course, entirely endocrine events. Hormones, as you'll see, determine the success of all of these things. So let's just get in.
- 6.2** The order of the lecture is going to be just talking about male reproduction first. It's very short and very effective. After all, men are involved in the process, albeit in a very glorious singular moment. Then women have the metabolic marathon of pregnancy, and so there's a lot more to discuss when it comes to fertility, including menstrual cycle, including childbirth, including lactation. The metabolic demands come on the woman. It's quite a responsibility. Then we'll talk about some reproductive disorders at the end.
- 6.3** Firstly, with male reproduction, let me just remind you of the axis where the hypothalamus is producing GnRH, and then the anterior pituitary is producing LH and FSH. FSH is having a specific effect at the Sertoli cells to promote spermatogenesis. At the same time, LH is having an effect on the Leydig cells, telling them to produce lots of testosterone. Even with spermatogenesis, testosterone is very important. Testosterone is a necessary signal helping promote sperm production. Of course, we've already discussed liberally the effect of testosterone on puberty, allowing all of the physical structures and changes to happen to even enable reproduction.
- 6.4** The process of producing sperm is a multi-step process, although it doesn't require the dynamic changes in hormones that we will explore in the female. It's just happening. If the signals are there, like FSH and testosterone, then you are developing from what's called the spermatogonia, the primordial little sperm, and it's maturing into the full-on sperm cell like we think of.
- 6.5** Now, there's one last aspect to this when it comes to male fertility. It's one thing to make the sperm, and then it's another when it comes to reproduction to actually get it where it needs to get. This involves a little mnemonic which is point and shoot. The point refers to the erection, and the shoot refers to the ejaculation, so the actual emission of the sperm and the contents of the ejaculate, because there's a little more than just sperm there.
- 6.6** Well, the P and the S of point and shoot is also a bit of a mnemonic because the P refers to the parasympathetic nervous system, which is the rest and digest part. We could add reproductive part. But then the S stands for sympathetic. So for normal

erectile function, the man must have functioning or have enabled the parasympathetic nervous system. That elicits the change in blood flow. When the parasympathetic nervous system predominates, it produces nitric oxide in the blood vessels, which allows normal erectile function with the big change in blood flow.

6.7 That's not the same as ejaculation. That actually requires the sympathetic nervous system to induce the contraction of things like the vas deferens and the seminal vesicles and the prostate, thereby ejecting the ejaculate and the sperm that goes with it. But both of those, even then, when you talk about parasympathetic and sympathetic nervous systems, you're talking about hormones—things like epinephrine and norepinephrine and different receptors. Remember those alpha and beta G protein-coupled receptors? They're predominating and explaining all of this phenomenon. Again, the point and the shoot—the simple, yet profoundly relevant part of males when it comes to reproduction.

6.8 But the male system is simple. It is, if you will, like a barbershop quartet. It sounds good, it's important, but gosh, it's very simple to put together. That would be like comparing now in the female, a 200-piece symphony. There are a lot of components to it, and they all need to be playing their part to work well. And again, the reasoning for this could be nothing more than it's the women, it's the female body that has to now carry this and grow the baby, carry it, and then feed the baby on the back end, historically. So it's not surprising that it's a little more complicated.

6.9 The complication is, in fact, at the level of an anatomical difference. Whereas the production of sperm is quite simple—the sperm are producing and moving through a lumen and then out. But when it comes to eggs, the eggs are surrounded by follicles. You'll recall when I talked about a follicle in the context of thyroid, I compared it to vesicles because a vesicle will just be a little bubble formed with fats and proteins. Those are not cells. A follicle is a bigger bubble formed of cells.

6.10 So you have cells that are called granulosa cells, forming the edge of this follicle. Granulosa cells are surrounding the oocyte, or the ova, the egg. Then outside of the granulosa cells are the theca cells. Now, throughout all of this, we have FSH, coming from the anterior pituitary, going to the granulosa cells, helping stimulate the growth of the ova. Then we have LH, going to the theca cells, stimulating the production of the sex hormones. Remember, it starts with testosterone, then the theca cell hands off the testosterone to the granulosa cell to turn it into estradiol, which is helping the development of the egg.

6.11 That's just general overview. We know the males now, we understand the dynamics of the ovaries a little bit, although it's about to get a lot more complicated. But now let's put this in the context of the female cycle.

6.12 There is a cycle where the female, for normal fertility, it involves, as we're going to learn, a lot of stimulus, getting the uterus ready for implantation. But that is a very

expensive process, if you will, that is dynamic. It'll grow and then it sheds, and it'll grow and then it sheds. There is a cyclical nature to this. As you'll see, the actual timing of it all is, in fact, narrow to actually enable pregnancy. But before we get into the nitty-gritty, the subcycles of that larger cycle, let's just walk through very lightly the anatomy of the female. This is not a class on anatomy. It's not even a class on physiology. It's endocrinology, but it's still helpful to just revisit these bases.

- 6.13** Firstly, we have the ovaries, which are the gonads that are tucked up in the lower part of the abdomen. They are filled with follicles. The follicle is going to be what can turn into eggs. Within every follicle, we have a mix of ova, eggs. We have granulosa cells, which are surrounding the follicles as they start to develop, when an egg starts to develop. Then we have the theca cells, which are even one side out of them, right on the outside of the follicle. Those are the three main cells that are present within the ovaries: the eggs, the granulosa cells, and then the theca cells.
- 6.14** What we need to have happen is throughout the cycles, as you'll see, one of these eggs —some eggs will start to develop into follicles, and one becomes the dominant, and we'll get to that in a moment. Beyond the ovaries is the ability of the ovaries to share the egg or put the egg where it needs to get.
- 6.15** Now, the ovaries, importantly, have blood flow. There are vessels coming in, there are vessels coming out. That allows the ovary tissue to be getting nourishment and getting hormone signals and be giving its own hormone signals out. But then, laying on top of the ovaries, if you will, are these little finger-like fimbriae coming from the fallopian tubes.
- 6.16** The fallopian tube will, almost like a little vacuum, it's going to want to pull in whichever follicle actually ovulates, which we've not talked about yet. But there will be one ovum, one egg, coming from one of the ovaries, and it's going to go into the fallopian tubes. So the fallopian tubes are just a few centimeters, about 10 centimeters or so, coming, then connecting in to the uterus. We'll get to this, but that's where fertilization must happen. Fertilization must happen in the fallopian tube. In other words, the sperm needs to meet the egg up closer to the ovary than most people think. So that's where fertilization, the first two coming in touch, has to happen in the fallopian tube.
- 6.17** But then we come to the uterus as we're coming out. The uterus is a thick, muscled organ. It has three distinct layers. The middlemost layer is the myometrium. Anytime you hear the word myo, think muscle. Yep, it's muscly because it may someday do a lot of work, which is pushing a baby out. So the middle layer is myometrium.
- 6.18** Inner to that is the endometrium, and it's the endometrium that experiences such remarkable growing and shedding, growing and shedding. So when it sloughs off, that is what we think of when we think of menstrual flow or the period. What's coming out is

all of the shedding and a little bit of bleeding as the blood vessels are exposed and then close, that's what's coming out with the menstrual flow or the period.

- 6.19** Then to wrap it all up, we have the cervix, so the end or the neck of the uterus, and then the vagina itself. Collectively, that's going to be the birth canal, which will become very important when we get to actual childbirth in just a bit. Before that, though, we need to discuss the menstrual cycle.
- 6.20** This is the typically, canonically, it's a 28-day—of course, in the chaos of biology, it's very common for gals to be a few days in either way of that— but a 28-day cycle, as it's typically presented, where you have within this broader menstrual cycle, two subcycles. One cycle that is based on the functions of the ovary, and then overlaying that is the cycle of the uterus, so what we could call the uterine cycle, namely the changes in the endometrium that I just mentioned a moment ago.
- 6.21** They are working in concert with each other because the same endocrine signals are resulting in this timed change. They need to be synchronized so that the ovarian cycle is happening at the right time, matched with the uterine cycle. Again, thankfully, it's hormones now acting as a bit of a central conductor, making sure that everyone is playing together as they ought to. It is these sex hormones like estrogens and progesterone that dominate.
- 6.22** Let's start with this dominant period with estrogen, and let's start by talking about the ovarian cycle. Just to clarify, this is going to be the changes happening in the ovary itself. Nothing to do with the uterus, just the ovary. It is separated into two primary phases. The follicular phase, that's the first phase, days 1 through 14-ish. Then, beyond that, the latter half of it, the next two weeks-ish, is going to be the luteal phase.
- 6.23** Each of those terms reflects what's actually happening, or rather, which part of the ovary is the dominant part. The first part is the follicle, as we'll discuss. Then the second part is the corpus luteum. It's the luteal phase, the follicle phase, then ovulation, and then the luteal phase. As we'll discuss, the corpus luteum is the remnant. You have the follicle, it's getting bigger and bigger and bigger. The egg is growing, it is ready, and it ovulates. And we'll talk about how that happens in a minute.
- 6.24** But then after that, that follicle is not gone. It actually continues to play an extraordinarily important role. And that remains for a period of time, as we'll discuss, as the corpus luteum. So the follicle phase, then ovulation, then the remnant of the follicle, namely the corpus luteum, which helps identify or determine the luteal phase. Again, it's all ovary that we're talking about.
- 6.25** So that first follicular phase, days 1 through the middle of the cycle-ish, is dominated by estradiol. Estradiol levels start at baseline, and they start getting higher and higher and higher and higher. Then at the end of that, we have an LH surge, which we'll come to. The LH surge is the critical event that actually tells the follicle, "Okay, you're good to go."

It's time to go out." And then actual ovulation occurs, the pivotal event of the entire menstrual cycle. It's all to get to ovulation.

- 6.26** Within that follicular phase, as I noted, it's dominated by estradiol. But where does the estradiol come from? The fact that the estradiol is getting higher and higher and higher is actually a product of that growing follicle. As the follicle is getting bigger and bigger and bigger, think about the cells that are determining the follicle. It's the granulosa cells. The granulosa cells are the cells making estradiol.
- 6.27** Now, of course, it has to get the building blocks for that from the theca cells. But as the follicle is getting bigger and bigger and bigger, it starts producing more and more and more estradiol. Then it gets to a point where what was once a negative feedback signal, which I'll talk about in more detail in just a second, once this estrogen—so normally, it's negative feedback. Estradiol is going up, it inhibits LH. But we need LH to go really high. In fact, we need it to surge. We need an LH surge because that's going to be the pivotal event that enables ovulation.
- 6.28** What is it that changes? Once again, it's this frequency signal that appears to be so important when it comes to sex and reproduction events. Here we have the follicle getting bigger and bigger and bigger. Estrogens—estradiol is coming up, it's coming up, it's coming up. And once it stays elevated for about 36 to 48 hours, it flips a switch in GnRH and then LH, where what was once negative feedback, now elicits a reverse effect. Now we start getting a lot of LH receptors and we start producing a lot more LH. And so the pulsatility of GnRH changes as the estradiol levels come up and reach a threshold and stay elevated for, again, about a day and a half to two days. That is the stimulus that results in this really big LH surge.
- 6.29** The LH surge is so important because when it goes up, it actually stimulates these cytokines. You wouldn't imagine they'd be relevant here. But once again, it's just a testament to how things get named something before we know all of which it does, like prostaglandins. When you hear the word prostaglandins, you may be thinking of an allergic reaction to something, when in reality it also has a role in ovulation.
- 6.30** So when LH starts to really surge, it induces this big load of prostaglandins. And then prostaglandins start to erode the follicular wall. So the follicle is getting bigger and bigger and bigger. LH spikes, and it produces these prostaglandins, and now we erode the follicular wall. As the follicle has grown, it now has broken apart and releases the egg into the fallopian tube.
- 6.31** It was the LH surge, and then the subsequent big production in prostaglandins, these inflammatory proteins. But that's just another word for saying it's a hormone. It's just a peptide hormone, to say it another way. It enables the actual event of ovulation, breaking through the follicle so the egg will move into the fallopian tube because we've had the prostaglandins break down the barrier between the two.

- 6.32** The egg is on its journey through the fallopian tube now. So it's out of the ovary, and we're only talking about the ovary now, so we're not following the egg's journey. Within the ovary, now we've ovulated, we have graduated into the luteal phase. This is, as I noted, defined by the presence of the corpus luteum. The corpus luteum is going to take over hormone production for a little time. It's going to be producing more estradiol. The corpus luteum, the small little follicle structure, starts really ramping up estradiol production.
- 6.33** At the same time, for essentially the first time in the cycle, now we have really high progesterone. That progesterone is coming from the corpus luteum, this remnant structure. It's amazing because progesterone is, of course, the hormone of gestation. It wants the body to get ready for a fertilized egg to get implanted. So it starts to change everything. When we get to the uterus, all of the changes in the uterus are happening at this point because of this progesterone, which is coming from the remnant of the follicle, namely the corpus luteum.
- 6.34** But very importantly, there is some critical feedback here where, when progesterone goes up, it is a very strong inhibitory signal to the brain, to the hypothalamus and the anterior pituitary saying, "Stop making LH and FSH because I don't want you to start telling other follicles to start developing right now." It's a way to prevent follicles from starting to develop off cycle—the corpus luteum making progesterone, progesterone telling the axis to quiet down, saying, "Hey, give me a couple of weeks. Let's just see what's going to happen here." Then if pregnancy doesn't happen, then we'll go back to where we were before. With the luteal phase, again, we have the corpus luteum making estrogens, estradiol in particular, and making a lot of progesterone, and progesterone keeping the pathway from starting over again.
- 6.35** Now, if pregnancy does not happen, if there's no implantation of the fertilized egg—which alone starts to initiate other events, which we'll come to—then everything resets. Then the corpus luteum, which has about a 7-10 day life cycle in this case, starts to erode. Then as the corpus luteum erodes, we lose the estradiol and the progesterone signal. As the estradiol and progesterone come down, the cycle resets. That's the end of the ovarian cycle.
- 6.36** So we went through follicle phase with the growing follicle, the higher and higher levels of estradiol, triggering a big LH. LH then triggering ovulation. Then the remnant of the corpus luteum defining the luteal phase and its higher estradiol and even higher progesterone. If implantation has not happened—and we'll talk about how it does in a moment—then those hormones go to normal as the corpus luteum has lived its life and gone away. Then the cycle starts again, the ovarian cycle. That helps us understand the changes in the ovaries during this time.
- 6.37** Now, let me just make one comment as an interesting aside about progesterone. The luteal phase of the ovarian cycle is interesting in the female body because

progesterone has other effects that go beyond the uterus, like promoting hunger. Progesterone actually activates hunger centers within the brain.

- 6.38** This could be why a gal may find it harder to fast. There are people who have very well-articulated ideas on fasting in females that are premenopausal, where fasting during the luteal phase may be harder than normal because of the increased progesterone. Progesterone has a central effect at promoting hunger, which also explains why mommy, during pregnancy with ever-higher levels of progesterone, as we'll get to, may find her appetite getting higher and higher, and maybe even a little weird.
- 6.39** All right, let's leave the ovary behind and come into the uterus, because as I said, the other subcycle of the menstrual cycle is the uterine cycle. That is defined by the changes in the endometrium. The myometrium and the other layers don't change. It's just that innermost layer that needs to get big and ripe for implantation or not. Then it sheds. That is the overall uterine cycle part of the menstrual cycle.
- 6.40** So this phase starts actually how it ends, which is the shedding. It's a bit of awkward timing here because I meant what I said, where the beginning to match it up with the ovarian cycle, it's not so tidy. But we're going to do it. We're going to let the ovarian cycle be the one that determines it because the ovarian cycle does determine the uterine cycle. So it's not wrong to show it this way.
- 6.41** But how it starts is actually the menstrual phase or menses or the shedding. But that's also how it ended, where the last day or two is actually the beginning of the menstrual phase, and that continues to the beginning. If we start the cycle synced up with the ovarian phase, the ovarian phase has started while the menstrual phase is still going. So we're shedding off the endometrium. And that's a result of the estradiol and the progesterone coming down. So as the corpus luteum eroded, its hormone products went away, and now what was sustaining the endometrium is gone. So the endometrium sheds. So that's the menstrual flow that we would typically think of here.
- 6.42** Once again, this actually has a part with prostaglandins that result in blood vessels constricting. So you basically deprive the endometrium of blood flow. It's been getting thicker and thicker with more and more blood. You come in and close those blood vessels off, and so the outermost part of the endometrium dies and it sheds out. There's a perfectly designed exit to come out through the vagina. This is just a few days, again, starting at the end of the ovarian cycle and in the first few days of the beginning.
- 6.43** Then we start to move back in to the normal cycle where the first big part of the uterine cycle is the proliferation phase. The proliferation phase is determined because of the growing levels of estradiol again. Remember, the estradiol is coming higher and higher from the follicle that we talked about earlier. Now the follicle is releasing estradiol, and now the innermost lining of the uterus, the endometrium, is starting to

grow. That's why it's called the proliferation phase. It's getting bigger and bigger and bigger, the endometrial lining. Again, estradiol is really driving that effect.

- 6.44** Then we have ovulation. If implantation happens, that's going to change the story, which we'll get to. But it goes into what's called the secretory phase to get ready for implantation. Because right around now, the female has just ovulated, it's coming through the fallopian tube, maybe sperm met it, and so the endometrium wants to be really big and ready. And so it is in this secretory phase because it's secreting a lot of molecules, not only hormones, but actually also glycogen.
- 6.45** It starts to load the lining of the endometrium with energy, with nutrients, calories, so that if the sperm meets the egg and it is being pushed down into the body of the uterus, if it settles there before it has all of its own blood flow through the placenta—it needs energy, and so it's secreting all of this glycogen to feed this little fertilized egg in the event it comes. So that is the secretory phase. Lots of blood flow, lots of glycogen being deposited and secreted along the endometrial lining, all to prepare itself for implantation if fertilization occurred.
- 6.46** Of course, if implantation did not occur, then the hormone levels sustaining it go away, namely that coming from the corpus luteum because progesterone and estradiol are keeping that proliferative phase going. Then as progesterone and estradiol fade, as the corpus luteum fades, or because the corpus luteum fades, then it's lost its sustaining signal. Then we have the end of the secretory phase as it progresses once again back into a menstrual phase and shedding it all out.
- 6.47** All right, that is in a way the most complicated part of all of it. You just have to appreciate the shift from negative feedback to positive feedback because of the growing follicle and its growing levels of estradiol, which then triggers the LH surge, ovulation, and then everything else.
- 6.48** The way I just got done describing it for the most part was the phase ending being that there was no implantation. So something interesting has to happen with implantation to keep that hormone profile going, namely the high estradiol and the high progesterone. Of course, the implantation itself actually triggers the changes, which I'll come to in a moment.
- 6.49** But back to what I mentioned earlier, the sperm has been ejaculated up into the vagina through the birth canal. It is swimming its way up, not knowing which fallopian tube to get to, because it could have come from either, but it needs to get through it. Lest you think that glycogen is fueling the sperm, it's not. Remember that glycogen that I mentioned earlier from the secretory phase? The sperm is not using it. It has its own modest little fuel supply, but it's not using the glycogen.
- 6.50** Swimming, swimming, swimming, not knowing which fallopian tube to go down, but doing both, and then bumping into the egg. That is, of course, fertilization. There are steps to that I won't get into because it goes a little outside of endocrinology and more

biochemistry, but the sperm meets the egg, comes in, and closes itself off so that other sperm don't get in.

- 6.51** Now, however, we have what's called a peristaltic motion. Just like how we have peristalsis from our intestines—we eat something, the intestines are rhythmically contracting to move it out—so too are the fallopian tubes rhythmically contracting and moving with little fimbriae, these little finger-like things, tickling the fertilized egg down the fallopian tube where it needs to get.
- 6.52** If the fallopian tube is not moving it, that is going to result in—you have to probably get that little baby out. If it stays in the fallopian tube, it could kill the mom. You need to get it. That would be an ectopic pregnancy—ectopic meaning the pregnancy is happening where it shouldn't happen. So we need to move it into the body of the uterus. The proliferative phase has been dominant so there's lots of energy, lots of blood flow, lots of hormones ready.
- 6.53** How does it settle in? This is actually, again, a progesterone-dependent effect, where once again, we have cytokines or these molecules that are capable of eroding a layer and adhering. There's adhesion molecules. There's hormones that are causing new blood vessels to grow, like vascular endothelial growth factors. And all of it is under progesterone's oversight.
- 6.54** Progesterone is monitoring, if you will, this fertilized egg saying, "I need you to get here, and I'm going to change the whole milieu of the uterus to help you get stuck right in the body of the uterus." It is genius. Suffice to say, progesterone initiates the erosion of the wall and then adhesion of this fertilized egg to allow the replication events to continue, and for this little fertilized egg to become the fetus like we think about.
- 6.55** In order to do that, we still need a lot of endocrinology involvement. So we start to form what looks like another follicle, but will ultimately become the placenta, where we have this one layer of cells surrounding the fertilized egg. That layer of cells is the cytotrophoblasts. Then the cytotrophoblasts are flanked by the syncytiotrophoblasts. So, cytotrophoblast, syncytiotrophoblast, collectively, that's sometimes referred to as the chorion or a chorionic layer. It's this future placental layer starting to wrap around the fertilized egg, or what's the embryo, what's becoming the little fetus. So the chorion is that layer.
- 6.56** I only mention that because some of the hormones we're going to say have the word chorionic in them. Now you'll know where that term comes from. It's derivative of these particular cells, the cytotrophoblasts, flanked by the syncytiotrophoblasts, and then together working.
- 6.57** Speaking of flanking, on each side of this now, we have what is going to become the fetal blood flow of the developing little fetus. That baby will have its own blood flow mingled through mom's placenta, acting as a bit of a lung or a filter, if you will, lung and liver in a way. Then we have the maternal blood flow. Both sides, maternal blood flow

and fetal blood flow, are interacting at this chorionic layer with the cytotrophoblasts and the syncytiotrophoblasts that will go on to define the placenta.

- 6.58** What's interesting is that we actually have a small version of the hypothalamic-pituitary axis, where some of the cells of the chorionic layer of the cytotrophoblasts are producing hormones that match the hypothalamus, like CRH, for example, or even GnRH, gonadotropin-releasing hormone. Yes, it's normally a product of the hypothalamus, but here we actually have it being produced in the cytotrophoblasts, and it is then releasing its products out.
- 6.59** This is very important because it gives rise to two very, very relevant hormones early on in the process of pregnancy, and that is hCG, human chorionic gonadotropin, and hCS, human chorionic—get ready for this one—somatomammotropin. Yes, I said that word correctly—human chorionic somatomammotropin, aka human placental lactogen. Let's talk about each of them.
- 6.60** hCG, you've heard of. Now, again, just to confirm, the origin of the hCG is from the chorion, this placenta. It's the interaction of the cytotrophoblast and the syncytiotrophoblast surrounding the fertilized egg, what's becoming the fetus. So now we have this infusion, this huge spike, and then it drops. It doesn't go away, but it goes from nothing to really, really high to still a little something. So it spikes and then falls.
- 6.61** But hCG is very important because it helps keep the corpus luteum alive because we need it to stay. That little corpus luteum—back to the ovary—is continuing to pump out hormones. Normally, without the implantation, estradiol and progesterone dropped. Well, we can't have that. We need progesterone and estradiol not only not to drop, we need them to continue to go up, so we need it to be uninterrupted. Thus, we need the corpus luteum to linger longer. What would have been a 10-day life cycle, we need it to be up about 10 weeks. Well, hCG helps that happen.
- 6.62** So hCG sustains the corpus luteum. It gives this survival signal, if you will, telling the corpus luteum, "I need you to stick around until the placenta is big enough to start making its own estrogen, estradiol, and progesterone. Until then, I need you to stick around." And hCG is a very important signal to force that process. And so, again, the purpose of this is to keep the corpus luteum around, to keep the corpus luteum producing estradiol and progesterone, which we need for pregnancy.
- 6.63** Now, the mentioning of hCG begs the question of the hCG diet. After all, I'm a metabolic scientist. Well, even though the hCG diet has fallen out of favor, it still exists, it's still used by some people, and it appears to not be any better than placebo. In fact, if there's any mechanism of action, it could be a mild sense of nausea in a body that shouldn't have any hCG. After all, hCG should only exist in the pregnant woman, but it might be part of what makes her sometimes feel a little nauseous. If you are injecting yourself with hCG and you feel a little nauseous, you're just going to eat a little less. That would be the mechanism of action for hCG diet.

- 6.64** Now I've mentioned another one, which is human chorionic somatomammotropin, a.k.a. placental lactogen. It is quite similar to prolactin, and so it's no surprise that it's going to be stimulating the growth of the mammary glands, the substance and indeed the purpose of the breast tissue to allow it to start making milk to feed the future little baby that will be dependent on mommy, historically, for all nourishment.
- 6.65** At the same time, an interesting effect of the somatomammotropin or the lactogen molecule is that it actually also causes a mild state of insulin resistance in the mom, which may be a way to help the glucose, rather than fueling mom's body, to fuel the placental growth instead. So there could be a very good reason for this, but at the same time, it's going to complement prolactin's effects in enriching the mammary tissue, the mammary glands, with milk. I
- 6.66** In contrast to hCG, which rose and then fell, the placental lactogen or the somatomammotropin goes up and stays up throughout the pregnancy. Then, of course, progesterone is underlying all of this, which is to say it is essential to keep gestation happening. It is, after all, the hormone of gestation.
- 6.67** Before we're done, just to mention one other placental hormone, and that is human placental growth hormone. What is so fascinating is that during pregnancy, as the placenta is getting bigger and bigger, it actually becomes the dominant source of growth hormone in the mom. It's certainly contributing to fetal growth hormone as well, helping the fetus grow. But even in mom, most of her growth hormone, if you did a blood sample from mom and measure her growth hormone, most of what you're measuring is actually coming from her placenta now, not her anterior pituitary. So it's a very relevant source. It continues to help mom's body grow and get ready, and it helps the placenta grow. It helps the breast tissue grow to prepare for nursing.
- 6.68** The end of it all will be parturition or childbirth. We've been growing the baby throughout this process. Now it would be time for the baby to come out. When it's time for the baby to come out, it's actually largely initiated by a process of just physical pressure. As the baby has reached a particular dimension, much to mom's chagrin, it starts to push enough on the cervix, the exit of the uterus, and that alone, as the cervix starts to get too much pressure, will initiate the production of oxytocin.
- 6.69** Now, you'll recall, we spoke about oxytocin a long time ago when we spoke about the posterior pituitary, that little extension of the hypothalamus. We not only have the stretch, eliciting a nervous stimulation to produce more oxytocin, but then we also can have, once the baby is born and suckling—the tactile stimulation of the breast with the newborn baby will further promote more oxytocin. Because as you will see, we typically think of oxytocin as being only relevant to contraction and labor, but it's also very important for lactation.
- 6.70** No surprise, estrogen plays a very important part here whereas estradiol has been going up over the course of the pregnancy, it starts to express oxytocin receptors in the

myometrium or the thick muscle part of the uterus. So we have this now ability where not only do we have some stimuli telling oxytocin levels that it needs to go up, like the cervical stretch, but then we also have the mounting estradiol levels telling the uterus to be more sensitive or more responsive to the oxytocin.

- 6.71** When oxytocin is getting released during the cervical stretching, the pressure of the baby getting so big now and weighing down on mom's cervix, it will actually result in the myometrium—so oxytocin will come to a G protein-coupled receptor and tell the cells of the myometrium, which is muscle cell, to get a lot of calcium. Remember what I said earlier? Anytime you have muscle cells that have a lot of calcium, you are contracting. So you start to have the contraction, the rhythmic contractions of that myometrium, that muscled layer of the uterus, the big, thick, strong muscle starting to push the baby the only way it can go, which is out through the birth canal.
- 6.72** Beyond the oxytocin is another hormone that's produced from the placenta, the syncytiotrophoblasts, called relaxin. And relaxin actually is going to have the opposite effect of oxytocin, but it's happening in different places. Whereas oxytocin was resulting in the tissue contracting, relaxin will actually act at the neck of the uterus or the cervix, which is functionally different from the rest of the uterus, and it will result in the production of nitric oxide, which I've mentioned before. Nitric oxide induces vasodilation. It wants muscle to relax.
- 6.73** The same thing happens at the cervix. While you have the contraction, the oxytocin-induced contraction of the uterus, because of all the calcium changes, you actually at the same time have relaxin relaxing the smooth muscle because of an increase in nitric oxide of the cervix. So that's a perfect complement. This is pushing, wanting to push out, and this starts to relax, enabling the overall exit of the baby through the birth canal.
- 6.74** With birth growth and the placenta loss, you, of course, immediately have the loss of hCG and then the somatomammotropin or the lactogen. Those are both coming from the placenta, so they drop to nothing again; they're gone. At the same time, progesterone has served its purpose, and there's no need. As hCG goes away, what happens to the corpus luteum and the placenta—or the corpus luteum already went away. The placenta was producing the progesterone because the corpus luteum only lasted for a few months. And then the placenta took over producing all the progesterone. The placenta is out, so progesterone goes away. Estradiol doesn't go away, but it comes down a lot, but it still stays higher than it did before.
- 6.75** Now, we come back to two other hormones that are very important as the woman would transition from being pregnant into lactation. There are two hormones that are important for lactation: oxytocin in an unexpected way, perhaps, and then prolactin, now, finally having its day in the sun. Both of these hormones play a part. Whereas prolactin is involved in the production of the milk in the mammary glands, oxytocin is

involved in the ejection, or the elimination of the milk, contracting the little muscle cells that line the gland, helping inject the milk. They're both very important.

6.76 With prolactin, it's not only activating lipolysis, so mobilizing mom's fat to enrich it in her milk, or activating the osteoclasts of her bone, telling her bones, "Hey, you have a lot of nice calcium here. We need some of it to enrich the milk for baby." And of course, a lot of other things are getting infused in that milk as well—vitamins, other fats, hormones— all of which is, by design, good for the baby. Prolactin is enabling all of that, taking everything it needs from mom's body and giving it into the milk, into the mammary gland.

6.77 At the same time, it's very important to acknowledge that prolactin inhibits GnRH. This is now prolactin's way. It's the lactation signal telling the brain, "Don't you dare ovulate again. We're busy. We don't have time for another baby. Stop ovulation." So as long as prolactin is elevated, it will inhibit GnRH, which will prevent LH and FSH from starting to play around and inducing a new menstrual cycle. So the menstrual cycle has stopped as long as prolactin is elevated.

6.78 Now, it was one thing to fill the mammary gland with the milk that prolactin put there, but it's another for the mammary gland to actually squeeze through its little layer of muscle and eject that milk. That's where, once again, oxytocin comes in because oxytocin promotes the contraction of what's called the myo—remember, anytime you hear myo, it's muscle. It induces the contraction of the myoepithelium. It squeezes the milk-filled glands or ducts to eject it so the baby can enjoy all of that nourishment.

6.79 What happens when an ovary has run out of follicles or has spent all of its eggs? That is menopause. As you'll recall, the development in the dynamic nature of the menstrual cycle really happened because a follicle would start developing. In the absence of the developing follicle, you lose a lot of—not only the eggs are gone, but remember, the granulosa cells played a role by creating the follicle. So the loss of eggs means a loss of follicle, which means a lot fewer granulosa cells.

6.80 You'll recall that the granulosa cell is the cell that actually makes estradiol from the ovary. It's the theca cell that makes testosterone. Theca cells are still there, so the ovaries can still produce testosterone, and indeed, they do. But now the ovaries are less capable at turning that testosterone into estradiol. The woman starts to have lower levels of estrogens, and thus she starts to manifest with the symptoms of menopause.

6.81 Because after all, the reason a woman is so bulletproof when it comes to metabolic health is because of estradiol. It helps the body store fat in a healthy way. It helps the body burn fat in a healthy way. It helps maintain neurons and neurogenesis. It helps stimulate mitochondrial biogenesis. This estradiol is why, largely, a female can be fatter than her male counterpart and be healthier in every single regard. Estradiol helps her be metabolically bulletproof. Now, with menopause, she is subject to the metabolic mortality that her male counterpart has been dealing with his whole life.

- 6.82** Again, the initiating event is the loss of eggs. The loss of eggs means no follicle, which is not enough granulosa cells to make estradiol. That does not mean she doesn't have hormones. Two things—one, the thecal cells are still there making androgens. At the same time, she also starts making androgens from her adrenal cortex. Remember the zona reticularis starts making more androgens. Then you would say, "Yeah, but I don't want the androgens. I want estradiol." After all, even a woman in menopause still has estradiol levels, albeit at a lower level. That's actually where the fat tissue comes in.
- 6.83** As I mentioned before, fat tissue, like ovaries, has a lot of aromatase. That is the enzyme that converts testosterone into estradiol. And that is a process called aromatization. So the quirk of menopause is that she's still making estradiol, but it's coming from her fat tissue now, primarily. That is the main source of estradiol in the menopausal woman.
- 6.84** Now let's talk about, let's end with some reproductive disorders, with precocious puberty and then male and then female. Precocious puberty is not totally defined because after all, some gals and boys will start puberty a little earlier, but it's mostly a problem in girls where the girl—and I'll explain why in just a moment—she is initiating puberty either in a partial way, meaning it's just perhaps the development of breast tissue, or a complete way, meaning it's both breast development and menstrual initiation or menses—that's a complete precocious puberty, but before she should.
- 6.85** Again, the reason it's hard is because not only are just people different, but it's also very different across ethnicities. For example, Hispanic gals will naturally initiate, it appears, puberty earlier than, say, Asian gals will. So there are some ethnic, genetic differences that are inherent, which makes it hard. But what we do know is that worldwide, puberty is starting at earlier and earlier ages. Within the past just about 50 years within the United States, puberty once started in girls at about 14 and is now starting at about 12. So it is shifting more and more.
- 6.86** Thus, there are lifestyle components to this, and the biggest is leptin. As you'll recall, for puberty to start, we have to have a change in the pulsatility of GnRH. That requires kisspeptin to be turned on, and one of the strongest signals for kisspeptin is leptin. Leptin comes from fat tissue. So the fact that we have kids that are getting a lot fatter a lot earlier, more fat tissue, as we will learn in the adipose lecture later—more fat tissue means more leptin. More leptin means an earlier initiation of the pulsatility of GnRH that leads to puberty.
- 6.87** So what is the main driver of precocious puberty? It is too much body fat, which leads to too much leptin. Now, why have I mentioned girls? I don't mean to call out the girls specifically, but females naturally are making more leptin than boys are. If you take out a gram of fat from a boy, a gram of fat from—well, let's say, male or female, because whether it's boy or man or girl or woman, it doesn't matter—then that given amount of fat from the female will be making a lot more leptin. This is already a body that's going to be having higher levels of leptin and indeed needs it.

- 6.88** This is why if you take a boy and a girl and put them to 5% body fat, the boy is fine. I mean, that's very low. He'll still be producing sperm just fine. If you put that girl at 5%, she now does not have enough leptin to ovulate anymore. That's again back to how important is leptin and it being coupled to how much fat a person has. Too little body fat in a body that needs more leptin, like the female form, will result in a lack of not only puberty, but stopping the reproductive cycle entirely. Then just, again, to put a fine point on it, a lack of leptin is a lack of puberty or fertility or a menstrual cycle. It will not happen, especially in women who just have a higher need, a higher signal when it comes to leptin and it driving puberty.
- 6.89** One other component could be the synthetic estrogens that are present within our environment. So whether it's plasticizers or detergents, some of these mimic—these molecules will mimic estrogens in the body and then initiate puberty a little earlier than normal. So there's a case to be made for it not even being nutritional, but just environmental as well.
- 6.90** Now let's talk specifically about male infertility or male reproductive problems, rather, and then we'll end with female reproductive problems. One problem in men is low testosterone. That is probably—well, it's one of the more common disorders these days, and it has everything to do with fat tissue, where he could have testes that are making a perfectly normal amount of testosterone, but as his fat mass is growing, more fat means more aromatase—that enzyme that converts testosterone into estradiol. So his testes are busy working all day and night to make testosterone, and the fat tissue is sabotaging the testes efforts to load the body with all of the manly testosterone, because the fat cell is taking in the androgens, which it does very easily and very well, and converts it to estradiol. So it's directly robbing the body of the testosterone.
- 6.91** This is an important paradigm that I believe shifts people's views in some way because we commonly hear the radio ad say, "Are you low T? Have you gained weight? Are you low? Are you tired at work? Have a hard time concentrating? You need more testosterone." Maybe some people need more testosterone, but in the average guy, the lack of testosterone isn't the problem. It's too much body fat.
- 6.92** Commonly the view is low T creates more body fat, and it does. If you have a guy who's true hypogonad, who has hypogonadism, that relatively lower level of testosterone will make it easier to store fat. You'll recall, when testosterone levels go up, it increases lipoprotein lipase in muscle. It wants to burn fat, not store it. So that view isn't wrong; it just doesn't explain the average guy nowadays.
- 6.93** The average guy has too much fat, which is breaking down, or converting, his testosterone into estradiol, not only robbing him of the potency of the testosterone, but also potentially with estradiol levels being higher, telling him to start storing more and more of that fat in prototypical female ways, so resulting in him having a chest that looks a little bit like breast. So as he has more and more of the estradiol, not only is he

going to be more inclined to store fat, but he will start storing it in a more prototypical female pattern as well, adding insult to injury.

- 6.94** One final comment on men in reproduction is this more and more common phenomenon of erectile dysfunction. Earlier, I talked about the point and shoot of male reproduction and how a part of that with the point aspect, the erection, we need a lot of nitric oxide to increase blood flow to enable the normal erectile function. What I didn't mention at the time was that insulin is actually an important part of this, where the hormone insulin, which we often just think of as a metabolic hormone, actually will flow through the blood, increase the expression of nitric oxide, and it's even relevant, yes, with normal erectile function.
- 6.95** As the endothelium, as the lining of the blood vessel becomes insulin resistant, insulin is less capable of inducing the production of nitric oxide. Less nitric oxide means vessels stay constricted. Constricted vessels mean less blood flow. Less blood flow means erectile dysfunction. So it's no surprise, with that view in mind, that some people consider erectile dysfunction to be the earliest sign of insulin resistance in otherwise healthy even young men. To say all that another way— if you have a young man who has erectile dysfunction, what might be causing it is, in fact, this underlying metabolic issue, where if you compromise the insulin sensitivity or function in the body, you compromise normal erectile function.
- 6.96** With women, we'll take time to discuss the most common form of infertility, which is polycystic ovary syndrome. It affects roughly 15% more, it seems, every year, of women. In order for the woman to be diagnosed with PCOS, she has to meet two of the three Rotterdam criteria, being: she has high testosterone, she's not menstruating or ovulating frequently, and she has cysts in her ovaries, as confirmed from some imaging technology like an ultrasound.
- 6.97** Two of those three would check the box of saying you have PCOS, and this can be substantial. Those follicles—in fact, it is the follicle. I'm getting a little bit ahead of myself, but over the course, in the process towards ovulation, there are multiple follicles developing. One becomes dominant. As that one becomes dominant and ovulates, the others all go away.
- 6.98** Well, what if they're growing but one doesn't ovulate because you don't have this big estradiol bump, which deprives you of the LH bump? So you have follicles that grow, but if one doesn't ovulate, they all stick around. And then another cycle starts, and then more come, and more come. And so the ovaries can go from something that would easily fit in the palm of my hand to being bigger than my entire hand. You can imagine how painful it could be then during the cycle if her ovaries are getting even bigger. You're just hoping that some of those follicles turned cysts will start to go away. But again, they're waiting for the signal of ovulation to know that they served their purpose, none of them were picked, and they all can just erode back into the substance of the ovary, go back into nothing.

- 6.99** Why would a metabolic scientist be interested in PCOS? Because it is a metabolic problem at its core. It is a disease of too much insulin. This hormone that is spiked when blood glucose goes up—when insulin is up, one of the unexpected aspects of insulin is that it will inhibit aromatase. You'll recall, aromatase is that enzyme that will take androgens and turn them into estrogens like estradiol. Insulin will turn that enzyme off.
- 6.100** So the higher insulin is getting, the more the ovaries from the theca cells produce androgens. They want to hand them off to the granulosa cells to turn it into estradiol, but that enzyme is blocked. And so not only is she getting less estradiol, certainly than she needs for ovulation—remember, we needed a really big bump. As the follicle is growing, it's making more and more estradiol unless insulin is inhibiting the production. So she fails to get that estradiol surge, which prevents the LH surge. The result is the follicles just stick around. They don't know where to go. We need ovulation to happen in order for the follicles to be degraded.
- 6.101** Just to show you how powerful this is—if you take human ovary cell tissue and incubate it with insulin, you can increase its production of testosterone by about 13 times, a huge increase in the amount of testosterone coming from those ovaries, not because you're stimulating it to make more sex hormones, but because you're preventing it from converting the testosterone into estradiol.
- 6.102** The good news of PCOS is that if you acknowledge the metabolic origins, namely the chronically elevated insulin, then you are well suited to help this woman reverse her PCOS, all without the need for drugs. This is just one of many papers, this one being a fairly recent one published within the past few years, finding that just by placing the woman on a low carbohydrate diet, she is able to bring down her insulin to less than half of what it was, and at the same time, bring down her testosterone and her free testosterone and enable the renewed production of estradiol, all of which you need in order to produce the normal changes in hormones in the female sex cycle.
- 6.103** To wrap all of this up, we have discussed the complicated process of reproduction, particularly in women, where it is a big multi-piece symphony, comparing that to the very simple barbershop quartet of male fertility, which is a very easy, simple process. Give it a little testosterone, functioning Sertoli cells, and you're ready to roll at any moment. But in the women, we need a defined period of time. It's a dynamic state where pregnancy can happen, and if it doesn't, we need to reset everything. If pregnancy does happen, then we have a new normal for the next period of time.
- 6.104** But at the end of it all, there are individuals who will struggle with reproduction, a very heartbreaking situation for those who desperately want to have children. It can manifest in men, with the most common one being erectile dysfunction, which has a metabolic origin, as you now know, and the most common in female, PCOS, which also has a metabolic origin. But that, to a degree, is some good news because if it's a metabolic problem, it can have a metabolic solution.

6.105 Thanks.

Lecture 7: Hormones of Metabolism

At a glance — Hormones of metabolism, centered on the pancreas.

- 7.1** Today, as we start discussing the pancreas, the pancreas is a bit quirky compared to the other endocrine tissues we spoke about in the previous lectures, because the other ones fit within the axis that we established in the second lecture. You'll recall that lecture number two was introducing you to the hypothalamic-pituitary axis, where there's this central conductor sending the signal down to whatever the endocrine organ of interest may be. Those lectures from before, it was the adrenal glands, it was the thyroid, and the gonads. Again, they're listening to a central conductor.
- 7.2** The pancreas is unique because it doesn't need to get a central signal. It is constantly sampling the body in a systemic sense. It's not relying on a central regulator. As we go through the pancreas today, you'll see that it's autonomous compared to the other ones. They're obeying a directive from the brain. The pancreas is working on its own, but it has an outsized impact, as you'll see.
- 7.3** The order of the discussion for the pancreas is going to be introducing you to the islets, these small little islands tucked throughout the pancreas, and their cells. Then we're going to get into the main pancreatic disorders, which is going to get us deep into the realm of diabetes—ones you're familiar with, and probably a form of diabetes you've never heard of.
- 7.4** Now, even though this is a course, a series about endocrinology, the actual bulk of the pancreas is what's called the exocrine pancreas. I used the term exocrine when I was describing thyroid physiology. But exocrine is when something's going out, out of the blood, out of the body, in this case. Where the pancreas actually, by mass, you could say, is a tissue that's more involved in digestion than it is in hormones, because over 90% of what we call the pancreas is actually cells that are making digestive enzymes that will be pushed into the intestines to help with digestion and then absorption, and it helps us get all our nutrition.
- 7.5** That's not what we're talking about, though. But I just want you to be familiar with the fact that there are two functional aspects to the pancreas: the exocrine pancreas involved in digestion, putting stuff into the guts; and then, of course, what we discuss, the endocrine pancreas. To understand the endocrine pancreas, we do zoom back into those little islands of cells called the islets of Langerhans.
- 7.6** The islets of Langerhans are small, distinct sections within the pancreas that have three predominant cells. There are actually a few more that are in there, but they're so irrelevant in the grand scheme of endocrinology that we won't get into them. There are three main cells, with two of them being the primary ones.

- 7.7** The first one is the alpha cell, and the alpha cell makes the hormone glucagon. The second is the famous beta cell, which makes the hormone insulin. And then lastly—it's one of the big three, but it's the lesser of the three—and that's the delta cells, which make somatostatin. So you need to appreciate these three home types within this little neighborhood, and they have their own little microenvironment.
- 7.8** As we get into talking about diabetes, particularly Type 1 diabetes, I'm going to reinforce this idea that the microenvironment has its own set of rules, like a frustrating HOA, if you will. But yet they still have a relevance in the grander scale of the human body, the city, if you will. But there's stuff that happens within the neighborhood that doesn't stretch into the broader systemic circulation of the body. So let's discuss how these cells and their products interact a little bit. And then again, we're going to revisit that in a pathological way when we discuss the relevance in Type 1 diabetes.
- 7.9** So firstly, the beta cell makes insulin. And insulin is a powerful hormone. But in this case, it's interesting because it wields that power in a bit of a bully-ish way. Because insulin inhibits both the alpha cell and the delta cell. So when insulin is going up within that little microenvironment, it will tell the alpha cell, "Make less glucagon." At the same time, it will tell the delta cell, "Make less somatostatin." So insulin wants to keep everybody else down. It's an angry neighbor that wants everyone just to be quiet. It's in charge.
- 7.10** Glucagon is the nice, happy neighbor that just wants to get along. So when the alpha cell releases glucagon, it actually has a stimulatory effect on the beta cell. It tells the beta cell to release more insulin. So again, beta cell sends its signal insulin to the alpha cell, telling it to stop. But the alpha cell sends its signal in the form of glucagon to the beta cell, telling it to turn on. So there is a bit of a mismatch in function.
- 7.11** But the delta cell with somatostatin decides to just play the same game that insulin is playing. And if insulin is going to tell the delta cell to be inhibited, well, the delta cell is just going to tell the same thing to the beta cell. Lots of alpha, beta, deltas here, isn't there? Then lastly, somatostatin, the hormone from the delta cell, also has a negative signal on the alpha cell with regards to glucagon. I believe I've said all of that correctly.
- 7.12** Suffice to say, the beta cell wants to inhibit both, the delta cell wants to inhibit both, and glucagon just wants to get along and ends up actually stimulating insulin. Now, this interaction, particularly between—as you're going to see—beta cell and alpha cell, so the interaction between insulin and glucagon, is going to become extremely important to understand the endocrinology as the product of the pancreas comes into general circulation.
- 7.13** Now, just one other point on this—and again, I'm going to revisit this when we talk about Type 1 diabetes— the microenvironment of all of these cells interacting is a much, much louder place than we see what ends up spilling into general circulation. In other words, when the beta cell is releasing insulin, it is as if the alpha cell releasing

glucagon is standing right next to a really loud speaker at a concert. It is hearing a lot of noise, or in other words, it's getting a tremendous amount of insulin being right next to the beta cell.

7.14 But then compare that to the amount of insulin that may, say, be flowing through our muscle cells. It's really loud right here within the pancreas itself. But by the time it spills into the circulation, it's as if the muscle is on the other side of the room rather than standing right next to the speaker. In other words, the sound with this metaphor is getting diluted. Back to the body, when we have the islets of Langerhans, the alpha cell is getting an enormous insulin load. It sees insulin at such a high concentration that no other cell in the body is going to see it. Because by the time the beta cell releases its insulin into general circulation, it's been diluted into several liters of blood. Does that make sense?

7.15 So you need to appreciate the microenvironment, which is really loud, to come back to that metaphor of standing next to the speaker. In contrast, the broader environment of the entire body, or what we would call the systemic circulation, that macroenvironment is much quieter. It has a much, much lower signal. And we'll get into some of the specific numbers in just a bit.

7.16 All right, now, the interplay between the alpha cell and the beta cell, and glucagon and insulin, is critically important because they're going to have such opposite effects. You're going to see this most obviously with glucose, but it also applies to other aspects, particularly of metabolism, which I'll show you in a minute. But at that most relevant, most famous aspect, it is the glucose control. And they have opposite effects and opposite stimuli.

7.17 When glucose is low, glucagon will be released from the alpha cell. When glucose goes high, insulin will be released from the beta cell. So they are a yin and a yang. And that's appropriate because glucagon, as you're going to learn, wants to increase blood glucose. And so it's no surprise that its primary stimulus would be a low glucose level. In contrast, insulin wants to lower blood glucose, and thus it's no surprise that its primary stimulus would be an increased glucose. So they are opposites. They really are a beautiful yin and yang, but it's more complicated than it seems.

7.18 Within both the alpha cell and the beta cell, we have glucose transporters: GLUT1 and GLUT2. And no, that has nothing to do with the big muscles on your backside. The glucose transporter one, glucose transporter two. What's interesting about these: two things to note. One, neither of them needs insulin. When we often think about a cell pulling in glucose, we often think, well, it must need insulin to do so. That's not true. Most cells of the body can just pull in glucose whenever the glucose is in the blood. It doesn't need a gatekeeper like insulin to tell it what to do. GLUT1 and GLUT2 fit that description. They do not need insulin to determine their function, but they are different.

- 7.19** GLUT1 is on the alpha cell, and it has what's called a low K_m . And K_m is a term used in chemistry to describe the action of a protein, or how readily it can turn on. What this means: by having a low K_m , that glucose transporter, GLUT1 in the alpha cell, has a high affinity for glucose. In other words, just a little bit of glucose in the blood is enough to go into the alpha cell. That's very, very important. Even if blood glucose is just low, it's still enough to be pulled in via GLUT1. GLUT1 is just that sensitive to the glucose, if you will. And so low glucose is enough to go into GLUT1, into the alpha cell, through GLUT1.
- 7.20** But the GLUT2 in the beta cell is what's called a high K_m , which means, perhaps paradoxically it sounds, it has a lower affinity, which is another way of saying it needs more glucose. The glucose needs to be higher for it to get through the GLUT2 into the beta cell.
- 7.21** So that right there helps us understand some of the yin-yang balance between the alpha cell and the beta cell, glucagon and insulin. The alpha cell can pull in glucose at low glucose levels, which would be too low for the beta cell to take it in. The beta cell needs a higher level of insulin before it gets its glucose. And that is important in both instances because the glucose actually acts as a trigger for both.
- 7.22** All right, so now that you can appreciate GLUT1 versus GLUT2, high affinity for glucose versus lower affinity for glucose, not needing as much glucose to work versus needing more glucose to work—no more comparisons to make there—let's just look at the alpha cell for just another moment.
- 7.23** So when glucose is at a low level, it's enough to go into the alpha cell. When that alpha cell is pulling in the glucose, it will metabolize the glucose via glycolysis, which is, of course, a topic I discussed in the Basics in Biology series, so I won't revisit that now. But the alpha cell's pulling in glucose, it's burning the glucose for energy, and that energetic state of the alpha cell is enough to elicit this series of events to kick out the glucagon. So the alpha cell pulls in glucose, metabolizes the glucose, and in the process is able to release glucagon. That's nice because glucose levels are low. And as you'll see, one of glucagon's primary effects is going to be to increase blood glucose.
- 7.24** Now, at that low level of glucose, it is not enough to activate the GLUT2. And so the beta cells don't even know that there's any glucose around, and they stay quiet; which is nice in this case because we don't want a lot of insulin if glucose is low. However, the script starts to change the moment glucose goes up. When glucose climbs, now it's enough to funnel into GLUT2. Now, the beta cell gets to do what the alpha cell just did. Now, the beta cell will undergo glycolysis and burn the glucose for energy, and in the process, stimulate the production of insulin and then release that insulin.
- 7.25** All right, so now you can see how low glucose can stimulate glucagon, but then how high glucose can stimulate insulin. But can you see the potential problem? That GLUT1 on the alpha cell doesn't turn off when glucose goes up. It is turned on at a low glucose level, but it stays turned on even though glucose levels are higher. So we're putting our

body in a bit of a metabolic conundrum, a paradox. We're high glucose, yes, now it is sufficient to stimulate the beta cell to release insulin, but it's still more than sufficient to stimulate the alpha cell to release glucagon. We don't want both hormones being elevated at the same time. That is going to be this, like I said, metabolic paradox.

- 7.26** So how do we reconcile this? Well, we come back to that interplay that I mentioned a moment ago. Because if insulin is getting released, it will have that paracrine effect on the alpha cell. So glucose is trying to turn them both on, but the beta cell releases its insulin right next door to the alpha cell and says, "I'm in charge right now. I'm the captain now. You inhibit. You go down. It's my turn to go up because glucose is up, so I need to be in charge."
- 7.27** So they had that common stimulus of glucose, but when push comes to shove, insulin takes over, by directly blocking the actions of glucagon. And you'll recall glucagon would in turn be stimulating insulin. Remember, it had that complementary stimulating effect. So this is how we reconcile the two: sharing a common stimulus by way of glucose, but when they're both turned on, one will make sure the other one turns off.
- 7.28** Okay, so let's talk about glucagon for just a moment, and then we're actually going to revisit some glucagon consequences in the next lecture. So with glucagon, it will be transcribed and produced; so there'll be a gene that turns on, that gene will be turned into a protein, and that protein will be glucagon. But when we make glucagon, something that's very interesting is that the alpha cell will make a molecule called proglucagon. And you'll [remember] in the very first lecture when we spoke about different hormones, peptide hormones undergo a process of snipping and folding and bending and twisting. In the process of snipping, folding, bending, twisting, in the alpha cell, the proglucagon will be trimmed accordingly to just give us glucagon.
- 7.29** However, when we talk about adipose tissue, I'll mention that there's a hormone that has tremendous effect on adipose tissue called GLP-1. And so the intestines has its own form of proglucagon and trims it its own way to not make glucagon, but to make GLP-1. But in the alpha cell, you have that same proglucagon molecule that you'd find in the intestines. But because of the post-translational modifications, the snipping and folding, it will only make glucagon. And then again, it's the intestine that will take that molecule and with its own changes will only make GLP-1. So don't confuse the two. They start with the same mother molecule, proglucagon, but because of the different trimming, they end up with different products. And again, in the alpha cell, we end up with glucagon.
- 7.30** Now, I mentioned how glucose levels are a trigger for glucagon, and that is right. That is a nutrient trigger, which is kind of uncommon. But there's also hormone triggers. Because if glucose levels are going low, one of the body's responses to that low glucose will be increasing epinephrine from the adrenal medulla. And epinephrine will come to the alpha cells, bind a G protein-coupled receptor, and get through cyclic AMP—this primary second messenger that I mentioned so long ago—and that cyclic AMP pathway

will tell the alpha cell to make more glucagon. So outside of the glucose regulation of glucagon production, we also now have a hormone regulation where epinephrine can do the same thing.

- 7.31** But you'll recall, I mentioned from the delta cells in those islets, that somatostatin doesn't like any other hormone to be released. And so when somatostatin is being released, it actually just inhibits that very process. So where epinephrine is trying to turn it on, somatostatin will try to turn it off. And then insulin will do the same thing, although they act at different sites. So the alpha cell is surrounded by bullies. The delta cell with its somatostatin is inhibiting part of that pathway. And the beta cell with its insulin is actually trying to inhibit a different part of that pathway. So they're ganging up on the alpha cell.
- 7.32** Now, let's mention, I've alluded to this, but very, very briefly: a theme of glucagon's effect is that it's going to be opposite of whatever insulin wants to do. And in fact, within their own way, they both are perfect representations of the two halves of what we call metabolism. When you think about the word metabolism, people often just think of that word as meaning you're breaking something down. And yet that's not accurate.
- 7.33** Metabolism is the balancing, the calculus, of both the anabolic reactions, the building up, and the catabolic reactions or the breaking down. So very briefly, anabolism is when you're taking simple building blocks and creating a more complicated structure from them. In contrast, catabolism is taking a big, complicated structure and trimming it back into its base units. Both of those are essential. Both of those come together to create what we call metabolism, in a very real biochemistry sense.
- 7.34** Insulin is a prototypical anabolic hormone. It hates breaking stuff down. It only wants to build things up. That's neither good nor bad. We have to have that happening. We just don't want to have it happening all the time. Similarly, we don't want to be breaking things down all the time, but sometimes we need to. And glucagon is one of the prototypical hormones that represent a catabolic process.
- 7.35** So just speaking about glucagon for another moment, if you think about nutrients—which again, as a metabolic scientist, I like to do—glucagon is going to want to break down fat. So it's going to activate lipolysis. It's going to want to break down glycogen, the stored form of glucose in the liver. It's going to want to create ketones from that fat burning, so it stimulates ketogenesis. Any of these processes of just burning fuel, glucagon is going to want to turn on. Most especially, again, it wants blood glucose to go up.
- 7.36** All right, that's it for the alpha cell and glucagon. Let's transition and talk about the big angry kid on the block, which is insulin coming from the beta cell. Insulin is produced when its gene gets turned on and transcribed, and it will, in its own form, come in a proinsulin, just like glucagon had a proglucagon. But in the beta cell, when the

proinsulin gets created, as it's making its way to the blood, it will get split apart. It will release both an insulin molecule and another sister molecule called C-peptide. It's interesting to look at how both of these are used clinically.

- 7.37** Some clinicians will rely on C-peptide as an indicator or as a marker, a surrogate marker, for insulin, which is a terrible mistake. Even though they are produced one-to-one, they have such fantastically different half-lives—in other words, how quickly they're cleared from the blood—that you can't use one to get a measurement of the other. Sometimes they will use it to—they'll look at just a measure of C-peptide and say, "Well, we know what insulin levels are."
- 7.38** Now, insulin gets cleared a lot faster than C-peptide does. So C-peptide should be used as an indicator of making insulin. If you have a person with Type 1 or Type 2 diabetes, and you want to know the degree to which they are making any insulin, you would say, "Well, just measure their insulin." Yeah, but what if they're injecting insulin as part of their therapy? Then what you're measuring could be the insulin you're injecting. But you want to know whether the beta cells are actually producing insulin. You can't measure insulin because of the noise of injecting it, so you have to rely on C-peptide. So C-peptide is not a surrogate marker for insulin levels, but it's a good indicator of insulin production.
- 7.39** Now, as I mentioned, glucagon wants to play nice. So the canonical, or the big most famous stimulus for insulin is going to be glucose levels going up. But there's also hormone signals that come in. And glucagon will come to the beta cell, bind to a G protein-coupled receptor, activate cyclic AMP, and through this pathway, induce the production of insulin. So there's a hormone-dependent mechanism. But once again, like we saw with glucagon, the other little neighbor in the islets of Langerhans, the somatostatin from the delta cell, does not want it to be released, and it will inhibit that process just like it did or attempted to do with glucagon.
- 7.40** But insulin has additional inputs, including something I'd mentioned in a previous lecture, called a neurocrine input, where the pancreas has a direct innervation from a nerve called the vagus nerve. This is coming all the way from the brain. This is a direct line, quite literally, from the brain going into the pancreas. When that vagus nerve comes, it will, at the end of this junction, release acetylcholine. Again, go back to my previous series when I spoke about the parasympathetic nervous system. At this synapse, we have acetylcholine that is now coming to the beta cell, and it will tell the beta cell to release more insulin. So the vagus can be a direct signal.
- 7.41** Just as a very interesting aside, there are some therapies where they will shock the vagus and in so doing, force the pancreas to produce less insulin, and they do that as a weight-loss tool. And that's just reflective of how powerful insulin is at dictating how much the fat cells are growing or shrinking. So we had the hormone signal coming from glucagon. We also now have a nervous signal, or a neurocrine signal, coming from the vagus, releasing its acetylcholine.

- 7.42** Now, what does insulin do? When insulin starts moving through the blood, it is an uncommon peptide hormone because it can affect every single cell of the body. Every cell of the body has insulin receptors. Again, that is uncommon when it comes to peptides. From brain cells to bone cells, from lung cells to liver cells, and every cell in between, insulin is going to have some effect. That's one of the reasons why problems of insulin can connect to so many problems throughout the body, whether it's liver problems or bone problems or brain problems or heart problems. Insulin tells all of those tissues to do something. When the signal starts to get disrupted, it's no surprise that some things start to go wrong.
- 7.43** Now, the insulin receptor is a tyrosine kinase receptor. It's two little receptors that will link together when insulin comes, then they will turn themselves on. I alluded to this in the endocrine basics lecture. A lot of people only think of insulin as being relevant to glucose handling. And yet insulin is related to dozens, if not well over 100 distinct processes within every cell. Insulin has its hand in countless effects. So just thinking of insulin as a glucose hormone is profoundly unfair. It is involved in a tremendous number of things, from inflammation, to blood vessel dynamics, to salt and electrolyte handling, and as we noted previously, sex hormone production and many, many more processes.
- 7.44** But it is most famous for its effects in metabolism. Again, I'm a metabolic scientist, so let's just take a moment to describe those. When insulin is moving through the body, it's going to have effects on essentially every nutrient you can think of. If it can be burned for energy in a cell, insulin is going to do something to it.
- 7.45** First and foremost, and most famously, insulin will activate GLUT4. Now, you will recall we spoke about GLUT1, we spoke about GLUT2. Those do not need insulin. In fact, none of the glucose transporters do, and there are many of them, well over a dozen. The only one that does is GLUT4. But GLUT4 is a big one because it's the one we have in our muscle and our fat tissue.
- 7.46** When the big hungry muscles, just think, if the muscles could just use any glucose they wanted any time, no one else would get any. It would hog it all up. It's very, very smart that we only allow the muscle to take it in when insulin has turned on. Because if insulin turns on from the beta cell, it reflects that there's a lot of glucose. There's plenty for everyone. We make sure that the big hungry beast that's going to eat the lion's share of glucose waits until everyone else has been able to eat, and then it is fed at the end because once it starts eating, it's going to eat it all. The most famous effect is the activation of GLUT4 on muscle cells and even fat cells, and then it starts pulling in that glucose, lowering blood glucose very quickly. But that becomes a problem in Type 2 diabetes, like we'll discuss in a bit.
- 7.47** At the same time, insulin is inhibiting anything that would be breaking down nutrients. So lipolysis, the breakdown of fat, insulin abhors that. In fact, not only does insulin turn off lipolysis, it actually turns on lipogenesis. It wants to make fat. It wants to store the

energy. Insulin wants to be miserly with any nutrients. What about glycogenolysis and the breakdown of glycogen from the liver to release it as glucose? Oh, no. Insulin inhibits glycogenolysis, and in contrast, will promote glycogenesis. It wants the liver to pull in any glucose and hold on to it. It does not want to break stuff down. It only wants to build things up.

- 7.48** So whatever the nutrient—ketogenesis I mentioned with glucagon. Glucagon wants to burn, and I spoke all about ketones in a previous series, so I'm not going to go through that metabolism now. But just as a brief primer, anytime the body is burning a lot of fat, it'll start making ketones from that fat. That's all ketones are, just products of fat burning. Nothing magical about them. But insulin doesn't want fat burning to happen. So as much as glucagon activates ketogenesis, insulin inhibits it. It does not want the fat to be burning up and turning into ketones.
- 7.49** Now, the muscle, I mentioned muscle a moment ago, and I mentioned how the muscle will only eat glucose when insulin is up. Insulin will only be up, commonly, if glucose has been elevated. Earlier, I'd shown you how the vagus nerve can come to the pancreas and tell the pancreas to release insulin. But that's a parasympathetic event—back to the parasympathetic nervous system lecture from a previous series.
- 7.50** The sympathetic nervous system has its own effect. Whereas the parasympathetic wants to release insulin, when we're resting and digesting. But if it's fight or flight, then the sympathetic nervous system is dominant, and it has the opposite effect. It has its own direct signal to the pancreas and will tell the pancreas, "Don't release insulin." Now, why am I even describing that when I'm talking about the muscle? Because if you get up and start exercising, that is a sympathetic nervous system moment. And so it's no surprise that as you start exercising, insulin levels are going to come down very quickly.
- 7.51** Well, then how are the muscle is going to eat? What a terrible situation to put the muscles in. The moment we're demanding more of them, we lower the insulin and we don't allow the muscle cells to eat? That's the way it sounds, right? And yet the muscle is too clever, and too hungry, and too demanding to allow insulin to deprive it of fuel.
- 7.52** So the muscle has this remarkable internal biochemistry that as it starts contracting and relaxing, like it is rhythmically during exercise, it has its own pathway, a back door, if you will, to open the GLUT4. So as much as I just got done saying the GLUT4 is unique because it has to have insulin as a gatekeeper to open it, yes, that's true, unless the muscle is exercising. If you can really change the bioenergetics of the muscle cell, it can turn on its own little back door to open the GLUT4, which is good, again, because during exercise, insulin is being inhibited through that neurocrine regulation of the sympathetic nervous system.
- 7.53** I'd mentioned the epinephrine effect sympathetic nervous system. Glucagon is a signal, and other signals are turning insulin on to be released from the beta cell. Then suffice it

to say, when epinephrine comes, it binds to a G protein-coupled receptor, the alpha-2 receptor, and will turn that whole process off. So a direct sympathetic nervous system, an epinephrine signal, that turns it off.

7.54 Okay, so at this point now, you have a unique appreciation for the microenvironment of the pancreas, of the islets. You also can appreciate the interplay happening, most especially between alpha cells and glucagon and the beta cell and insulin, with the delta cell and somatostatin just being on the side. Those are the three main cells that are relevant. But with an appreciation of all of those, actually, let's now transition out of endocrinology per se and more into the pathological side and discuss some pancreas disorders.

7.55 Very briefly, I'll mention insulin resistance because I've talked about that in a previous series, in fact, multiple times. But then we'll talk about Type 2, how insulin resistance gets—it flips a switch and becomes Type 2 diabetes. Then Type 1. And then a version of Type 1 that no one's ever heard of, but in fact happens in people; and if it's misdiagnosed, it's mistreated, so it's worth describing.

7.56 Insulin resistance, I have devoted my career to it. My initial Peterson Academy series of Why We Get Sick is completely devoted to it. But why would I devote my career to understanding insulin resistance? Because it is the most common health disorder worldwide. Whether it is in the Middle East or East Asia, Southeast Asia, South Asia, South America, this is the most common health disorder. And beyond the prevalence is the relevance, where every single chronic disease is in some way impacted by insulin resistance.

7.57 So very, very briefly, insulin will come to a cell and tell a cell to do something. And there are many, many some things that insulin will tell the cell to do, as I mentioned. The most famous something would be to pull in glucose at some tissues, as you'll recall. That's the one thing that people associate with insulin, and often nothing more.

7.58 But what if that signal starts to decay? So once upon a time, insulin would come and knock on the door because it's a water-soluble peptide, and so it needs to knock on a receptor, it can't go in. Insulin comes and knocks, and then the cell would respond very, very well. But over time, or due to various noxious stimuli, that signal can break down. The cell isn't responding as well as it used to. That would be insulin resistance. We have some consequences or some adaptations. Then if one polite molecule of insulin isn't enough to get the job done, well, then we need an angry mob of insulin pounding on the doors of the cell to try to get the cell to restore its response.

7.59 To understand insulin resistance, you need to understand two things. One, insulin isn't working as well as it used to at certain cells of the body. That is critical. Now, you'll notice how I said it. Some cells of the body are responding to insulin perfectly fine. It's not a universal effect. Some cells aren't responding, but some cells are. But that would be insulin resistance.

- 7.60** But that becomes particularly relevant in light of the second part of this problem, this two-part pathology, which is that insulin levels are higher than they were before. A condition called hyperinsulinemia. Hyperinsulinemia being in the blood. So blood levels of insulin are higher. And those two together are what we call insulin resistance. And it's understanding the two together that helps us understand how this problem contributes to so many chronic diseases.
- 7.61** Now, I've gone into all of those mechanisms and consequences in the Why We Get Sick series. Let's bring it back to Type 2 diabetes, because Type 2 diabetes exists and is built on a foundation of insulin resistance. If you wipe away the insulin resistance, there is no Type 2 diabetes. The house of cards just tumbles to nothing. To say that again, Type 2 diabetes does not exist in the absence of pre-existing, and worsening, insulin resistance.
- 7.62** Now, over time, a person may be coming in to their physician getting blood tests every year, and they're always measuring glucose, and glucose is stable, stable, stable. It's getting higher, higher, higher. Now, it's getting really high. Unfortunately, because of our glucose-centric view, we don't measure insulin. We're so focused on glucose that we fail to see anything else. If we broadened our perspective, or even just had more of an insulin-centric paradigm, we would detect that insulin levels had been coming up potentially decades before the glucose ever did. So it's a much earlier signal.
- 7.63** In that initial state of high insulin, but normal glucose, is insulin resistance. But then, again, unfortunately, because we're only looking at the glucose, we only start to detect the problem once the glucose starts to climb. And once the glucose starts to climb, now we say it's Type 2 diabetes, which is accurate. Insulin resistance is high insulin, but normal glucose, Type 2 diabetes is still elevated insulin—even if it's come down a little, it's still much higher than it used to be—but also with elevated glucose now. High insulin, high glucose is Type 2 diabetes. And that is a fundamental feature.
- 7.64** Sometimes people will say in Type 2 diabetes, or when describing Type 2 diabetes, they'll say, "the beta cells are insufficient," or "insulin is insufficient." Well, that's a relative term. Sure, it means there's not enough insulin because it can't control the glucose anymore. But that's not a problem of having a lot of insulin. It's because the insulin isn't working very well, it's not because there's not enough. But that is a wrong view.
- 7.65** In Type 2 diabetes, insulin never goes to zero, ever. Some people like to think it does. They like to describe it like it does. In Type 2, insulin is always elevated, and there's way more beta cells. If you actually look at the pancreas, the beta cells of people, the amount of beta cells in someone who has Type 2 diabetes, there's way more. They're not dying out like you have happened with Type 1. It's not the same problem. And in fact, it's the exact opposite problem. A lot of beta cells producing a lot of insulin.

- 7.66** All right, now, what initiates that transition? When I described it earlier, insulin resistance is when insulin is high but glucose is normal. Well, what initiates the series of events that enables the glucose to start climbing? It's when the insulin resistance spreads to three important tissues. In no particular order, it would be: the pancreas, specifically the alpha cells, the liver, and the muscle.
- 7.67** Let's just start with the muscle, because by mass, it's the biggest, most important tissue here. In fact, in this regard, as I noted, the muscle eats the lion's share of glucose. When it eats, it feasts. If you were to ever look at your glucose level, you eat something starchy and sugary, glucose levels spike considerably, and then it goes down. 80% of what brings it down is by what's coming into the muscle. 80% of the glucose getting cleared from your blood at any moment is going into the muscle.
- 7.68** This is why if you eat something starchy and sugary and you get up and just go on something as mild as a 10-minute walk, your glucose levels will come down significantly faster. It'll be about 50% of the curve you would have gotten otherwise, in the absence of just getting up and going on a mild walk, because you're telling the muscle to eat even more of the glucose.
- 7.69** But what if the muscle becomes insulin resistant, now this tissue that was eating the lion's share is only eating the cubs share of the glucose. It's not eating nearly as much. If the muscle can't eat as much of the glucose, the glucose is just lingering in the blood. No surprise, blood glucose levels will be higher. When the muscle starts to become insulin resistant, the individual will absolutely start to see higher glucose levels, thus pushing them into a deeper state of Type 2 diabetes.
- 7.70** Now, what about the liver? I often refer to the liver as the soccer mom of nutrient metabolism. It knows what to do with everything. Ketones, lactate, glucose, amino acids, fats, it doesn't matter. The liver can handle all of it. But in this context, its most important effect is handling the glucose. As I noted earlier, when insulin is up, which will normally happen when glucose is up, it's going to tell the liver, "Hey, I got a lot of glucose. I want you to hold on to that glucose and turn it into glycogen."
- 7.71** Now, what's important is that the liver does not have GLUT4. It does not have GLUT4 translocator, so it doesn't need insulin to take in the glucose, but it still needs insulin to know what to do with the glucose. So even when insulin isn't dictating the uptake of the nutrient, it is still dictating the use of the nutrient. And that is a thematic effect. If we zoomed back out at the whole body, and looked at all the cells in all their variety, if you were to say, well, what is the one thing insulin is doing in all of them? It's dictating metabolic processes. And in the liver, it's activating glycogenesis. So it wants the liver to hold on to its glycogen, to hold on to the glucose as glycogen.
- 7.72** But what if the liver becomes insulin resistant? Now, even though glucose may be high, insulin is high, it's telling the liver to store glucose as glycogen, but the liver is not listening anymore. The message has decayed, it's broken down. In the absence of the

signal working, telling it to hold on to the glycogen, the liver ends up dumping it out as glucose, thus compounding the entire hyperglycemic problem and glucose levels go high again. When the liver starts to become insulin resistant, you're going to start to have higher and higher blood glucose levels because it just keeps pumping out glucose, even though there's already plenty in the blood, because insulin can't tell it to hold on to it anymore.

- 7.73** Now, at the end of it all, to understand Type 2 diabetes, the last piece is in a way the most complicated and perhaps the least appreciated, which is back to the microenvironment of the islets of Langerhans. Because you'll recall that when the beta cell makes insulin, it has an inhibitory effect on the alpha cell, telling the alpha cell to make less glucagon. That signal allows insulin to beat, to win that tug of war with glucose, which is to say: when insulin and glucagon are both trying to go up, insulin will push the glucagon away and thus be able to bring the glucose levels down, acting as a hero in this story, lowering blood glucose.
- 7.74** But the alpha cell itself can become insulin resistant. Now, even though glucose levels are up, which should have enabled insulin to be the dominant one, in part because insulin is going to come to the alpha cell and tell the glucagon to go away. Insulin comes to the alpha cell, it says, "I need you to inhibit glucagon," but the alpha cell says, "I can't hear you." The signal is lost. It's become insulin resistant. It's become deaf to insulin's demands.
- 7.75** Insulin's up, it's trying to lower blood glucose, but at the same time, now the alpha cell is continuing to release more and more glucagon. This is a fundamental feature of Type 2 diabetes, and as you'll see, Type 1 as well. At any moment, an individual with Type 2 diabetes, even though they have hyperinsulinemia, because the alpha cell is insulin resistant, they will also have hyperglucagonemia. And then glucagon's pushing the liver to make more and more glucose and release that glucose into the blood, compounding the problem. There's already a lot of glucose in the blood, and the insulin-resistant alpha cell keeps pushing the problem out.
- 7.76** All right, that is the ideology or the background, the causal events of Type 2 diabetes. You think it's built on a foundation of insulin resistance, but it's particularly when the insulin resistance spreads to those three tissues in any particular order: the alpha cells of the pancreas, or the liver, or the skeletal muscle.
- 7.77** Now, even though these diseases are considered to be part of a family, I actually consider that to be unfortunate because Type 2 diabetes is a disease of too much insulin. Too much insulin drives insulin resistance. Go watch the Why We Get Sick series to learn more. In Type 1 diabetes, it's a disease of too little insulin. But they do have one thing in common, which is glucose levels will be high. I actually think that's the wrong view. That's reflective of the glucose-centric view. They're not related diseases. They're opposite diseases. Again, one of too much, one of too little.

- 7.78** With Type 1 diabetes—everyone's familiar with it—it is the autoimmune destruction of the beta cells. The beta cells of the pancreas have been targeted for destruction from the immune system. And as the beta cells are destroyed, insulin production is destroyed. We've come in and bombed the factories. I'm actually using that metaphor deliberately because when we compare it to the other type of diabetes, the factories haven't been bombed, even though the disease looks the same. But in this case, we've bombed the factories. We've destroyed the beta cells.
- 7.79** And since we've destroyed the beta cells, we've lost insulin's ability to control metabolism, and the most famous effect is going to be that glucose levels are high. There's a lot of hyperglycemia. There are other consequences, particularly at fat cells and things that we'll discuss in the fat tissue lecture. But for now, we'll just focus on the glucose.
- 7.80** The only treatment in someone with Type 1 diabetes is to give them insulin. But just as an interesting aside, and to whet your appetite for the fat lecture coming up: why do we tell a Type 1 diabetic to rotate their injection sites? Why put it in the belly right here, or put it in the thigh, or put it on the low back on one side and then rotate it around? It's because insulin is so good at telling fat cells to grow, it's like fertilizer on fat cells, that it will result in a dramatic, exaggerated, almost even grotesque growth of fat cells if you just give those same fat cells the same insulin all the time.
- 7.81** It is this weird little growth. It's like a garden and you're only nourishing one little part. Everything's flourishing here, and all the other plants are suffering. That would be like, all these other fat cells are really, really small, not getting this growth signal, and these fat cells, which only get the growth signal all the time, are getting really, really big.
- 7.82** Now, one of the problems with Type 1 diabetes is the accidental over-injection of insulin, leading to a problem called diabetic shock. Now, there's an interesting metabolic explanation here, and let's just reason through it in a moment. And it has everything to do with the brain. The brain has a very high metabolic rate, and it needs to get nutrients all the time. It relies primarily on two nutrients. It relies on glucose, and it relies on ketones, and it doesn't care. It will happily use both. It is opportunistic.
- 7.83** There's a mistaken view that the brain requires some amount of glucose, and as if it will only use ketones in desperation—absolutely false. In fact, if anything, when the brain has equal access to both, it actually prefers ketones dramatically over glucose. So if the brain has a preferred fuel, it's the ketones, and that's been made very, very clear in the evidence. But still, it's opportunistic. It will take either glucose or ketones.
- 7.84** But what if neither is available? And that's what happens with an over-injection, an overdosing of insulin. Because when the individual puts in too much insulin, both of those fuels are gone. When it comes to glucose, the person with the over-injection of insulin has forced it all into the muscle. The muscle consumed it all. So they become hypoglycemic, potentially dangerously so.

- 7.85** Now, frankly, the brain won't panic if there's ketones to use because the brain will happily just use the ketones. But remember, what does insulin do to ketogenesis? Not that we've talked about nutrient biochemistry here, but I mentioned it. Insulin inhibits ketogenesis. So with an over-injection of insulin, we've deprived the brain of both of its fuels. Glucose has come down, ketones have come down, and then the brain will be so hungry or even starving that it shuts off. Now the person has gone into this coma or what we call a diabetic shock. All right.
- 7.86** Now, I'd mentioned that there's a little bit of a different view of Type 1 diabetes. The traditional view being, and it's not wrong, that it's a disease of too little insulin. But the alternative view is that it's a disease of too much glucagon. And earlier I had mentioned this metaphor of standing next to the speaker. So when we think about the microenvironment of the islets of Langerhans, glucagon is used to seeing insulin in what we can call a millimolar range. A millimolar being a unit of concentration. So it's hard for you to put that into perspective at the moment, but millimolar is the range, is the dose of insulin or the concentration of insulin within that microenvironment.
- 7.87** By the time the insulin is in the blood, it's in a picomolar range, so it's literally less than a thousandth of the same concentration. So if standing right next to the speaker is some unit of volume, imagine being 1,000 times further away, or however we would have to quantify that, to hear the volume at one thousandth the level. That's what the muscles are used to hearing. And so they're primed to hear it at that. Does that make sense?
- 7.88** And so imagine the Type 1 diabetic who's injecting insulin at picomolar concentrations, because the individual injects insulin at a concentration sufficient to match the systemic load, not sufficient to match the local microenvironment load. And so the injected insulin is enough to tell the muscle what to do and the other tissues, but it's never enough to tell the alpha cell to stop making glucagon. So the alpha cell will never get enough of a signal from insulin to stop. And so in Type 1 diabetes, but for very different reasons than we saw in Type 2, it will always be a disease of hyperglucagonemia.
- 7.89** So Type 1 will always have too much glucagon, and that explains part of the never-ending roller coaster of glycemia, because you're constantly battling. You're putting enough insulin to tell the muscle to take it in, or the fat tissue to take it in, but you're never putting enough in to tell the glucagon to stop. And so glucagon is always high. It's always telling the liver to keep dumping out glucose. It makes it for a very frustrating disease.
- 7.90** I would encourage anyone who wants to learn more to look up the work of Dr. Roger Unger, who posited this initial idea of the bihormonal nature of Type 1 diabetes. That as much as it is a disease of too little insulin, it's also a disease of too much glucagon. In his hands, he has shown that if you can block glucagon action, you can make the

disease much, much easier to control. Hopefully, that will become mainstream at some point in the future, because it ought to.

- 7.91** All right, now, the last form of diabetes is MODY. It's M-O-D-Y, Mature Onset Diabetes of the Youth, or of a young person. With MODY, the problem with it is it acts like Type 1. It seems like Type 1. The patient will come in and they will have all the signs and symptoms of a person with Type 1 diabetes, and yet it's not. Whereas Type 1 diabetes is the autoimmune destruction of the beta cells, that's back to bombing the factories.
- 7.92** The factories are there. If you actually did a little biopsy of the pancreas in someone with MODY, the beta cells are there. But there is one single gene, there's one mutation that is stopping the beta cells from making the insulin. This would be back to the metaphor. The factory is there, all the machinery is there to make the insulin, but there's one single cog missing. That's it. There's one single gene and its protein product that just hasn't been produced well or at all, but everything else is there.
- 7.93** In fact, in this case, whereas Type 1 diabetes is an autoimmune disease, MODY is an autosomal dominant disease. Which is interesting because that means there's going to be a particular inheritance pattern to it—this goes back to Basics in Biology, which I spoke about then. But if it's an autosomal dominant disease, that means it affects both sexes and it's not skipping generations.
- 7.94** Anyone listening to this, if you have Type 1 diabetes and you have a parent who has it, you have a grandparent who has it, et cetera. You have siblings who have it. It's very, very possible that you actually have MODY and not Type 1. Type 1 is not genetic, but MODY is. I have now known, at this point in my life, multiple people who were told they had Type 1, and yet they actually had MODY.
- 7.95** That is life-changing because once a person knows they have MODY, they actually can treat this problem by just either adopting a low carb diet. That the reality is they still do make some insulin, it's just not enough to handle the high carbohydrate, high starch, high sugar, constant consumption diet that we eat nowadays.
- 7.96** But in other instances, it can be solved by just taking one single pill, a type of medication in the family called sulfonylurea. It's a type of medication called an insulin secretagogue, and it's enough to overcome the absence of that one little cog in one of the machines, if you will, that's producing insulin. That's why I noted that the therapy becomes life-changing.
- 7.97** Rather than having to wear an insulin pump or constantly pricking and injecting insulin, they can take one pill, and it's sufficient to overcome that genetic absence, and the beta cells are able to produce insulin like normal. So again, if someone listening notices this familial frequency, it's possible that you have MODY, and then you just have to get a genetic test in order to confirm that.

7.98 So in conclusion, you are now experts in understanding the dynamics between the alpha cells and the beta cells, the two predominant players within the endocrine pancreas. With that knowledge, you're pretty well-armed to appreciate the main forms of diabetes— where they come from and what to do about it. Thanks.

Lecture 8: Functions of Fat

At a glance — Functions of fat — adipose tissue as an unexpected endocrine organ.

- 8.1** I left the "chubbiest" topic for last. It's also a bit of an unexpected one because when you think about the endocrine system, you don't often think about fat tissue as being a hormone-producing organ, and yet it is very active in the realm of endocrinology. In fact, this is one of those tissues that's relatively newer to the endocrinology realm. In fact, if you were to look at an endocrinology textbook from just a few years ago, it won't include the adipose tissue or fat tissue at all. It's one of the newer ones.
- 8.2** But as a personal aside, when I was a young graduate student, about 20 years ago, it was me finding out that fat tissue is an endocrine organ that actually put me on the path that I still find myself on to this day. It was an extraordinarily fortunate event in my life, and it has left me with, to this day, a fascination and a particular affection for fat tissue. But to say that all another way: fat tissue is far more than just an inert storage organ. It does far more than just store calories in moments of plenty for when we may need them later. As you'll see, it's very, very active in producing hormones and then sending signals throughout the body.
- 8.3** The order of this lecture will be touching on a topic I've discussed previously here, which is the nature in which fat tissue even grows. How does fat tissue grow? Then how does that growth in turn influence its production of hormones? Because it does have an effect. The fat tissue is unlike any other tissue, where the bigger it gets, the more its hormone profile starts to shift. That's probably because other hormone-producing tissues don't change in size to the degree that the fat cell can change—because indeed, no tissue can change that much.
- 8.4** Then we'll look at some of the hormones from the fat tissue. Then at the very end, once again, we'll just come back to some fat tissue disorders. Then, because I've never been able to talk about this in a Peterson Academy lecture, I'm going to just spend a few minutes describing some of these new famous weight loss drugs because they do have an effect on promoting the loss of fat from the body.
- 8.5** All right, so I noted a moment ago that fat mass, when the fat tissue is growing, it can grow through two different ways. One way is through a process called hypertrophy, where each individual fat cell is getting bigger and bigger and bigger. You could have one fellow, one version of me who's gained 10 pounds of fat through just hypertrophy. I don't have any change in my number of fat cells, but the fat cells are larger.
- 8.6** Another version of me could have fat cells that stayed the same size, but I still gained 10 pounds of fat, but it's because my fat cells were replicating. So they're storing more. I have 10 more pounds of fat, and yet I have more fat cells. So one is a matter of size, hypertrophy. The other is a matter of number, hyperplasia. As you'll come to find out,

hyperplasia is actually a healthy way to store fat. If you have more but smaller fat cells, you will have healthier fat cells.

- 8.7** Now, there are some other terms that's used here. Throughout the course of this series, I've used the word lipogenesis a few times, the synthesis of new fat. Well, that refers to the fat molecules that are stored within a fat cell. When a fat cell is undergoing hypertrophy, part of what's fueling or supplying the substance for that growth is lipogenesis— making new fat within the fat cell. That's not the same term to use to describe the hyperplasia here, when fat cells are multiplying, that's adipogenesis.
- 8.8** Just to have some new terms get worked in here, hypertrophy is linked to lipogenesis, the synthesis of new fat. Hyperplasia is linked to adipogenesis, the synthesis of new adipocytes or new fat cells. So new fat cells versus new fat, but being stored in existing fat cells. That's the difference.
- 8.9** Now, with those differences in mind, coming back to the sex lecture from before and all of the endocrinology, of what makes the little boy look like a man and what makes the little girl look like a woman. I mentioned to you that sex hormones influence where the body stores fat, but also how the body stores fat.
- 8.10** Because as much as estradiol is stimulating fat storage at the prototypical female sites, breasts, hips, and buttocks, it actually does so by activating adipogenesis. And so the woman will have more, but smaller, fat cells than her male counterpart. This is why, invariably, a female will have more body fat than the male, but be healthier in every metabolic marker because the fat cells are small. As we'll come to, it's large fat cells that promote metabolic problems, not small fat cells, even if there are more of them.
- 8.11** Now, underlying all of this is this lingering appreciation of the fact that none of this can happen unless insulin is elevated. There is no such thing. That's a point I'm going to make multiple times in this lecture. There is an endocrinology to fat storage that goes well beyond just counting calories. Insulin is an absolutely essential signal. All right, I'm going to come back to that in a moment.
- 8.12** But let's come back to the nature of fat cells growing. I'm going to be a little brief in this because once again, this is a topic I have touched on previously. Remember, fat tissue can grow through hypertrophy or through hyperplasia, more cells growing. There are two things that go wrong when fat tissue is growing through hypertrophy. The first one is that the fat cell actually gets to a point of maximum dimension. This is enormous. The fat cell can grow about 10 to 20 times in volume. This would be taking a small little balloon that we only filled up with water to this size and then allowing this naughty little child to fill it up to 10 to 20 times the volume.
- 8.13** Now, as you can imagine, the more that's getting filled up, the closer and closer we're coming to the membrane not being able to hold it together anymore. And thus, we run the risk of popping the water balloon and making a big mess. To the fat cell, that same

thing can happen. A cell can only maintain its integrity to a certain point. And so as the fat cell's undergoing this mounting hypertrophy, it starts to, if you will, sense that it's reaching this maximum dimension, and that if it continues to grow, it will start to degrade and burst out of its cell membrane, which would be a very messy, very bad, very inflammatory process. It would not be ideal.

- 8.14** Now, what is the signal telling the fat cell to grow? It's not calories. It's that little petite peptide, that humble hormone, insulin. Insulin is the signal, the singular signal that is necessary to tell the fat cell to grow. I'll come back to this again in a moment. I keep saying that, but there's still a lot to come, I guess, if you will. I'll say it that way. Where calories matter, but only in so far as the fat cell knows what to do with those calories. Insulin is the signal telling it to store it.
- 8.15** So if insulin is continuing to try to force-feed the fat cell, which has already reached its maximum dimension, the fat cell can't tell insulin to stop coming. It can't lower insulin levels, but it can lower its sensitivity to it. So what does the fat cell do to ensure it stops growing? It becomes insulin resistant. Now, what normally is happening is insulin is coming and telling the fat cell to take in fat and hold on to it by inhibiting lipolysis, we end up with this quirky metabolic milieu where insulin is still high because it always is in insulin resistance, but now free fatty acids are also going up. This is a strange thing.
- 8.16** Remember, free fatty acids are the product of lipolysis. When you have a triglyceride molecule, which is the stored form of fat, and when you're breaking down that fat, you're pulling off individual free fatty acids one at a time. And so when insulin is up, normally we inhibit lipolysis. High insulin normally relates to low free fatty acids because you stop fat breakdown. When insulin is low, like during fasting or low carbohydrate diet, then free fatty acids would be up because you're breaking down the fat more readily.
- 8.17** But as I noted, we have a paradox here because insulin is high because they have insulin resistance, but the fat cell has gotten so big, it can't hold on to it anymore, and it starts leaking its free fatty acids. That is something you can measure clinically where they have high insulin and high free fatty acids. That is proof positive that the fat cells have gotten too big and that they are reaching this hypertrophic limit. That's the first problem. It's essentially the fat cells way of saying, "I can't fit anymore. I can't eat anymore. Insulin, stop force-feeding me. I'm going to stop listening." The first adaptation to stop the growth is to become insulin resistant. But we're not done.
- 8.18** I said there are two problems. When the fat cells have gotten so big, they've created a blood flow problem. Normally, a cell must be within just a few micrometers of a capillary, and its life-giving blood. The capillary is the unit of the blood vessel that actually perfuses through tissues. You'll have an artery going to the muscle, and then before it gets to the vein coming back up, it will go through this series of capillaries. That's where you're moving things in and out. Capillaries are really porous. It's giving up

oxygen, for example, and nutrients, and pulling in metabolites and waste products and CO₂ to take back to get rid of it.

- 8.19** Normally, the cell has to be very close to the capillary. But what happens when the fat cell has expanded in its 10 to 20 times volume? Now what happens is the fat cells start to get pushed too far from the capillaries, and they start to become what's called hypoxic. That's a term to just say that the cells are suffocating. They can't get enough oxygen. Now, we don't want the fat cell to die. That would be a messy, inflamed process. And so the fat cell thinks, "I need to tell those capillaries way over here to start growing this way towards me so I can get some of the life-giving blood and oxygen and nutrients that it has."
- 8.20** To send that signal, like a trail of breadcrumbs, it starts releasing these pro-inflammatory hormones that we often call cytokines. Cytokines, and another molecule being C-reactive protein—because you can measure it, it's often measured in clinical lab tests—they are simply hormones of inflammation and immunity. But they are hormones by every metric. So cytokines are hormones.
- 8.21** So the hypertrophic fat cell becomes inflamed, not to try to beat some infection, but because some of these pro-inflammatory hormones will act like a trail of breadcrumbs. The vessel, the capillary will sense it, and it will start budding off a capillary to come and feed this area, which is like an oxygen or blood desert, and bringing its life-giving water in the form of blood.
- 8.22** Now we have inflammation spreading throughout the body. But pity the poor fat cell, because as the fat cell underwent hypertrophy, it became insulin resistant to prevent further growth, lest it explode. It became pro-inflammatory to correct the hypoxia, lest it suffocate to death. But these two things that the hypertrophic fat cell had to do to survive now end up creating metabolic havoc throughout the rest of the body. Insulin resistance starts to spread. The inflammation, as I noted in a previous series, is another cause of the insulin resistance. And so the rest of the body will start to suffer all so that the fat cells can survive.
- 8.23** All right, so now you know why growing fat cells are a problem. Let's get in more back to the endocrinology—even though there has been a lingering endocrinology here. Remember, you cannot have fat cells grow unless insulin is stimulating that growth. Don't forget that. That's the entire reason I even could justify talking about fat mass expansion in an endocrinology lecture.
- 8.24** Now, when we talk about the endocrinology of what's coming from fat, we need to look at the two distinct fat depots. I've mentioned both of them. We have brown adipose tissue, which is tucked around the clavicular, thoracic area. Then we have white adipose tissue, which is any of the "jiggle and pinchable" fat, that's going to be white fat.

- 8.25** But there's also white fat that is deep within the visceral space called visceral adipose tissue. When we talk about white fat, it can either be subcutaneous, what we pinch and jiggle, or it can be visceral, which is going to be tucked beneath the muscle layer. That would be like the guy who has the big hard beer belly. He's pregnant with visceral fat. He looks like he's pregnant with a baby, but there's no darling little cherub that's about to come out. It's just a bunch of visceral fat.
- 8.26** Now, there is a mistaken belief when it comes to visceral fat. Many people think that visceral fat is inherently harmful. That's not true. The problem with visceral fat is the limitations of the space, the volume. Subcutaneous fat can grow out and it can dangle and hang around. So it can grow more through hyperplasia. So there's more production of new fat cells. So if a person is gaining more fat in their subcutaneous space, there's generally going to be more fat cells, but smaller, so healthy.
- 8.27** Visceral fat cannot allow such growth because it is within a very narrow realm, the visceral cavity. There's intestines, there's kidneys, there's the adrenal glands, the liver. All of these tissues are right there. If we allowed the fat mass to just grow without any checking of it, it would start to compress that space so much that we would die. It would compress the blood vessels and all these tissues.
- 8.28** So visceral fat cells tend to be a problem, not because they're there. They actually play an important role in immunity. So it's there for a reason. But when they are growing because of lots of insulin, stimulating the growth and calories to fuel that growth, they can only grow through hypertrophy. And so visceral fat cells are more problematic than subcutaneous fat cells because they can only grow through hypertrophy. And we now know the two relevant mechanisms that explain why hypertrophic fat cells are so problematic.
- 8.29** Okay, so let's come back to the brown adipose tissue and then the white adipose tissue, whether it is subcutaneous or visceral. Now, I mentioned before the brown adipose tissue is unique because it also produces a lot of heat. When you look at brown fat, it's brown because of how much mitochondria it has. White fat is very white because it has almost no mitochondria. It has a little, but very, very little. Almost everything you see is just one single big lipid droplet. But the tissues look fantastically different, have fantastically different metabolic rates, where brown fat can have a metabolic rate that's easily 20 times higher than the metabolic of the white fat. That's because of this process called mitochondrial uncoupling. I used the metaphor of revving the engine in a previous lecture.
- 8.30** Now, my lab actually published some pretty cool papers, and so I just want to mention these, where we found that ketones are capable of making white fat behave more like brown fat. But in contrast, we found that insulin makes brown fat behave more like white fat. Insulin wants to slow the revving of the engine down in brown fat, whereas ketones want to accelerate the engine of the very sedentary white fat. So ketones make white fat behave more like brown fat.

- 8.31** Insulin makes brown fat behave more like white fat. There are some interesting metabolic consequences to this. Then, of course, underlying all of even the ketone part of it is, if insulin is elevated, it's inhibiting the production of ketones. So insulin is going to dominate that environment. If insulin is up, you're having the brown fat behave more like white, slowing it down, and you're inhibiting the production of ketones, making sure that nothing's going to interrupt that tendency.
- 8.32** Now I noted, at the beginning, that we often think about fat tissue as just being a storage organ. I've said multiple times that insulin is essential to that phenomenon, that if you want fat to be stored in fat cells, insulin must be elevated. That is in effect of insulin's signaling.
- 8.33** When we look at the endocrinology of insulin coming to the fat cell, every single gene involved in telling that fat cell to make fat is turned on. Every single gene will be turned on the moment insulin comes and knocks. Insulin knocks on the door of the fat cell, then the fat cell says, "I know what to do now. Guys, mobilize all the machinery to turn on every enzyme that's going to make us get fatter." Every enzyme, and there are multiple, is going to be turned on in response to insulin.
- 8.34** To just bring this down to its simplest level, in my lab, at any moment, we're growing fat cells in little petri dishes. Right now, even back in my lab. These fat cells are swimming in a sea of calories. Tons of glucose, tons of fats. All the calories a little fat cell needs to become a big fat fat cell and realize its potential and all its glory. However, it isn't getting fat. When we look at these fat cells day, after day, after day, they stay very small. They're not even recognizable as fat cells. They're not, in any practical sense, until we add one thing.
- 8.35** The moment we spike in insulin into that culture, into that little petri dish, now the fat cells know what to do with that energy, with those calories. They start taking in the calories and storing it as triglycerides, the stored form of fat. If we check those fat cells just about 6 hours later, already, we'll start to see a big fat bubble or a lipid droplet, which identifies the cell as a fat cell. But until then, nothing happens. The fat cells just stay these skinny little depressed fat cells. Only when we spike in the insulin do they now get big and juicy and happy, if you will. All right.
- 8.36** This phenomenon is sometimes abused in people with Type 1 diabetes, where a Type 1 diabetic, prior to their diagnosis, will be experiencing this metabolic paradox of an insatiable appetite while wasting away and getting ever thinner. Now, imagine, get into the head of a young teenager, perhaps, who's just been diagnosed, because it helps us then have some empathy with the temptation here. But they will—again, they're used to eating whatever they want and being super, super thin.
- 8.37** Then they get diagnosed with Type 1 diabetes, thankfully, otherwise they would die. Now, by the time they leave the hospital, maybe having, depending on how serious

things got, just days later, they have gained so much fat that they can't wear the same clothes they came in with.

- 8.38** So some Type 1 diabetics refer to insulin as their fat shots. Because even if they don't understand the biochemistry—they might not be able to envision the endocrinology and the signaling in the fat cell of insulin's effects—they know that the moment I started injecting with this, I got fatter. And paradoxically, I started eating less. So calorie consumption went down, and now fat mass is going up, as opposed to eating as many calories as you want, and you can't store a single speck of fat.
- 8.39** So here's the temptation, here's where it can get abused. There is a condition called diabulimia, and you can hear in there a hint of an eating disorder. But imagine the temptation. They don't need to starve themselves. They don't need to vomit. They can eat whatever they want and enjoy all the satisfaction of chewing and swallowing and letting it sit in their stomach. All they do is deliberately underdose their insulin. They'll be as skinny as they want.
- 8.40** Of course, they're going to be massively hyperglycemic. They may be getting ketone levels so high that they get into ketoacidosis. So they will feel miserable. They'll be lethargic, they'll have headaches, and they will die young, but they'll look great in their swimsuit or whatever it is they want to wear.
- 8.41** But just imagine the temptation to abuse that fact where they just would think, "Wait, you're telling me all I need to do is not poke myself with a needle and inject this insulin? Okay, maybe I'm just going to do that. And then I can be as thin as I want." That's diabulimia. It is one of the things I point to to just really, I would imagine, settle the debate. A lot of people will say, "No, fat cells are just a matter of calories in, calories out." No, because the calories need The fat cell needs direction, and it's insulin that dictates the use of those calories. That's not to say calories don't matter, but the fat cell needs to know what to do with those calories, and that's where insulin is so important.
- 8.42** All right, so let's talk about some of these hormones, and let's start with the brown fat tissue, the brown adipose tissue. Brown adipose tissue makes a lot of hormones. I want to highlight what I consider to be the biggest two. The first one is T3, the dominant form of thyroid hormone. Now, we got done earlier talking about thyroid hormone and its place in the hypothalamic-pituitary axis and all the effects of thyroid hormone, how it's made.
- 8.43** Interestingly, brown fat does not have the follicle, it does not have the colloid. It has none of the machinery to make thyroid hormone. And yet when the body is cold and you start shivering, the brown fat becomes the dominant source of the active thyroid hormone in your blood. So if you measure T3, the active form of thyroid hormone, most of it's coming from brown fat. So temperature starts to drop. Brown fat is going to

start to make more energy because brown fat makes heat. And so you can stimulate it with cold.

- 8.44** But thyroid hormone also increases metabolic rate. Remember, in people who have hyperthyroidism, they feel hot. Well, if you're shivering because you're in an ice bath or you're out in the cold, then thyroid hormone will go up. It wants to help with that process. But interestingly, what starts to happen is the brown fat starts pulling in T4. You'll remember, the thyroid gland makes 20 times more T4 than it makes T3. But T3 is the active form of the hormone. T4 starts going into the brown adipose tissue. And when the brown adipose tissue gets stimulated because it's cold, it starts converting the T4 into T3. It becomes the dominant source of that monodeiodination that we talked about. So we coupled the diiodotyrosines to make T4. Now, the brown adipocyte, the brown fat cell, has stripped off one of those and turned that T4 into the very, very active T3.
- 8.45** During cold exposure, the brown fat tissue becomes the dominant source of active thyroid hormone, more than even, well, any tissue, including the thyroid gland itself. That's a helpful thing in that moment because one of the effects of the thyroid hormone will be to stimulate the production of new mitochondria and mitochondrial uncoupling to help the body have more heat.
- 8.46** Brown fat tissue releases another hormone called FGF21, fibroblast growth factor 21. What's interesting about this is once again, it will stimulate— the brown fat will be stimulated to release this with cold. Because when we get cold and start to shiver, we activate the sympathetic nervous system. We are alert. We are ready for fight or flight, even though we're half naked sitting in a pool of ice, of icewater or something. Anytime we're cold and shivering, we activate the sympathetic nervous system. Through a neurocrine signal, that nerve, that sympathetic nerve, is coming to the brown fat and telling it to release more FGF21.
- 8.47** But you'll recall, what's interesting here is the autocrine effect. Because while that FGF21 can flow through the blood and have effects at the liver or the muscle, it also comes right back to the brown fat that made it. So we have that autocrine signal. You'll recall the very first topic we discussed: endocrine signaling when the hormone goes into the blood; paracrine signaling when it goes from one neighbor to the next, like in the microenvironment of the islets of Langerhans of the pancreas.
- 8.48** Now we have autocrine, where a hormone is being made from a cell and it comes right back to that cell. It's like I toss a ball up and I catch it. The fat cell will release FGF21, the brown fat cell, and it will come back in turn and tell it to make more mitochondria. It'll tell it to burn more fat, burn more glucose, have more mitochondrial uncoupling to rev the metabolic engine even hotter than before.
- 8.49** All right, now, we don't have too much brown fat, but we all have some, and we can develop more, and we can activate it more with things like cold exposure. The

predominant fat, of course, is white fat. Now, everything I'm going to talk about now applies just as much to visceral fat as it does to subcutaneous fat. But when I talk about the differences with hypertrophy—because that changes the profile of the hormones that we make—you'll have to just remember that visceral fat undergoes hypertrophy more readily than subcutaneous fat does.

- 8.50** But there's three hormones in particular I'm going to mention. We'll spend most of our talking about leptin and adiponectin. You've heard of leptin, you've probably not heard of adiponectin. Then I'm going to mention briefly this poster child for pro-inflammatory hormones—remember, those are called cytokines—called TNF- α . Because when the fat cell gets big, it starts making a lot of TNF- α . That's one of those pro-inflammatory hormones that's starting to get dumped from the fat cell when it gets really big and inflamed.
- 8.51** So firstly, leptin, it's the most famous. And I spoke about leptin more in the Fundamentals in Physiology course, so I don't want to repeat the very fascinating history of it, but it is interesting. And suffice it to say what was thought to be a very promising weight-loss therapy was an absolute bust because most people not only have plenty of leptin, most obese people have more than plenty. So it's not a matter of leptin not being there.
- 8.52** Now, what tells the fat cell to make leptin: the strongest signal is insulin itself. When insulin comes to the fat cell, binds its receptor—one of its many, many pathways, and there are dozens to hundreds—it will be to induce the production of leptin. Now, there are other signals that will tell the fat cell not to release leptin, like norepinephrine, one of those catecholamines, but that's a more modest signal. The strong signal is stimulatory. Insulin wants the fat cell to make more leptin.
- 8.53** And then leptin has various effects throughout the body. But before I mention those, it's important to note that the more body fat is going up in an individual, the more leptin is going up. And this is reflective of what I mentioned a moment ago, that far from leptin being a therapy, when they started to test this in humans, they found that it had no effect whatsoever because bigger people have more leptin already. It's just a drop in the bucket. It's just not working very well.
- 8.54** However, having said that, there are people who can be born with actual leptin mutations, and there's nothing subtle about it. If a person has an actual leptin mutation, they will become fantastically obese in very early childhood. It will not be subtle. And there's multiple things that leptin is doing that in its absence enable this massive fat gain, and we'll get into those in a moment.
- 8.55** But just importantly, I don't mean to imply leptin isn't important for body fat regulation. It is because if you can't make it, or you can't sense it, then you get none of the benefits of leptin, and the person will become fantastically obese. In that situation,

by restoring the leptin in the event the person is not making it, you do correct the weight gain immediately. They'll immediately start losing weight.

- 8.56** Let's just go through some of leptin's effects from top to bottom. We'll talk about brain, fat cells, beta cells—because to understand why an absence of leptin leads to such obesity, you have to appreciate what is now not happening with insulin control. Then we'll mention the muscle, too, because leptin affects all of those. When most people think about leptin, though, they only think about the brain and they only think about satiety. I actually think that's not fair to leptin, because leptin does a lot, as we'll get into in a moment, affecting tissues throughout the body.
- 8.57** But leptin is one of many satiety signals. But it is a satiety signal. When leptin comes to a distinct part of the hypothalamus called the arcuate nucleus, it will induce the production of something you've heard of before, pro-opiomelanocortin. Do you remember where I've mentioned that previously? Where did we talk about pro-opiomelanocortin? It was in the CRH-ACTH-cortisol signaling.
- 8.58** I mentioned that you'll recall with Addison's disease, when the person is not making cortisol, there's no negative feedback and they're making a lot of ACTH. Remember, JFK had a famously coppery complexion because he had Addison's disease? When the anterior pituitary makes POMC, it splits it into ACTH and MSH, melanocyte-stimulating hormone. Does that sound familiar, now?
- 8.59** We have the exact same molecule, but it's in a totally different place and does not have any effect on melanin systemically. But it's the same molecule, but not in the anterior pituitary. It's in a specific nucleus of the hypothalamus. So when leptin comes to the hypothalamus, it activates the production of pro-opiomelanocortin. And pro-opiomelanocortin will have a strong satiety signal in the brain.
- 8.60** Now, again, it's one of many. I mentioned a moment ago that we're going to talk about weight loss drugs, the famous GLP-1 drugs. GLP-1 also has a satiety signal. Insulin has a satiety signal. Ghrelin has a satiety signal. There are a lot of satiety hormones. Leptin is just one of many, which is why I consider it to be somewhat unfair to just talk about leptin in the realm of satiety. Not that it doesn't matter, but that actually is not what's explaining the weight gain. What explains the weight gain is going to be the insulin effect in just a moment.
- 8.61** But leptin has an effect at the fat tissue. So fat tissue makes leptin and it responds to leptin. Can you remind me? What is that type of signaling called? Not endocrine, not paracrine. Autocrine. Yeah, it's self-activating. So it has an autocrine effect where when leptin comes to the fat cell, it's basically turning on any of the pathways related to lipolysis. So leptin will directly stimulate fat breakdown at fat cells. Of course, in that sense, leptin becomes very helpful for maintaining a healthy body fat level, wanting there to be a little less body fat at any given moment.

- 8.62** Further helping that is what leptin does at the muscle, because it's one thing to break down the fat, but then it's another thing to burn the fat. The muscle is the main fat burner. It's the big hungry engine that wants to take in any fuel it can. Thankfully, as much as leptin is telling the fat cell to share its fat, it's also enabling the muscle cell to take in and burn that fat. We have a place, we have a destination and a metabolic outcome with that fat. We're breaking it down and we're burning it. Well, that's also very favorable to maintain a healthy body fat level.
- 8.63** But let's come to the pancreas, because this is where we actually can get a better understanding of the consequences of not having leptin. You would say, "Well, Ben, it's easy. If you don't have leptin, then you're eating more. If you don't have leptin, then you're not breaking down as much fat or burning as much fat." Yeah, but insulin is the one hormone to rule them all when it comes to fat metabolism. But even here, it makes some sense.
- 8.64** Earlier I mentioned that insulin is a signal that stimulates the production of leptin. Leptin in turn is a signal that inhibits insulin production. If you've lost leptin, what happens to insulin levels? They go really high. And in fact, the first scientist who discovered the effect of leptin, and all the focus was on leptin, he noted that insulin levels—it's his term—goes through the roof.
- 8.65** And so anytime a person doesn't have leptin, their insulin levels are way higher than they need to be. And that high insulin is pushing the fuel into the fat cell and constantly wanting glucose levels to drop. And so what is a person wanting to do? They're wanting to restore those glucose levels so they're eating all the time. Just the high insulin effect is muddying the conclusion that it's all just because of no leptin. Lack of leptin means a lot of insulin, and a lot of insulin will mean fat storage and hunger because you're tucking all the calories away in the fat cell and the rest of the body, the brain included, is saying, "Well, I need some calories, so I guess I'm just going to have to eat some."
- 8.66** Then as a reminder, whenever you have too much of a hormone, including leptin, because the person has too much fat mass, the signal starts to go down. The same thing that happens with insulin—that I spoke about in *Why We Get Sick*—too much insulin is the main driver of insulin resistance. Similarly, too much leptin is the main driver of leptin resistance. And this explains why people that are overweight will invariably have more leptin than someone who's lean.
- 8.67** Okay, now let's talk about a close sibling to leptin that's coming from the fat cell, which is adiponectin. Adiponectin is the hero from the fat cells that most people have never heard of. And it actually goes in the opposite direction from leptin. With leptin, the more fat mass a person is accumulating, the more leptin is being released. But in contrast, the more fat mass a person is accumulating, the less adiponectin is coming out. This is why I noted, that as an endocrine organ, fat is so unique. There's nothing else that changes its levels like this based on its size. But again, the more fat a person has, the less adiponectin, and you'll see why that's a problem.

- 8.68** Just like with leptin, adiponectin will be created from the fat cell in response to insulin. Insulin will bind its receptor. It will initiate a signaling cascade that will result in the production of adiponectin, then adiponectin will come out and be released. However, as the fat cells expand more and more and more, we have another autocrine signal. The fatter the fat cell gets, you'll recall, the more it starts producing pro-inflammatory hormones like TNF- α .
- 8.69** TNF- α is one of the strongest pro-inflammatory signals in the body. I actually have used it in my own research during my postdoctoral fellowship days, causing insulin resistance. We could take fat cells—which is what we're discussing here—incubate it with a little bit of TNF- α in this little bath, and within just 30 minutes, measure how responsive it is to insulin and detect that the insulin signal had decayed significantly. We could make them very rapidly insulin resistant.
- 8.70** Part of the problem is here, where when TNF- α starts to be expressed from the swelling, fattening fat cell, it will come back to the fat cell and inhibit the fat cell's ability to produce adiponectin. So it blocks that adiponectin production. That is going to be a problem, as I'll show you in a moment, when we talk about how adiponectin helps with insulin sensitivity.
- 8.71** But before we do that, let's just go through some tissues: adipose, liver, and muscle. So once again, we have an autocrine signal where the fat cell is producing adiponectin, and the adiponectin is coming right back. Well, that has a lot of favorable effects at the fat cell. It'll tell the fat cell it helps it pull in glucose. It helps the fat cell burn more energy. So a fat cell has mitochondria, and so it can burn fat. And so it stimulates a little bit of fat burning within the fat cell. At the same time, it inhibits inflammation by producing anti-inflammatory proteins.
- 8.72** In the muscle, these beneficial metabolic effects continue. Once again, your adiponectin is stimulating glucose uptake. It is also stimulating the production of new mitochondria. And within those mitochondria, it's stimulating the burning of fat. And again, just like it did with the fat cell, it even tells the fat cell to burn more fat, and it's also telling the muscle cell to burn fat.
- 8.73** Now, within the liver, the liver is involved, as I've noted, in handling of all nutrients. It does so much. Once again, adiponectin is having this catabolic effect. It wants to break down. Whether it's glycogen, adiponectin wants it to break down glycogen. Fat burning and lipolysis, adiponectin wants the liver to be breaking down and burning fat. So all of those effects, once again, the catabolic balance of metabolism is dominating in response to adiponectin.
- 8.74** Now, a moment ago, I'd mentioned that one of adiponectin's effects, and in fact, this appears to be an effect throughout the entire body, not specific to any of the cells I just went through, not limited to those either. When adiponectin comes and binds, it activates an enzyme called ceramidase. Now, we haven't, in the context of this series, I

haven't mentioned the word ceramides. However, in *Why We Get Sick*, I spoke about how ceramides are the main molecular mediator of insulin resistance.

- 8.75** No matter how a cell is becoming insulin resistant—and at the time, I described some main causes like stress hormones, inflammation, too much insulin— all of those cause insulin resistance because they promote the synthesis and accumulation of ceramides within the cell. Then ceramides block the insulin receptor. It's like sticking the stick in the spokes of the bike. It has stopped.
- 8.76** But adiponectin activates an enzyme called ceramidase, and ceramidase will take the ceramide and break it apart. Now, where you have more adiponectin coming to a cell, you will invariably have less ceramides in those cells. Where you degrade ceramides, you restore insulin sensitivity. We have one last profound feather in the cap that is a benefit of adiponectin in a very broad, very relevant metabolic sense.
- 8.77** Now, collectively, these two hormones actually once again enable us to get some insight into the fat cell because I described that when a fat cell is growing, what's happening to leptin production? It's going up. In contrast, what's happening to adiponectin production? It's going down. Now, clinically, unlike in my lab, you could not go to your clinician and say, "I want to know how big my fat cells are." You can't do that. We do it, as you'll recall, going through that weird process of taking out fat biopsies, sectioning them, fixing them, staining them, and looking at them under a microscope, and we can quantify the literal cross-section area down to micrometers. A clinic can't do that.
- 8.78** But the leptin to adiponectin ratio, the ratio of these two hormones actually is reflective. So a person could get their leptin measured, they could get their adiponectin measured. And as you see the leptin continuing to go up and the adiponectin continuing to go down, that's going to be very strong evidence of the fat cell getting bigger— the hypertrophic fat cell. And the reason that matters is because that, in my view, is the origin of so much metabolic mayhem.
- 8.79** As much as other tissues can start to suffer, and we talked about some today, liver, fat tissue, alpha cells of the pancreas, they're not the first ones to go. I very much have a fat-first focus. If we're stacking up these tissues like dominoes, the first one is going to be the fat cell, and it's because it's undergoing hypertrophy. So that leptin and the adiponectin ratio can be somewhat revealing when it comes to understanding the size of the fat cell.
- 8.80** All right, that ends our formal discussion of the endocrinology of adipose tissue. But there is a fascinating disease, a disease of not being able to make fat cells. It's a condition called lipodystrophy. Doesn't it just sound glorious? We curse our fat tissue so much, and yet people who can't make it suffer from horrible consequences. But think, if you will, for a moment, how will a person look? Imagine if you cannot make fat tissue. Then think of the body— even a lean person has subcutaneous fat beneath their

skin to some degree, they have some fat within their cheeks—but this person has none. So they look fantastically ripped. They are exceptionally lean to the point that you can see the very ripples of their muscle. You can see the blood vessels. This person cannot make fat tissue.

- 8.81** Now, there's two different forms this takes. Generalized can be either congenital, they're born with it, or it can be a result of infections, like HIV can contribute to this too. But they will have no fat anywhere. They have destroyed their fat cells. That's a generalized lipodystrophy. The more common form is actually partial, where an individual will have this oddly patterned fat, where—usually in a woman, but a man can have this too—the individual may have a lot of fat from the buttocks, hips, down through the legs, and little to no fat from waist up. Or it may all be waist up, and then there's none on the buttocks, hips, and legs.
- 8.82** So suffice to say, with general lipodystrophy, it's everywhere, no fat cells. With partial, there's lots of fat cells in some sections and no fat cells—I mean, exceptionally lean, the same striated, ripped muscle view—in other places. So very unique and very paradoxical.
- 8.83** In the individual who can't make any fat cells, they won't have any adiponectin being produced, literally none. They won't have any leptin. Now, think about the consequences of the absence of leptin. Do you remember why leptin is relevant to fertility? Leptin is going to be a signal that comes to the brain, activates that kisspeptin protein, and enables the pulsatility of GnRH to enable normal puberty and even normal fertility, because that pulsatility is relevant, than just puberty itself. It maintains a relevance throughout adulthood.
- 8.84** In the absence of leptin, you have an absence of sufficient GnRH pulsatility, and so the person will be infertile. And in fact, they won't even experience normal puberty, male or female. You have to have leptin. Now, the good news is you can give them leptin injections. That can be a treatment to give them leptin to help them restore that function and have puberty and fertility, et cetera.
- 8.85** But at the same time, throughout this, if you can't store fat in fat cells, you're going to start storing it in other places. So the person will often have a lot of fat stored in their muscle. They'll often have fatty liver disease, creating liver problems as well. So they actually start to have some characteristics of the overweight Type 2 diabetic, where they have this phenomenon called ectopic lipid storage. Fat is meant to be stored in the fat cells. If you can't make the fat cells, then you're going to store that fat somewhere else. So one therapy is giving them leptin.
- 8.86** Another therapy is giving them a class of drugs to force the production of new fat cells, a process we've not discussed, but it basically mimics the adipogenesis. You can force that to happen with certain drugs these days. So someone with lipodystrophy could be

taking leptin as long as they need to. And in some instances, even this other drug class called PPAR- γ activators can also facilitate, you can force them to make new fat cells.

- 8.87** But as we near the end, what if a person has a lot of fat and they want to get rid of it? I've already, in the Why We Get Sick series, talked about the balance— and I've alluded to this—the balance of insulin and energy. Calories matter, but the fat cell has to know what to do with those calories, and that's where the insulin comes into play. I'm not going to revisit that now because I think I addressed that pretty well previously. But suffice it to say, I believe changing diet needs to be the main mechanism of promoting weight loss.
- 8.88** But what if a person just wants an easy way out and they want to undergo liposuction? You think about liposuction, and what's interesting about it is, where are you sucking the fat from in a person with liposuction? It is invariably going to be from subcutaneous fat. You're taking the fat from the belly or the fat from the buttocks. It's going to be subcutaneous.
- 8.89** Subcutaneous fat is healthy fat. It's usually more fat cells. They're smaller, and so they're healthier, but there's more of them, so you may have more fat mass there than you want. But remember, those are nice fat cells that are helping the body be insulin sensitive. So what happens when you go and suck those fat cells out? You will have lost fat. In fact, maybe even a lot of fat. You may have lost 20 or 30 pounds of fat, and yet not a single health marker is better.
- 8.90** This study that was published a few years ago documented a 44% reduction in abdominal fat. They came in and took out 44% of the person's fat from their belly, then they documented whether they were a Type 2 diabetic or not, and not a single health outcome got better. You'd say, "Well, they lost a lot of weight. They should be healthier." Yeah, but remember, it's not the number of fat cells that matters most for these things. It's the size of the fat cell. Remember, the female is healthier than the male metabolically because she has small fat cells. Even though she has more, it's not the number that mattered. It's that the fat cell is big or small. So when you suck out fat cells, you're not shrinking the fat cells. You just have fewer of them.
- 8.91** In fact, that creates a problem because guess what happens as the person starts to regain weight, never having learned how to change their lifestyle to lose that weight in the first place? If you suck those fat cells out, as they start gaining weight back, it starts to gain back in the fat cells that are remaining. And so if you sucked out fat cells here, they'll just get fat here now. If you've sucked out fat cells here, they're just going to get fatter here. And so the fat cells will start to get bigger, and they may paradoxically end up actually with worse metabolic health than they had before they ever started the process.
- 8.92** Okay, as we wrap up all of it, I thought at this moment in time in the world, there is a global obsession with a class of weight-loss drugs that are mimicking a hormone. It's

appropriate for me to insert this at the end because of its popularity and relevance right now, but also because it's a weight-loss drug—shrinking fat cells, and shrinking fat cells is a good thing—but it's also a drug that is mimicking the effects of a hormone. Of course, that hormone is GLP-1, glucagon-like peptide 1.

- 8.93** Now, GLP-1 is a hormone we make, but at the moment, most people think of it in the form of many, many drugs. There are lots of different drugs, with lots of different names, suffice it to say, if it ends with the suffix -ide, or especially a -glutide or an -utide, it's going to be in these classes of drugs. There are lots of different forms of it nowadays with lots of different names, with drug names. But back to the body and the endocrinology of it all.
- 8.94** There are cells within the intestine called L cells. These L cells will have the same proglucagon molecule that we had in the alpha cell that we discussed just moments ago, but it will undergo a different post-translational processing. Rather than being cleaved up to produce glucagon, with the remnants just being discarded, this will be split apart to create GLP-1, most especially—and some GLP-2 as well, which has its own more modest effects. But here we have proglucagon from the L cells of the intestine getting split apart to give us a GLP-1.
- 8.95** Anything that goes into the intestines will have some effect, if it's some bulk, it will have some effect on inducing some GLP-1 release. This is a study to simply show that you can eat glucose, or protein, or fat, and all of them will have some degree of inducing some GLP-1 release. The glucose, and the protein, the glucose will have a stronger effect at the front end and then taper off. Fat will have a delayed effect and then it will go up. Then protein has a little bit of an effect throughout the entire time. But anytime you're eating something like this, you're going to have some GLP-1 response.
- 8.96** Now, what does GLP-1 do? Well, on one hand, when it's acting in its normal levels, it will slow the stomach down. It's sending this signal saying, "Hey, let's slow down. We don't need to eat again." So food will stay a little longer. And by slowing down the movement, it is maybe helping us digest the food. But by having more food in your intestines, you're going to be less inclined to start eating again. And so that bulk of the food you just ate will take a little longer to get through you. And so that'll make you feel somewhat full because there's something in your stomach, something in your intestines.
- 8.97** At the same time, that feeling of satiety is amplified because of what GLP-1 is doing when it comes to the brain. Because GLP-1 has a central effect at promoting satiety. It tells the brain that we don't need to eat. And so the combination of those two things is a very strong effect to say, "I'm done. I don't need to eat anymore." We have the intestine slowing down, and we have the brain telling us we're full. That's a good combination to promote satiety.

- 8.98** When it comes to just weight loss and how the drugs were originally used, which is just as an anti-diabetic—that's how I learned about them 20-ish years ago as a graduate student. When I first learned about GLP-1 as a hormone and then a drug, it was only discussed in the context of controlling blood glucose. If you're eating less, little surprise your glucose will come down. But at the same time, if glucose is coming down, then insulin will come down, and it'll be easier to burn fat, too. That'll be a good thing.
- 8.99** However, in 1996, there was a fascinating paper that was published, which has been totally overlooked. I consider it a matter of divine intervention that I even found it. It is fascinating and I think revealing, suggesting a relevance to GLP-1, even maybe as a therapy, but with a more specific use than it is used nowadays in just a general weight-loss drug sense.
- 8.100** In 1996, they took people and they gave them a load of fat, and they said, "Here is your fat meal." Then they found that lean and obese individuals had a roughly similar degree of GLP-1 response. They eat this fatty meal, and for the most part, the GLP-1 is the same. You can see these data that the GLP-1 response was a little lower in the obese group at the end, but for at least the first part of it, it's matched. This would suggest that whether you are lean or obese— but that's how they noticed the differences—the lean group and the obese group will eat a fatty meal and they will have a roughly equal degree of satiety. Just based on GLP-1 alone. So the GLP-1 response will be pretty similar between the two.
- 8.101** However, the difference became shocking when they gave them a high carbohydrate meal. When they gave them a high carbohydrate meal, the lean group had a significant increase in GLP-1 response, suggesting a very high degree of satiety. But the obese group had no functional response. At no point did it reach a degree of statistical significance as an increased response. And in fact, at around two and a half hours, it actually dropped below where it started. So rather than getting a satiety signal, they have no satiety signal. And in fact, it may actually promote more hunger because again, at around two and a half hours, it started to drop below where it even started.
- 8.102** This matters, I believe, because you could have two people. One eats a given load of a carbohydrate-heavy meal, has a big GLP-1 response, the delayed gastric emptying, the satiety signaling, and they say, "I'm done. I don't need anymore. I'm good for a while, actually." Whereas the other individual eats that same amount of food and doesn't get any satiety signal whatsoever. Not surprising, they want to go back and start eating more.
- 8.103** What if the lack of GLP-1, specifically in response to carbohydrates, is one of the main drivers of someone overeating? Remember, when you eat carbohydrate, that will get digested to glucose. What is high glucose going to do to insulin? Of course, it will increase it, telling the body to store more fat. I'm going to revisit that idea in just a moment because I think that one paper, identifying that some people don't get a GLP-1 response to carbohydrates, reveals the way forward with these drugs.

- 8.104** But as these drugs are currently used and at the doses they're currently used, there are consequences that are not often discussed. One is that when you turn up the GLP-1 levels so high, like you do with these drugs at the doses they're used, you may slow the intestines down too much. This would be why we see case studies that have been published documenting that people will actually experience a paralysis of their intestines, which can become very dangerous, very lethal, and kill the individual quickly.
- 8.105** At the same time, a paper has been published that's showing over two years of use, about 40% of the weight that people lose is coming from fat-free mass. If a person is rejoicing in the fact that they've just lost 100 pounds, 40 pounds of that weight loss will be coming from stuff that is not fat. Now, you would say, "Well, that's worth the trade-off." It might not be, as we'll come back to in a moment. At the same time, it's having perhaps too much of an effect on the guts, too much of an effect on lean mass like muscle and bone.
- 8.106** There's also a profoundly relevant effect at the brain. That individuals who are on the drugs for about two years, this paper that was just published in 2025, found that these individuals on these weight-loss drugs had an almost 200% increased risk of major depression and an over 100% risk of increased anxiety and 100% increased risk of suicidal behavior. This is very relevant, for how common these drugs are being used, that perhaps we're using too much of a good thing. I do think GLP-1 is a good thing. I think it's a hormone we want to try to take advantage of.
- 8.107** Now, people will often report that, "When I take the drug, for the first time it has quieted my food cravings." Let's pause on that topic for just a moment, because what do people crave? There are three macronutrients, three main things we eat: fats, proteins, and carbohydrates. Have you ever been sitting down on a Friday night and craved a plate of bacon and eggs? You just had to have bacon and eggs? That doesn't really happen.
- 8.108** There's no neurobiology in humans that suggests a craving or an addiction for fat and protein. It's always carbohydrate. When you look at cravings in addictive food eating, it's always surrounding carbohydrate. That's not my evidence. It's decades worth of studies. And so when they report that their food cravings have gotten better, it's not that they were craving a juicy steak. It's that they were craving something sweet and gooey, or salty and crunchy. And it works.
- 8.109** This paper that was published documented that within the first few months of being on the drug, that these individuals reported that their cravings for sweet things like ice cream, and chocolate, and candies, was way down. That would absolutely prove this idea that their cravings are getting better.
- 8.110** However, when they measured it one year in, it was still good, but there was a direction change. So up through one year, the cravings were still solid. They could

report that their cravings were still quite quiet. But one year later, at two years, despite the ongoing use of the drug, their cravings were right back where they started. So the cravings, the drug, as happens so often, has experienced this diminishing returns, and now their cravings have come back.

- 8.111** The tragedy in this situation is the person starts to feel helpless, and then they start to self-administer the drug, and will sometimes increase their dosing on their own, which can lead to the intestines slowing down too much, like I mentioned a moment ago. A total return of cravings at two years. This could be why about 70% of patients get off the drug at two years. They just don't want to be on it anymore because it's not working and they feel a little nauseous.
- 8.112** And so what happens when they get off? This now makes that 40% fat-free mass loss so relevant. Because when you suddenly remove the drug, what do you think happens to weight? It, of course, comes right back. But one of the quirks of being human is that our fat mass can restore very, very quickly. We have no problem storing more fat.
- 8.113** And so when you consider what the person is regaining, we have a very hard time restoring things like muscle and bone mass—especially if this is an older individual, which, let's face it, is most of the individuals on this drug. The fat mass will come right back. So if they've lost 50 pounds and they say, "I've gained back 40 within the following six months," they may somewhat say, "Well, at least I still have 10 pounds less." Yeah, but you're probably fatter than you were before because what came back was just fat mass, and probably none of the fat-free mass. That's gone and may never come back, depending on your age.
- 8.114** To me, in light of that 1996 study, where there are some people who eat carbohydrates—which, let's face it, is the thing people crave, that's what they're addicted to—and they don't get that satiety signal. To me, if there is a best use case for these GLP-1 drugs, it is not at these really high doses and just saying, "It's a weight-loss drug," because that convinces the person that they actually don't have to change anything, that the drug is just going to magically help them lose weight.
- 8.115** To me, the best use of the drug should be, "You need to eat fewer carbohydrates because that will bring down your insulin. Insulin being down will help you shrink your fat cells and lose weight." "Okay, thanks. I can't do that because every time you tell me to eat less carbs, I do it fine for a day or not even. I just can't quit. I'm addicted, after all." And so let's have some empathy for the addict. Food addiction is a real thing. "You're addicted. Here's a drug that can help you sense eating that better. You're going to have a little bit of GLP-1."
- 8.116** My view is these GLP-1 drugs shouldn't be prescribed for weight loss. They should be prescribed to help a person learn to control their carbohydrate cravings. The drug should be used to control their cravings for carbohydrates. That then quiets the food noise, but it should be used at a much lower dose. Again, with the deliberate intention

of controlling their carbohydrates. Be liberal with fat and protein because they don't really impact insulin anyway.

- 8.117** Now the person is in a better position and with better ongoing monitoring. Maybe you see the patient again four weeks later, "Hey, how are your carb cravings coming?" "Oh, it's going really great." "Awesome. Let's keep it up." Then maybe a month or two later, they report, "Boy, I've lost 20 pounds. This is amazing. I've never felt so liberated with regards to my cravings for carbohydrates."
- 8.118** Then we say, "Okay, great. Let's see if these habits have stuck, and let's cycle you off the drug." It was already a low dose, so we were microdosing the drug, a fraction of the normal dose, literally. Now, let's not give it to you with the intention of you being on it forever. It's an effort to help you learn what it feels like and what it looks like to not have that constant craving for carbohydrates. So we started you on a low dose, we're going to cycle you off.
- 8.119** And maybe the person finds that after 90 days, which is long enough to change habits, they're done and they never need to get on the drug again. Some people may find the habits coming back. All right, let's address the habits and cycle you on again, maybe, albeit still at a microdose level.
- 8.120** So here we have it at the end, this fascinating interaction of a gut hormone being leveraged, in some instances, perhaps appropriately, where people aren't getting a GLP-1 response to certain foods, but all with the intention of shrinking fat mass, and in the process, not only improving the health of the fat, but the very hormones that it produces.
- 8.121** In this lecture, you now have some greater experience with the endocrinology of adipose tissue. What are the hormones that the fat cells produce? How does the profile of those hormones change based on the growing or the shrinking of those fat cells? And even a little bit of what to do about it.
- 8.122** You talked about some adipose disorders. In terms of lipedema— which is not really regular fat—how does that, if in any way, react with insulin, and does insulin resistance affect the lipedema tissue?
- 8.123** Yeah, it's a good question. Lipedema is, in fact, not a fat tissue disorder—well, it is, but it's not because of fat cells, it's because of collagen. Lipedema is a collagen problem, where the collagen—fat cells will naturally, like all tissues, there will be little fibrous connective tissue woven through the tissue, helping it maintain its substance.
- 8.124** The problem with lipedema is now you have this connective tissue getting too rigid. As the fat cells are trying to expand, it's pushing against this high pressure, rigid wall, which can create the discomfort, the pain, and the mottled appearance. Again, it's not because the fat cells are broken, it's because the collagen has become too rigid, so it's a collagen problem. This is one of those instances where liposuction helps. As much as I

have a very cool-ish view of—I don't look at it warmly—with liposuction, when it comes to lipedema, it is something that has been shown to help.

- 8.125** With hypertrophy of a fat cell, you talked about insulin being needed for the response. What about the hyperplasia? The creation of new fat cells?
- 8.126** Yeah, awesome. Excellent question. In fact, one version of this lecture, I actually talked about the biochemical signaling, and I took it out last night thinking, "That's too much information. It's going to take too long." Insulin is necessary for both of them, and there are both some environmental and genetic components that play into it.
- 8.127** On the adipogenesis side, insulin will promote the fat growth, and it will do the fat cell hyperplasia by activating that PPAR- γ gene. When I talked about the lipodystrophy patient, how they'll take a PPAR- γ activator, PPAR- γ will stimulate adipogenesis. In contrast, on the other side, promoting just hypertrophy, it's the lipogenic signal that comes from a protein called SREBP-1c. Insulin will still activate that. Insulin activates both. But some people have variants in PPAR- γ expression that won't allow them to make fat cells as well.
- 8.128** This starts to explain the ethnic differences that we see globally. In South Asians and Hispanic individuals, there is known to be a higher prevalence of PPAR- γ variants that prevent adipogenesis and hyperplasia. This explains, in fact—again, one version of this—I actually show you a study that took Caucasian European and South Asian Indian men of the same body size, same body fat, and did fat biopsies. The fat cells from The South Asian guys are about five times bigger diameter than the fat cells of the Caucasian guys. East Asians are going to have the same thing, Japanese people.
- 8.129** It's interesting to compare the United States. With our ethnicities, we're one of the fattest countries in the world, and yet our Type 2 diabetes isn't even in the top 70 countries. In Japan or Singapore, where I lived and did my old postdoc studies, they're very lean East Asians. It's very uncommon to see a really chubby East Asian, they maybe are a little chubby. And yet their diabetes rates they're in the top 10. Singapore, I think, is number 9. Japan's in the top 20. And yet you're walking on the streets thinking everyone's so lean because they don't make new fat cells.
- 8.130** Caucasians and blacks, especially, women more than men, even within ethnicities, some ethnicities will make more fat cells. And then paradoxically, having more fat cells, if they're smaller, is a way to be metabolically healthy. In fact, the PPAR- γ drugs are sometimes given to people with Type 2 diabetes. You'll recall, if a fat cell is growing through hypertrophy, it becomes very insulin resistant. The more insulin resistant they become, the more Type 2 diabetic they will be.
- 8.131** Think about the irony here. You have a Type 2 diabetic, very insulin resistant. The physician gives them a PPAR- γ drug, and in the process makes them fatter than they were before. But you're forcing their fat cells to multiply. So even though their fat mass has gone up, what's happened to the size of their fat cells? It's gone down, and in fact,

they get better. They're more insulin sensitive than they were before. Their glucose levels come down, all while they're getting fatter. Within six months, they can easily gain over 50 pounds. That's a hell of a trade-off. But it's part of this paradox of: it's not the mass of fat that matters most, but the size of the fat cell.

8.132 But back to the question, explicitly, insulin will drive both of them. One of the reasons I think seed oils are problematic—and I say that they're an indirect cause of insulin resistance—because when a seed oil is converted into a peroxide metabolite called 4-HNE, it prevents adipogenesis and only leaves lipogenesis. So it forces the fat tissue to grow through hypertrophy, which, of course, will then be a cause of insulin resistance.

8.133 Could you just expand on why depression increases 20% on GLP?

8.134 Oh, yeah. It was over 200%. Yeah, not 20%.

8.135 I thought I said 200. Yeah.

8.136 Yeah, okay. You might have. I think you said 20. Yeah. Depression increases dramatically. Actually, I don't know, I can only speculate. I can only speculate. I don't know of the direct effect of it. I'm just not enough of a brain guy to know the answer to that. I'm speculating, and it's even almost a philosophical speculation in a way.

8.137 We brag about the benefits of these GLP-1 drugs being, "I don't crave these foods anymore." What if, however, the bigger effect is, "I don't crave anything anymore." You do have reports of people noting, describing their depression. You have a guy who used to get together and play pool with his buddies. "I don't really want to go." A gal would go play pickleball or go on a walk with her girlfriends. "I don't really want to go." What they'll report is that their enthusiasm for, their cravings for life, is just gone. I think that is actually more accurate than it may seem. That as much as we focus on the reduced food cravings, I think that might actually just be one reflection of, "I don't really crave anything anymore."

8.138 One reason I feel so strongly about those drugs is because as a college professor, I now have met multiple perfectly young, adorable, darling little girls at school, who have parents that have somehow enabled this behavior, who are on these drugs, and they are already thin. It has become a way for someone to enable an eating disorder.

8.139 That worries me profoundly because if you have a gal who already has a low self-esteem and is constantly trying to lose weight, and now you're compounding this, it scares me a lot. These drugs, I think they're so powerful that they need to be used more carefully than they are. I think they have a potential, but it's not how they're being used currently.

8.140 What role, if any, does cortisol play in the development of visceral fat and/or ectopic fat?

8.141 Yeah, great question. Little role with ectopic fat, just to get that one out. But it does play a role in promoting visceral fat. The problem with interpreting the effects is that a

lot of the studies that have been done showing cortisol's effects on visceral fat come from rodent models. And humans, we're not rodents, but of course, that's a pithy answer because sometimes we can really rely on the rodent data. But when it comes to cortisol and its metabolites, there are other corticosteroids, and rodents make different ones than humans do. It's not as clear-cut as some people think—will point to a study, and yet it's a rodent study, and there's no comparable effect in humans.

8.142 Now, having said all of that, cortisol will have a general effect at promoting, as I noted, more central fat, subcutaneous and visceral, even in humans. So someone with Cushing syndrome and Cushing disease, high cortisol, will start to store all their fat here, and we do know the visceral fat mass will expand as well.

8.143 But remember, there's nothing inherently pathogenic about the visceral fat, but it's only a problem because of how it grows, where, as I noted earlier, this fat can only grow through hypertrophy because it wants to limit its own growth, lest it start to crush everything in that space. Whereas the subcutaneous fat will both grow through some hypertrophy and, to the degree that a person has the genetics to allow it, some hyperplasia as well. So the fat will start to bubble down or be jiggly. But here, they'll have an expansion and it'll be more hypertrophic, which is a problematic fat growth.

8.144 Thanks, guys.